

Dilation Rigidity of Surreal Numbers and Omnific Integers

Autonomous Mahler equations, finite support, sharp algebraic degrees,
a higher-order hierarchy, and the support rank of dilation orbits

Research article prepared with ChatGPT for Vladimir Reshetnikov

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and two of batch 33, pinned to `efc5446` and `958b5c4`

Abstract

For an integer $d \geq 2$, exponent dilation sends $\sum a_\gamma t^\gamma$ to $\sum a_\gamma t^{d\gamma}$. We prove that a nonconstant Hahn series algebraically dependent with its own dilate over the constant field must belong to a single ordinary Laurent series subfield. This holds in characteristic zero over every ordered abelian value group, without an Archimedean or divisibility assumption. If $F(y, S_d y) = 0$, the necessary ramification is at most $\deg_y F$. The proof combines Newton–Puisseux branches with a formal Böttcher coordinate and a least-support-term argument; every evaluation is justified by strong Hahn summability, not by a contraction principle.

For the negative-support ring this becomes an exact classification: first-order autonomous algebraic dependence occurs precisely for polynomials in one monomial. The least degree in the dilated variable is exactly the degree in the primitive monomial. Consequently every infinite-support omnific integer is algebraically independent of each of its nontrivial integer dilates. We also obtain a finite-profile description for a fixed algebraic equation, an exact Newton-weight existence test, complete rational-map solutions, and a sharp infinitesimal-tail obstruction to omnific solutions. Higher order behaves differently: an infinite-support omnific integer has exact autonomous order two, and examples realize every finite order. A research agenda identifies the remaining classification, effectivity, positive-characteristic, and formalization problems.

[merge] Two later manuscripts (Sections 19–27) prove that support rank at least m forces any m distinct positive dilates to be algebraically independent, so that the autonomous order of a finite-support omnific integer is its rational support rank, in every rank.

Research status. The proposed contributions are the arbitrary-rank classifications and their consequences, not the classical Newton–Puisseux or Böttcher theorems. Complete written arguments are supplied relative to the stated classical inputs. The results have not been refereed or Lean-verified, and priority is not certified. No named published conjecture is claimed solved. Exact finite checks accompany the article but do not verify the infinite theorems. An AI-assisted research draft. Every label carries the prefix `adr:`. Conventions, the relation to the collection as it now stands, the ledger of non-claims and the provenance are in sections 1.4, 1.5 and 18 and Appendix C; text added when the manuscript joined the collection is marked [write] where it sits inside the source’s text. Sections 19–27 print two later manuscripts (Appendix C.5); text added in that merge is marked [merge].

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1 Research position and the questions answered

An omnific integer can have a transfinite normal form, infinitely many nonzero coefficients, and exponents at mutually incomparable Archimedean scales. A polynomial in one monomial has none of this support complexity. The principal question of this article is whether an algebraic relation between a number and its own exponent-dilated copy can distinguish these situations.

The dilation is

$$S_d \left(\sum_{\alpha} a_{\alpha} \omega^{g_{\alpha}} \right) = \sum_{\alpha} a_{\alpha} \omega^{d g_{\alpha}}. \quad (1.1)$$

It is a field automorphism on **No** and **No**[i], and fixes ordinary constants. It is *not* the map $y \mapsto y^d$. For example,

$$S_2(\omega + 1) = \omega^2 + 1, \quad (\omega + 1)^2 = \omega^2 + 2\omega + 1.$$

The question is about polynomial equations over the ordinary constant field, with S_d additionally available. It is not a statement that an infinite omnific integer is algebraic over $\mathbb{R}$.

1.1 The main answer in omnific notation

For $x \in \mathbf{Oz} \setminus \mathbb{Z}$ and $d \geq 2$, we prove the equivalence

$$\begin{aligned} & x \text{ and } S_d x \text{ are algebraically dependent over } \mathbb{R} \\ \iff & x = P(\omega^{\delta}) \text{ for some } \delta > 0, \quad P \in \mathbb{R}[T], \quad P(0) \in \mathbb{Z}. \end{aligned} \quad (1.2)$$

The same statement holds for Gaussian omnific integers with $\mathbb{C}$ and $\mathbb{Z}[i]$ in place of $\mathbb{R}$ and $\mathbb{Z}$. Thus an infinite normal-form support is an unconditional obstruction to *every* first-order autonomous algebraic relation. A finite support can also obstruct a relation: its positive exponents must be commensurable, not merely comparable in size.

There is an exact quantitative refinement. Normalize δ so that the nonzero positive exponents of x generate $\delta\mathbb{Z}$. Then, writing $x = P(\omega^{\delta})$ and $M = \deg P$,

$$\min\{\deg_Y F : 0 \neq F \in \mathbb{R}[X, Y], F(x, S_d x) = 0\} = M. \quad (1.3)$$

In particular, the answer is independent of the chosen integer $d \geq 2$. This is a classification of a substantial class of equations, rather than one specially selected identity.

1.2 Relation to the repository

The repository was inspected at commit

b895e8672990e8b5a56f97dd7246f9dd5f86c771.

The main points of contact are the report on a single dilation [2], the dynamics report [3], and the omnific and Hahn-support infrastructure catalogued in the repository [1]. In particular, the single-dilation report already uses the recognizer $S_2 w = w^2$ for canonical monomials and treats linear and multiplicative difference equations. That recognizer is a premise to extend, not a novelty to reclaim. We reprove its elementary characteristic-zero version for general d .

The present extension is from this one special equation to *every* constant-coefficient polynomial relation $F(y, S_d y) = 0$, including ramified algebraic correspondences. It supplies support bounds, an exact omnific

classification, and sharp degrees. The inspected repository excerpts do not supply this classification. That comparison is not an assertion that every line of every repository manuscript has been independently reviewed. [write] This subsection is the source's, written at its pin. The relation to the collection as it now stands, with exact labels, is in section 1.5.

1.3 Literature and the novelty boundary

Hahn summation and its support calculus are classical [4]. Newton–Puiseux factorization is classical; Mannaa–Coquand give a modern constructive treatment [5]. Formal Böttcher coordinates are also classical; the needed characteristic-zero theorem is recorded explicitly by Salerno–Silverman [6]. We use these inputs openly and give an explicit coordinate construction adapted to strong Hahn evaluation.

There is an established literature on autonomous Mahler equations, including Nishioka–Nishioka [7], and on nonlinear formal Mahler solutions [9]. Chyzak–Dreyfus–Dumas–Mezzarobba [10] and Faverjon–Roques [11] address linear Mahler equations and generalized series. The later works of Faverjon–Poulet [12] and Faverjon–Roques [13] develop basis algorithms and generalized Mahler-series structure for linear systems; they are cited here for context from their abstracts, not used as proof dependencies. These works prevent any claim that Mahler equations, Böttcher coordinates, or non-Puiseux solutions are new topics. The contribution proposed here is the *arbitrary-rank support-collapse theorem*, the resulting exact omnific criterion, and their quantitative synthesis. [write] The full text of [7] was read when the manuscript joined the collection. It records a 1983 criterion of Mahler [8] for the existence of a convergent power-series solution of a constant-coefficient relation $P(f(z), f(z^d)) = 0$, a rank-one, positive-support antecedent of the existence statements of section 7; see Appendix C.3.

For omnific arithmetic, L’Innocente–Mantova [14] provide the relevant published factorization framework. Our result is not a factorization theorem and does not settle the remaining general factorization conjectures. The user’s encyclopedia link [21] is useful orientation, but none of the proofs relies on an encyclopedia assertion in place of the classical inputs above.

1.4 Notation and conventions in this report

[write] This subsection and the next were written when the manuscript joined the collection. No symbol of the source was renamed and no normalization was changed; the meanings below are fixed once for the whole report ([merge] Sections 19–27 add their own symbols, fixed in section 19.2), and paths such as `surcomplex/single-dilation-hahn-support` are relative to `docs/`.

- *The dilation.* S_d is the exponent dilation (2.1), with $d \geq 2$ an ordinary integer; in section 11 and section 14 also S_q for rational $q > 1$. It is the map that the single-dilation report writes S_q at $q = d$ (`dsup:sec:conventions`) and that the surcomplex-automorphisms report writes S_a at $a = d$ (`saut:eq:dilation`). On **No** it is the canonical Hahn lift of the value-group dilation $\gamma \mapsto d\gamma$ in the exponential-automorphism report (`prop:hahn-lift`; that report writes the dilation T_r and its lift $\widehat{T}_r$). The tempting false readings, beside the true one: S_d is not $y \mapsto y^d$ (the two agree exactly on the monomials of lemma 3.3); it is not the argument dilation $f(z) \mapsto f(\lambda z)$ of formal functions in the holonomic-rigidity report (`hol:sec:qnecessity`); and it is not an exponential automorphism (section 1.5).
- *Fields and rings.* $K_\Gamma(k)$ is a set-sized full Hahn field over an arbitrary field k , of characteristic zero wherever the main results are stated; the notation guide `NOTATION.md` writes K_Γ for the case $k = \mathbb{C}$ and F_Γ for $k = \mathbb{R}$. The ring $A_\Gamma(k)$ of (2.3) is the *negative-support* ring, support in $\Gamma_{\leq 0}$; it is not the valuation ring $\mathcal{O}$. Under (2.4) it is the ring of forms with nonnegative growth support, which the notation

guide records as the Diophantine report's $\mathcal{A}_k(\Gamma)$. The symbol $A_\Gamma(k)$ is local to this report. $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ are as in (2.5), in agreement with the guide's $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$.

- *Order.* The autonomous algebraic order of section 13 is the least r for which $y, S_dy, \dots, S_d^r y$ are algebraically dependent over k . It is not the order of a linear Mahler equation, and not the differential order of the holonomic-rigidity report. ord_Z is the Z -adic order of an ordinary Laurent series.
- *Local letters.* $M(y)$ is the primitive degree of section 9; it is not the monomial formula $M(m)$ of the single-dilation report (`dsup:eq:monomialformula`). The letters M (an auxiliary series in the proof of lemma 5.1) and M_c (a chart in theorem 11.1) are local. M is also a polynomial degree in theorems 15.1 and 15.2. D is an integer degree in lemma 4.1 and theorems 6.1, 9.2, 15.1 and 15.2 and an auxiliary series in the proof of lemma 5.1; it never denotes a derivation. H_a is the normalized inverse Böttcher coordinate of proposition 5.2, an ordinary power series; it is not the convex subgroup H_α of the notation guide. L is a series in lemma 5.1 and in Appendix B, and a field in corollary 7.2 and theorem 9.2. T is a monomial variable, $t^{-\delta}$ or ω^δ , or an ordinary polynomial variable, and $T_j = \omega^{g_j}$ in theorem 13.3.
- *Strong* means preserving Hahn summation, not convergence of partial sums (section 3). *Cyclically supported* means support in one cyclic subgroup $\delta\mathbb{Z}$, bounded below (section 2).

1.5 Relation to other reports in the collection

[write] The source compared itself with selected reports at its pin (Appendix C.2). Checked against the collection at the placement commit, the relations are these.

Single dilation (surcomplex/single-dilation-hahn-support, prefix `dsup:`). This is the nearest report: the same operator and the same recognizer.

1. *Recognizer.* Lemma 3.3 at $d = 2$ is `dsup:lem:monomialrecognition`, where $a = a^2$ forces coefficient one, with the same least-term proof; the general- d statement is the source's. The equality $\text{Fix}(S_d) = k$ of lemma 3.2 is `dsup:eq:dilationfixed`. The report there also has the power-equation recognizer `dsup:lem:rationalrecognition` (item 3 below).
2. *Linear and multiplicative first-order equations* are treated there (`dsup:thm:dilationcohom`). The boundary equation $S_d u_d - u_d = \omega^d$ of section 14 is, at $d = 2$, the Conway form of that report's example `dsup:ex:opposite`: under (2.4) its second series becomes $y = \sum_{n \geq 1} \omega^{2^{-n}}$ with $S_2 y - y = \omega$, and $u_2 = S_2 y$, so applying S_2 gives $S_2 u_2 - u_2 = \omega^2$. For general d the same telescoping applies. That report presents the example as an illustrative calculation; the source credits the series pattern to [10, 9]. Only the exact autonomous order of theorem 13.2 is claimed here.
3. *Nonintegral rational dilations.* The source leaves implicit relations $F(y, S_q y) = 0$ with $q \in \mathbb{Q}_{>1} \setminus \mathbb{Z}$ open (section 14 and the question on nonintegral rational dilation in section 16). The single-dilation report already exhibits nonconstant solutions of such relations. For divisible Γ , $q = a/b$ with positive integers a, b and $n = |a - b|$, `dsup:lem:rationalrecognition` shows that the nonzero solutions of $(S_q x)^b = x^a$, that is of $F(X, Y) = Y^b - X^a$, are exactly the elements of $\mu_n(k) t^\Gamma$. These solutions are monomials, so they satisfy the cyclic-support conclusion trivially. There is no contradiction with theorem 11.1: since $q \notin \mathbb{Z}$ one has $b \geq 2$, and t^{qg} is not a rational function of t^g for $g \neq 0$. This is a partial datum for the question, not an answer: whether every solution of every such relation has controlled support remains open, and the caution at the end of section 14 stands.

4. *Decidability.* The undecidability of that report (`dsup:thm:undecidable`) concerns the full first-order theory of the expanded field; theorem 15.2 decides one specified equation. The source says there is no contradiction (section 14), and there is none.
5. *Its questions and non-claims.* None of the three questions of `dsup:sec:questions` is answered. In particular its question on definability from other exponent automorphisms asks which ordered exponent automorphisms θ admit polynomial equations recognizing canonical monomials; this report treats only $\theta =$ multiplication by an integer $d \geq 2$. Its non-claim that no solvability theorem for arbitrary variable-coefficient Mahler systems is inferred (item 7 of `dsup:sec:nonclaims`) stands: this report is about constant-coefficient nonlinear relations, and section 14 shows that the collapse fails with nonconstant coefficients.

Surcomplex field automorphisms (`surcomplex/surcomplex-field-automorphisms`, prefix `saut:`). S_d is its S_a at $a = d$ (`saut:eq:dilation`), built by its monomial construction (`saut:thm:monomial`). None of the questions of `saut:sec:questions` concerns algebraic relations between an element and its dilate, and none is addressed.

Exponential automorphism rigidity (`surreal/exponential-automorphism-rigidity`, bare labels). On **No**, S_d is the canonical lift `prop:hahn-lift` of the value-group dilation by d , and `thm:dilation` shows that for rational $r \neq 1$ the value-group dilation by r has no exponential lift at all. So S_d does not commute with the surreal exponential. That report's subject, which value-group automorphisms lift to exponential automorphisms (`q:image`), is not this report's subject: nothing here bears on it, and the source's last question, on the interaction of the finite-profile classes with the surreal exponential, remains open and is consistent with that theorem.

Holonomic rigidity (`surcomplex/holonomic-rigidity-for-entire-hahn-functions`, prefix `hol:`). Its dilations are argument dilations $f(z) \mapsto f(qz)$ of entire Hahn functions by a scalar q of the field, in linear q -difference equations (`hol:main:q`, `hol:sec:qnecessity`). They are not the exponent dilations of this report, whose equations are between numbers, not functions. No statement of that report is used or extended, as the single-dilation report also records.

Dynamics (`surcomplex/dynamics-and-normal-forms`). The source lists it as a point of contact. It studies multipliers, linearization and normal forms of germs; the superattracting germs and the Böttcher coordinate of section 5 occur in no other report of the collection, and no statement of that report is used.

Formalization. No `adr:` label has a Lean mapping in `FORMALIZATION.md`; no theorem of this report is Lean-verified.

The batch-33 additions. [merge] Two later manuscripts, sources 02 and 03, were written against this report and are printed in Sections 19–27; section 19.1 lists what they settle here (Research question 1 in every rank, the rank part of Research question 2, a rank datum for Research question 5, the relative part of Research question 8). They use none of sections 4 to 6. Their fresh-parameter constructions (section 25) separate a set-sized base by the same support localization as the independent-copies report (`surreal/independent-surreal-copies`, prefix `isc::isc:cor:parameters`), whose subject is independent copies of **No** and their intersections, not dilation orbits; neither report uses the other. No other report treats algebraic independence of exponent dilates.

2 Hahn fields, normal forms, and the two support directions

Definition 2.1. Let k be a field and Γ an ordered abelian group. The full Hahn field $K_\Gamma(k) = k((t^\Gamma))$ consists of formal sums

$$y = \sum_{\gamma \in \Gamma} a_\gamma t^\gamma, \quad a_\gamma \in k,$$

whose support is a well-ordered subset of Γ . For $y \neq 0$, set $v(y) = \min \text{supp } y$ and let $\text{lc}(y)$ be the coefficient there. Put $v(0) = +\infty$.

Throughout the main classification, $\text{char } k = 0$ and d is an ordinary integer at least two. No order on k is required. Dilation

$$S_d\left(\sum a_\gamma t^\gamma\right) = \sum a_\gamma t^{d\gamma} \quad (2.1)$$

is a strong injective field endomorphism; it is an automorphism when Γ is d -divisible. Its elementary valuation law is

$$v(S_d y) = d v(y). \quad (2.2)$$

The valuation ring and its maximal ideal are

$$\mathcal{O} = \{y : v(y) \geq 0\}, \quad \mathfrak{m} = \{y : v(y) > 0\}.$$

For $y \in \mathcal{O}$, its residue is its coefficient at zero, an element of k . In contrast, define

$$A_\Gamma(k) = \{y \in K_\Gamma(k) : \text{supp } y \subseteq \Gamma_{\leq 0}\}. \quad (2.3)$$

This is a subring, but it is *not* the valuation ring. A nonconstant element of $A_\Gamma(k)$ has negative valuation. The absence of *positive* exponents, not just a bound on the leading exponent, will be decisive.

Definition 2.2. For $\delta \in \Gamma_{>0}$, the notation $k((t^\delta))$ means the image of the ordinary Laurent series field $k((Z))$ under $Z \mapsto t^\delta$. Thus its elements have support contained in $\delta\mathbb{Z}$, bounded below in the integer indexing. A series with this property is called *cyclically supported* here.

Cyclic support is stronger than support in one Archimedean class and stronger than a one-dimensional rational span. The set $\{-1, -1/2, -1/4, \dots\}$ spans a one-dimensional $\mathbb{Q}$ -space but is not contained in any cyclic subgroup of $\mathbb{Q}$.

2.1 Translation to surreal numbers

Conway normal forms use decreasing powers of ω . The correspondence with the increasing Hahn convention is

$$t^\gamma \mapsto \omega^{-\gamma}. \quad (2.4)$$

For any set-sized ordered additive subgroup $\Gamma \subseteq \mathbf{No}$, it identifies $\mathbb{R}((t^\Gamma))$ with the corresponding normal-form subfield of $\mathbf{No}$; complexifying gives $\mathbb{C}((t^\Gamma)) \subseteq \mathbf{No}[i]$. Every individual surreal or surcomplex number is contained in such a workspace: take the divisible subgroup generated by its set-sized support. The usual normal-form arithmetic is the algebraic input here [14, 1].

The omnific integers have the description

$$\mathbf{Oz} = \left\{ n + \sum_{\alpha < \lambda} r_\alpha \omega^{g_\alpha} : n \in \mathbb{Z}, r_\alpha \in \mathbb{R}, g_\alpha > 0, (g_\alpha)_{\alpha < \lambda} \text{ strictly decreasing} \right\}, \quad (2.5)$$

with set-sized support and zero coefficients omitted. Thus the only additional restriction beyond the real negative-support ring is the integrality of the coefficient at zero. For $\mathbf{Oz}[i]$ the coefficients are complex and that constant lies in $\mathbb{Z}[i]$.

2.2 Size and topology conventions

All field-theoretic proofs below take place in a set-sized Hahn field. The statements for **No** and **No**[i] are then elementwise consequences. No proper-class polynomial ring, class-sized tensor product, or class-indexed analytic sum is required.

Also, $v(x) > 0$ need not imply that $v(x^n)$ tends cofinally to infinity in Γ . For example, in the lexicographically ordered group $\mathbb{Q} \oplus \mathbb{Q}$, the multiples of $(0, 1)$ never reach $(1, 0)$. Nevertheless the geometric series in x is strongly summable. This separation between support summability and topological convergence is essential throughout.

3 The support calculus and a monomial recognizer

A family (f_i) of Hahn series is *strongly summable* when the union of its supports is well ordered and every exponent occurs in the support of only finitely many family members. Its sum is then defined coefficientwise.

Lemma 3.1 (Formal evaluation). *Let $x \in \mathfrak{m}$. Every formal series $f(Z) \in k[[Z]]$ can be evaluated at x as a strong Hahn sum. This gives a k -algebra homomorphism. For $x \neq 0$ it extends to an injective map $k((Z)) \rightarrow K_\Gamma(k)$, and*

$$v(f(x)) = \text{ord}_Z(f) v(x) \quad (f \neq 0). \quad (3.1)$$

Formal composition is respected whenever the inner series has zero constant coefficient. Moreover,

$$S_d(f(x)) = f(S_d x). \quad (3.2)$$

Proof. The Neumann support lemma says that the monoid generated by a well-ordered subset of $\Gamma_{>0}$ is well ordered and that each exponent has only finitely many representations by finite words from that subset, allowing all word lengths [4]. Apply it to $\text{supp } x$. It justifies the family of powers and the regroupings for products and formal composition. There are only finitely many negative powers in a Laurent series, so these cause no extra infinite-sum issue.

If the initial term of f is aZ^m , its evaluation has initial valuation $m v(x)$: every later power has strictly larger valuation. This proves (3.1) and injectivity. Exponent dilation preserves well ordering, finite occurrence, and coefficientwise sums, proving (3.2). $\square$

Lemma 3.2 (Constants are relatively algebraically closed). *Every element of $K_\Gamma(k)$ algebraic over k belongs to k . In particular, every nonconstant Hahn series is transcendental over k , and $\text{Fix}(S_d) = k$.*

Proof. A nonzero algebraic element z satisfies a polynomial with nonzero constant and leading coefficients in k . If $v(z) > 0$, the constant term is the unique term of least valuation; if $v(z) < 0$, the highest-degree term is. Both are impossible. Hence $v(z) = 0$. Let $c \in k$ be its residue. If $z - c \neq 0$, this is an algebraic element of positive valuation, again impossible.

For a nonconstant fixed element y , use $x = y - c$ if $v(y) \geq 0$, with c its residue, and $x = 1/y$ if $v(y) < 0$. In both cases $x \neq 0$ has positive valuation and is fixed. The equation $d v(x) = v(x)$ contradicts $d > 1$ in an ordered group. $\square$

Lemma 3.3 (Monomial recognizer). *Suppose $w \neq 0$ and $S_d w = w^d$. Then*

$$w = at^\eta, \quad a \in k^\times, \quad a^{d-1} = 1, \quad \eta \in \Gamma. \quad (3.3)$$

Conversely every such monomial is a solution.

Proof. Write $w = at^\eta(1 + \varepsilon)$ with $\varepsilon \in \mathfrak{m}$. Initial coefficients give $a = a^d$. Dividing the original identity by $at^{d\eta}$ gives

$$1 + S_d \varepsilon = (1 + \varepsilon)^d.$$

If $\varepsilon \neq 0$ and $\rho = v(\varepsilon) > 0$, then the nonconstant part of the left side has valuation $d\rho$, whereas that of the right side has valuation ρ : its initial coefficient is multiplied by the nonzero scalar d . Since $d\rho > \rho$, this is impossible. $\square$

The least-term proof eliminates possible tails at *all* other scales at once. It neither examines the terms sequentially nor assumes that an iteration approaches its limit in the ambient valuation topology. The $d = 2$ instance is the repository's monomial recognizer [2]. [write] That is `dsup:lem:monomialrecognition`, proved there by the same least-term comparison; the equality $\text{Fix}(S_d) = k$ of lemma 3.2 is `dsup:eq:dilationfixed` (section 1.5).

4 The classical ramification input and its degree bound

We use the Newton–Puisseux theorem in the following standard form: if k is algebraically closed of characteristic zero, then

$$\bigcup_{e \geq 1} k((U^{1/e}))$$

is an algebraic closure of $k((U))$ [5]. Thus every nonzero polynomial in $k((U))[V]$ splits into linear factors there. The next lemma spells out both the quantitative point and its transport into a Hahn field.

Lemma 4.1 (Evaluated Puiseux branch). *Assume k algebraically closed, Γ divisible, and $x \in K_\Gamma(k)$ with $v(x) = \gamma > 0$. Let $0 \neq G(U, V) \in k[U, V]$ have V -degree D , and suppose $G(x, z) = 0$. Then there exist an integer $1 \leq e \leq D$, an e th root u of x in $K_\Gamma(k)$, and $\phi(Z) \in k((Z))$ such that*

$$z = \phi(u), \quad G(Z^e, \phi(Z)) = 0. \quad (4.1)$$

Moreover $v(u) = \gamma/e$, and $v(z) = \text{ord}_Z(\phi) \gamma/e$ when $z \neq 0$.

Proof. Apply Newton–Puisseux to $G(U, V)$ over $k((U))$. In a common extension $k((U^{1/N}))$ it factors as a nonzero leading coefficient times finitely many factors $V - \psi_j(U^{1/N})$.

An N th root of x exists in the Hahn field without using its full algebraic closedness: write $x = at^\gamma(1 + \varepsilon)$, choose $b^N = a$, and use

$$bt^{\gamma/N}(1 + \varepsilon)^{1/N},$$

where the binomial series is evaluated by lemma 3.1. Evaluation at this root sends the Puiseux factorization to a genuine factorization in $K_\Gamma(k)$. Its leading coefficient is nonzero by the injectivity in lemma 3.1. Consequently z equals one of the evaluated roots.

Choose the smallest denominator e for that root. Its series can be written as $\phi(U^{1/e})$, with $e \mid N$; the corresponding power of the chosen N th root gives u . Minimality means that the automorphisms $U^{1/e} \mapsto \zeta U^{1/e}$, for $\zeta^e = 1$, give exactly e distinct conjugates of this series. Indeed a stabilizing ζ must satisfy $\zeta^j = 1$ for every exponent j occurring in ϕ , and minimality says their common divisor with e is one. All these conjugates are roots of G . Hence $e \leq D$. The valuation statements follow from lemma 3.1. $\square$

Remark 4.2. The bound is on the denominator of *one selected branch*, not on a common denominator for every root. A common splitting denominator can be larger than D . Confusing these two denominators would invalidate the sharp degree theorem below.

5 A support-safe Böttcher construction

The formal-coordinate theorem in this section is classical [6, Proposition 1]. We include a construction so that no unstated convergence hypothesis enters its Hahn-field application. All sums in the construction are first formed in the ordinary formal ring $k[[Z]]$.

Lemma 5.1 (Normalized formal coordinate). *Let $h(Z) = Z^d U(Z)$, where $U \in 1 + Zk[[Z]]$ and $\text{char } k = 0$. There is a unique $b(Z) \in Z + Z^2 k[[Z]]$ with*

$$b(h(Z)) = b(Z)^d. \quad (5.1)$$

Its compositional inverse H satisfies

$$H(Z^d) = h(H(Z)). \quad (5.2)$$

Proof. Put

$$L(Z) = \sum_{n \geq 0} \frac{1}{d^{n+1}} \log U(h^{on}(Z)), \quad b(Z) = Z \exp L(Z). \quad (5.3)$$

The n th summand has order at least d^n unless it is zero, so each coefficient receives only finitely many contributions. Reindexing gives

$$dL(Z) = \log U(Z) + L(h(Z)).$$

Therefore

$$b(h(Z)) = Z^d U(Z) \exp L(h(Z)) = Z^d \exp(dL(Z)) = b(Z)^d.$$

The linear coefficient of b is one, so it has a unique compositional inverse, which gives (5.2).

For uniqueness, any normalized coordinate has the form $Z \exp M(Z)$. The difference $D = L - M$ satisfies $D(h) = dD$. A nonzero D of positive order would have different orders on these two sides, namely $d \text{ ord } D$ and $\text{ord } D$. Thus $D = 0$. $\square$

Proposition 5.2 (All Hahn solutions of a superattracting germ). *Let $h(Z) = AZ^d + O(Z^{d+1}) \in k[[Z]]$, $A \neq 0$, and let $x \neq 0$ have positive valuation. Then $S_d x = h(x)$ is solvable with prescribed leading coefficient $a \in k^\times$ and valuation $\delta \in \Gamma_{>0}$ if and only if*

$$Aa^{d-1} = 1. \quad (5.4)$$

For each such pair there is exactly one solution:

$$x = aH_a(t^\delta), \quad (5.5)$$

where H_a is the normalized inverse Böttcher coordinate of $h_a(Z) = a^{-1}h(aZ)$.

Proof. Leading coefficients force (5.4). With this condition, $h_a(Z) = Z^d + O(Z^{d+1})$. Write $u = x/a$, and let b_a be the coordinate of lemma 5.1. Strong evaluation and its commutation with S_d give

$$S_d(b_a(u)) = b_a(S_d u) = b_a(h_a(u)) = b_a(u)^d.$$

The left-hand argument $b_a(u)$ has valuation δ and leading coefficient one. By lemma 3.3 it is exactly t^δ . Inversion yields (5.5) and uniqueness. Conversely, (5.2), evaluated strongly at t^δ , verifies that this expression is a solution. $\square$

This is an exhaustive statement in the full Hahn field. The coordinate series is ordinary, but the element to which it is applied may initially have arbitrary support. The recognizer proves that every extra support scale disappears after the coordinate change.

6 The arbitrary-rank support-collapse theorem

Theorem 6.1 (First-order autonomous support rigidity). *Let k be any field of characteristic zero, Γ any ordered abelian group, and $d \geq 2$. Suppose $y \in K_\Gamma(k) \setminus k$ and*

$$F(y, S_dy) = 0, \quad 0 \neq F \in k[X, Y]. \quad (6.1)$$

Define a positive-valuation coordinate

$$x = \begin{cases} y - c, & v(y) \geq 0, \quad c = \text{res}(y), \\ 1/y, & v(y) < 0, \end{cases} \quad \gamma = v(x) > 0. \quad (6.2)$$

Then for some integer e with $1 \leq e \leq \deg_Y F$, one has

$$\delta = \gamma/e \in \Gamma, \quad y \in k((t^\delta)). \quad (6.3)$$

In particular, the support is finite or has order type ω and is contained in a cyclic subgroup of Γ .

Proof. First assume k algebraically closed and Γ divisible. Let $D = \deg_Y F$. Since y is transcendental over k , $D \geq 1$. The equation becomes $G(x, S_dx) = 0$ for a nonzero polynomial G with $\deg_V G \leq D$. In the finite case use

$$G(U, V) = F(c + U, c + V).$$

In the infinite case, if $a = \deg_X F$, use

$$G(U, V) = U^a V^D F(U^{-1}, V^{-1});$$

monomial factors can be removed. These substitutions give nonzero polynomials, and x, S_dx are nonzero.

Apply lemma 4.1. There are $1 \leq e \leq D$, an element z with $z^e = x$, and a Laurent series ϕ such that

$$S_dx = \phi(z), \quad v(z) = \gamma/e. \quad (6.4)$$

But $v(S_dx) = d\gamma$. Consequently

$$\text{ord}_Z \phi = de, \quad \phi(Z) = BZ^{de}(1 + u(Z)), \quad B \neq 0, \quad u \in Zk[[Z]].$$

Choose an e th root of B and form a formal e th root of this series. Every such root has the form

$$h(Z) = AZ^d(1 + u(Z))^{1/e} \in k[[Z]], \quad A^e = B.$$

Since $(S_dz)^e = S_dx = \phi(z)$, the ratio between S_dz and one evaluated root $h(z)$ is an e th root of unity. Multiplying A by that root of unity gives an h with the exact identity

$$S_dz = h(z). \quad (6.5)$$

By proposition 5.2, $z = aH(t^{\gamma/e})$ for an ordinary formal power series H and a nonzero constant a . Thus $x = z^e$ belongs to $k((t^{\gamma/e}))$, as does $y = c + x$ or $y = 1/x$. This proves the assertion in the auxiliary algebraically closed, divisible setting.

For general k and Γ , embed the Hahn field coefficientwise and exponentwise into

$$\bar{k}((t^{\Gamma_{\mathbb{Q}}})), \quad \Gamma_{\mathbb{Q}} = \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$$

with its natural ordered divisible hull. The proof gives an integer $e \leq D$ and support contained in $(\gamma/e)\mathbb{Z}$. The intersection

$$\Gamma \cap (\gamma/e)\mathbb{Z} = m(\gamma/e)\mathbb{Z}$$

for a positive integer m dividing e : it is a nonzero subgroup of a cyclic group and contains γ . Put $e' = e/m$. Then $1 \leq e' \leq D$, $\gamma/e' \in \Gamma$, and the original support is contained in $(\gamma/e')\mathbb{Z}$. Its coefficients are still in k , by uniqueness of Hahn coefficients. The support is bounded below in that cyclic group. Renaming e' as e proves (6.3). $\square$

Remark 6.2 (What the theorem does not assert). Membership in a cyclic Laurent subfield is a necessary condition for (6.1), not a claim that every ordinary Laurent series satisfies some autonomous Mahler equation. Nor does the support conclusion turn strong evaluation into ambient valuation convergence. The exponent group can have arbitrarily high rank even when the individual solution is forced into one cyclic subgroup.

7 Finite profiles, coefficient descent, and an existence test

The proof gives more than a necessary support condition. For a fixed polynomial equation it supplies a finite collection of ordinary formal shapes; the only remaining continuous scale parameter is the monomial at which a shape is evaluated.

Theorem 7.1 (Finite-profile classification). *Let k be algebraically closed of characteristic zero, $d \geq 2$, and $0 \neq F \in k[X, Y]$. There is a finite, possibly empty collection*

$$\mathcal{P}(F, d) \subset k((Z)) \setminus k$$

such that for every nonzero divisible ordered group Γ ,

$$\{y \in K_\Gamma(k) \setminus k : F(y, S_d y) = 0\} = \{f(t^\delta) : f \in \mathcal{P}(F, d), \delta \in \Gamma_{>0}\}. \quad (7.1)$$

Each profile is constructed from a Puiseux branch and a normalized Böttcher coordinate. A profile at infinity has pole order at most $\deg_Y F$. The list need not be free of duplicates.

Proof. Remove all factors $Y - X$ from F , obtaining F_0 . A nonconstant solution cannot satisfy $S_d y = y$ by lemma 3.2, so this removal does not change the nonconstant solution set. The polynomial $F_0(X, X)$ is not identically zero. Consequently there are only finitely many possible finite residues c , namely its roots.

For each such c , form $G(U, V) = F_0(c + U, c + V)$. Also form the reciprocal polynomial at infinity as in the proof of theorem 6.1. For each of these finitely many polynomials, choose its finitely many Puiseux roots $V = \phi(U^{1/e})$. Keep only the nonzero branches satisfying $\text{ord}_Z \phi = de$. For each kept branch, there are finitely many formal e th roots

$$h(Z)^e = \phi(Z), \quad h(Z) = AZ^d + O(Z^{d+1}),$$

and finitely many constants a with $Aa^{d-1} = 1$. Let H_a be the coordinate in proposition 5.2. Put on the list

$$f(Z) = c + (aH_a(Z))^e \quad \text{or} \quad f(Z) = (aH_a(Z))^{-e}, \quad (7.2)$$

according to the finite or infinite chart.

The proof of theorem 6.1 shows that every nonconstant solution has one of these forms after evaluation at a positive monomial. Conversely let $z = aH_a(t^\delta)$. Then $S_d z = h(z)$, so with $x = z^e$,

$$S_d x = h(z)^e = \phi(z), \quad G(x, S_d x) = 0.$$

Undoing the chart gives the original equation. The pole-order bound is the branch denominator bound $e \leq \deg_Y F$. $\square$

Corollary 7.2 (Coefficient descent). *Let $k_0 \subseteq L$ be characteristic-zero fields and $0 \neq F \in k_0[X, Y]$. If $y \in L((t^\Gamma)) \setminus L$ satisfies $F(y, S_dy) = 0$, then every Hahn coefficient of y is algebraic over k_0 . In particular, an equation over $\overline{\mathbb{Q}}$ admits no nonconstant solution having a coefficient transcendental over $\overline{\mathbb{Q}}$.*

Proof. Work in an algebraic closure $\overline{L}$ and let k' be the algebraic closure of k_0 inside $\overline{L}$. Construct the finite profiles over k' . Their Puiseux factorizations still split the relevant polynomials over $\overline{L}((U))$; the roots of unity and leading-coefficient roots are already in k' . Thus the same exhaustion proof gives profiles with all coefficients in k' , even for a solution originally in the larger coefficient field. Evaluation at a monomial changes exponents, not coefficients. The original coefficients lie in $L \cap k'$. $\square$

This is a descent statement for a *fixed coefficient field of the equation*. It does not contradict the existence of $a\omega^\delta$ as a first-order solution over $\mathbb{R}$ for arbitrary real a : the corresponding equation generally contains a among its coefficients.

Theorem 7.3 (Exact local Newton-weight criterion). *Let k be algebraically closed of characteristic zero and $\Gamma \neq 0$ divisible. For*

$$0 \neq G(U, V) = \sum g_{ij} U^i V^j \in k[U, V],$$

there is a nonzero $x \in \mathfrak{m}$ satisfying $G(x, S_dx) = 0$ if and only if

$$\min\{i + dj : g_{ij} \neq 0\} \quad \text{is attained by at least two monomials.} \quad (7.3)$$

Proof. If $v(x) = \gamma > 0$, the valuation of a term is $(i + dj)\gamma$. A unique least term cannot cancel, proving necessity.

For sufficiency use the ordinary Newton polygon, or equivalently the Puiseux factorization. The weighted initial polynomial

$$I(W) = \sum_{i+dj=w} g_{ij} W^j, \quad w = \min(i + dj),$$

has at least two terms. Distinct active monomials have distinct j . After its monomial factor is removed, it has a nonzero constant term and positive degree, hence a nonzero root over k . Factoring G into Puiseux linear factors shows that these nonzero roots are exactly the leading coefficients of the branches whose U -order is d . More explicitly, a factor with root order below d contributes a nonzero constant to the weighted initial product, one above d contributes W , and one equal to d contributes $W - A$ with $A \neq 0$. The product has a nonzero root precisely in the last case.

Thus some branch is $V = \phi(U^{1/e})$ with $\text{ord}_Z \phi = de$. The construction in theorem 7.1 gives a nonzero positive-valuation solution at every chosen positive monomial scale. $\square$

Corollary 7.4 (Scale-independent existence). *For a fixed F over an algebraically closed characteristic-zero field, existence of a nonconstant solution is the same in every nonzero divisible Hahn value group. It is decided by finitely many tests (7.3): the finite diagonal-residue charts after removal of $Y - X$ factors, and the chart at infinity. For algebraic-number coefficients this is an effective decision procedure.*

Proof. The charts and their number are as in theorem 7.1. Each chart has a solution exactly when theorem 7.3 applies. The test uses polynomial arithmetic, roots of one ordinary polynomial, and comparison of finitely many integer weights. With exact algebraic coefficients, all the required equality tests and root calculations are effective. A nonzero divisible ordered group always has a positive element. $\square$

The criterion decides *existence*, not whether an arbitrary encoded infinite Hahn series satisfies the equation. It also does not make an arbitrary complex coefficient into computable input.

8 The exact classification in the negative-support ring

Theorem 8.1 (Polynomial-in-one-monomial criterion). *Let $\text{char } k = 0$, $d \geq 2$, and $y \in A_\Gamma(k) \setminus k$. The following are equivalent:*

- (i) y and S_dy are algebraically dependent over k ;
- (ii) $y = P(t^{-\delta})$ for some $\delta \in \Gamma_{>0}$ and $P \in k[T]$;
- (iii) the support of y is finite and its nonzero exponents generate a cyclic subgroup of Γ .

If a relation $F(y, S_dy) = 0$ is given, the polynomial in (ii) can be chosen of degree at most $\deg_Y F$.

Proof. Since $y \notin k$ and its support has no positive exponent, $v(y) < 0$. Put $\gamma = -v(y)$. By theorem 6.1, for some $e \leq \deg_Y F$ its support lies in $(\gamma/e)\mathbb{Z}$. It also lies between $-\gamma$ and zero. There are only $e + 1$ points of this cyclic subgroup in that interval. Consequently

$$y = \sum_{j=0}^e a_j t^{-j\gamma/e},$$

which proves (i) $\Rightarrow$ (ii) with the degree bound.

For (ii) $\Rightarrow$ (i), write $T = t^{-\delta}$. Then $y = P(T)$ and $S_dy = P(T^d)$ belong to the one-variable rational function field $k(T)$, whose transcendence degree is one. More explicitly, the nonzero polynomial

$$R_P(X, Y) = \text{Res}_T(P(T) - X, P(T^d) - Y) \quad (8.1)$$

vanishes on (y, S_dy) . Its degree in Y is $\deg P$, so it is not identically zero. Conditions (ii) and (iii) are equivalent by taking a positive generator of the subgroup generated by the finitely many support exponents. $\square$

Corollary 8.2 (Dilation independence). *For $y \in A_\Gamma(k)$, dependence of y, S_dy for one integer $d \geq 2$ is equivalent to dependence for every such integer. If y has infinite support, every pair y, S_dy is algebraically independent over k . The same independence conclusion holds for a finite support of rational rank at least two.*

Proof. The support condition in theorem 8.1 does not mention d . For a finite subset of a torsion-free abelian group, the generated subgroup is cyclic exactly when its rational span has dimension at most one. $\square$

Remark 8.3 (Why the support restriction is indispensable). A Laurent series can have infinitely many positive exponents even when it is cyclically supported. Intersecting with $A_\Gamma(k)$ cuts off all of them. Thus the theorem is not a cardinality argument. Its force comes from a *discreteness conclusion* followed by a *two-sided support bound*. A merely bounded, nondiscrete support can still be infinite.

9 Sharp algebraic degrees and recovery of a monomial

Definition 9.1. For a nonconstant finite-support $y \in A_\Gamma(k)$ satisfying the criterion, let $\delta > 0$ be the positive generator of the group generated by its nonzero exponents. Write $y = P(t^{-\delta})$. The polynomial P is *primitive in its exponents*: the greatest common divisor of its positive supported degrees is one. Define $M(y) = \deg P$.

Theorem 9.2 (The exact relation degree). *For every $d \geq 2$,*

$$\min_{\substack{0 \neq F \in k[X, Y] \\ F(y, S_dy) = 0}} \deg_Y F = M(y). \quad (9.1)$$

Moreover, if $T = t^{-\delta}$ is the primitive monomial, then

$$k(y, S_dy) = k(T). \quad (9.2)$$

The irreducible relation has bidegree $(\deg_X, \deg_Y) = (dM(y), M(y))$.

Proof. Let $M = M(y)$ and let a relation of Y -degree D be given. The degree-bounded conclusion of theorem 8.1 supplies $y = Q(t^{-\delta'})$ of degree $e \leq D$. The support-generated subgroup $\delta\mathbb{Z}$ is contained in $\delta'\mathbb{Z}$, so $\delta = m\delta'$ for an integer $m \geq 1$. Comparison of the largest support magnitude gives $M\delta = e\delta'$, hence $e = Mm \geq M$. Therefore $D \geq M$. The resultant (8.1) gives the reverse inequality.

For the field statement, the extension $k(T)/k(P(T))$ has degree M . One elementary proof is that $P(T) - X$ is irreducible in $k(X)[T]$: its ideal in $k[T, X]$ has quotient $k[T]$, and Gauss's lemma applies. If $L = k(P(T), P(T^d))$, the just-proved minimum degree says

$$[L : k(P(T))] = M.$$

The tower law then forces $L = k(T)$. Finally $[k(T) : k(P(T^d))] = dM$, which is the degree of the irreducible relation in X ; its degree in Y is M . $\square$

Corollary 9.3 (An ordinary polynomial field identity). *Let $P \in k[T] \setminus k$, and let $g = \gcd\{j \geq 1 : [T^j]P \neq 0\}$. In characteristic zero,*

$$k(P(T), P(T^d)) = k(T^g) \quad (d \geq 2). \quad (9.3)$$

Proof. Write $P(T) = Q(T^g)$, with Q primitive in its exponents, and apply theorem 9.2 to Q in an ordinary Laurent Hahn field. $\square$

This corollary can be read entirely in classical rational-function algebra. We give it as a consequence and consistency check of the support argument, not as a certified priority claim in polynomial decomposition theory. The restriction to polynomials is real: for $P(T) = T + T^{-1}$ one has $P(T^2) = P(T)^2 - 2$, and the generated field is $k(T + T^{-1})$, a proper subfield of $k(T)$.

Example 9.4 (A genuinely quadratic relation). For $y = T^2 + T$ and $d = 2$, the primitive degree is two. The relation

$$\begin{aligned} 0 &= Y^2 - (2X^2 + 6X + 2)Y \\ &\quad + X^4 + 2X^3 + 2X^2 \end{aligned} \quad (9.4)$$

is obtained by elimination and has the least possible Y -degree. There is no rational function $R \in k(X)$ with $S_2y = R(y)$. Thus the implicit algebraic classification is strictly broader than the explicit rational-map classification in section 11.

10 Consequences for surreal and Gaussian omnific integers

Theorem 10.1 (Omnific dilation classification). *Let $d \geq 2$ and $x \in \mathbf{Oz} \setminus \mathbb{Z}$. The following are equivalent:*

- (i) x, S_dx are algebraically dependent over $\mathbb{R}$;
- (ii) x, S_dx are algebraically dependent over $\mathbb{C}$ inside $\mathbf{No}[i]$;
- (iii) for some $\delta \in \mathbf{No}_{>0}$, $x = P(\omega^\delta)$ with $P \in \mathbb{R}[T]$ and $P(0) \in \mathbb{Z}$;

(iv) x has finite normal-form support, and its positive exponents are commensurable.

With the primitive exponent step, its least relation degree is $\deg P$. For $x \in \mathbf{Oz}[i] \setminus \mathbb{Z}[i]$, the identical classification holds with $P \in \mathbb{C}[T]$, $P(0) \in \mathbb{Z}[i]$, and dependence over $\mathbb{C}$.

Proof. Choose a set-sized divisible subgroup of $\mathbf{No}$ containing the exponent support, and use (2.4). Apply theorems 8.1 and 9.2. In the real case, a complex polynomial relation splits into its real and imaginary coefficient polynomials; at least one is a nonzero real relation. Conversely a real relation is a complex one. The coefficient at zero imposes exactly the integer or Gaussian-integer condition and no condition on the other coefficients. $\square$

Corollary 10.2 (All distinct forward dilates). *If x is an infinite-support omnific or Gaussian omnific integer, then for every $d \geq 2$ and every pair of integers $0 \leq a < b$,*

$$S_d^a x \quad \text{and} \quad S_d^b x$$

are algebraically independent over $\mathbb{C}$ (and over $\mathbb{R}$ in the real case).

Proof. The first member still has infinite support and the second is its $S_{d^{b-a}}$ dilate. Apply theorem 10.1. $\square$

Example 10.3 (Finite but incommensurable). The number

$$x = \omega^{\sqrt{2}} + \omega + 1$$

is an omnific integer, and $x, S_d x$ are algebraically independent for every integer $d \geq 2$: the positive exponents $1, \sqrt{2}$ are $\mathbb{Q}$ -linearly independent. This particular x is also prime by the published theorem of L’Innocente–Mantova [14, Theorem B]. Its primality is cited, not proved or claimed as a contribution here. The dilation independence is a different property.

Example 10.4 (Comparable scales do not suffice). Both $\omega^{\sqrt{2}} + \omega$ and $\omega^{3/2} + \omega$ have two finite real exponents in the same Archimedean class of the exponent group. Only the second is a polynomial in one monomial: with $T = \omega^{1/2}$ it is $T^3 + T^2$. Its least relation degree is three; the first has no relation at all.

Example 10.5 (One rational span, unbounded denominators). The valid normal form

$$u_d = \sum_{n \geq 0} \omega^{d^{-n}}$$

has infinite support of order type ω . All its exponents lie in $\mathbb{Q}$, but not in any cyclic subgroup. It is algebraically independent of $S_d u_d$. In section 13 we show that its first three transforms satisfy an exact polynomial relation. Pairwise independence must not be confused with independence of a longer tuple.

Example 10.6 (A coefficient-field obstruction). Let $a \in \mathbb{R}$ be transcendental over $\mathbb{Q}$ and $\delta > 0$. Although $x = a\omega^\delta$ satisfies $S_d x = a^{1-d} x^d$ over $\mathbb{R}$, it has no first-order autonomous polynomial relation over $\mathbb{Q}$. This follows from corollary 7.2. The constants allowed in a relation materially change the answer.

11 Complete solutions for rational maps

We now classify the explicit equations $S_q y = R(y)$, with $q \in \mathbb{Q}_{>1}$, Γ divisible, and $R \in k(Z)$ nonconstant. The polynomial relation theorem allows implicit ramification; in this explicit case there is no such support ramification.

Theorem 11.1 (Rational-map classification). *Assume k algebraically closed of characteristic zero and $\Gamma \neq 0$ divisible. A nonconstant solution of*

$$S_q y = R(y) \quad (11.1)$$

exists exactly when R has a fixed point $c \in \mathbb{P}^1(k)$ whose local degree is q . In particular, q must be an integer $d \geq 2$. For every such fixed point, let

$$M_c(Y) = Y - c \quad (c \in k), \quad M_\infty(Y) = 1/Y,$$

and write

$$h = M_c \circ R \circ M_c^{-1} = AZ^d + O(Z^{d+1}).$$

All nonconstant solutions associated with c are exactly

$$y = M_c^{-1}(aH_a(t^\gamma)), \quad \gamma \in \Gamma_{>0}, \quad Aa^{d-1} = 1, \quad (11.2)$$

where H_a is the normalized inverse Böttcher coordinate of $a^{-1}h(aZ)$. Constant solutions are simply the finite fixed points of R where it is defined.

Proof. For $v(y) \geq 0$, let c be its residue. A pole of R at c would give $v(R(y)) < 0$, impossible. Reducing the equation modulo the maximal ideal therefore gives $R(c) = c$. For $v(y) < 0$, the equation requires ∞ to be a fixed point. In either case $u = M_c(y)$ is nonzero of positive valuation, and

$$S_q u = h(u).$$

If $m \geq 1$ is the local degree, comparison of valuations gives $q v(u) = m v(u)$, hence $q = m$. Since $q > 1$, this is an integer $d \geq 2$. The germ now has the form in proposition 5.2, which gives all solutions and their converse. Because k is algebraically closed, the required leading coefficients a exist. $\square$

In particular, not every nonlinear rational map admits a nonconstant solution. The relevant property is a *superattracting fixed point of exactly the dilation degree*, not just a nonlinear term somewhere in the formula. Over a non-algebraically-closed field the same description applies whenever the fixed point and the required leading-coefficient root lie in that field. For real solutions, these are genuine real-root conditions, not automatic consequences of the complex theorem.

Theorem 11.2 (Rational-map rigidity in the negative-support ring). *Let $\text{char } k = 0$, $y \in A_\Gamma(k) \setminus k$, and $d \geq 2$. If $S_d y = R(y)$ for $R \in k(Z)$, then*

$$y = \alpha t^{-\gamma} + b, \quad R(Z) = \alpha^{1-d}(Z - b)^d + b, \quad (11.3)$$

with $\alpha \in k^\times$, $b \in k$, and $\gamma \in \Gamma_{>0}$. Conversely every such pair satisfies the equation. For omnific solutions $b \in \mathbb{Z}$, and for Gaussian omnific solutions $b \in \mathbb{Z}[i]$.

Proof. Writing $R = A/B$ in reduced form, the equation $B(y)S_d y - A(y) = 0$ has degree one in the dilated variable. The denominator is nonzero since y is transcendental over k . By the degree bound in theorem 8.1, y is affine in one monomial, giving the first formula. Substitute this into the equation, and use transcendence of the monomial to obtain the displayed rational-function identity. The converse is direct substitution. The constant-term restrictions are exactly the definitions of the two omnific rings. $\square$

Corollary 11.3 (Simultaneous rational equations). *Suppose the same nonconstant $y \in A_\Gamma(k)$ satisfies $S_p y = P(y)$ and $S_q y = Q(y)$ for integers $p, q \geq 2$ and rational functions P, Q . Then, with one common α, b, γ ,*

$$y = \alpha t^{-\gamma} + b, \quad P(Z) = \alpha^{1-p}(Z - b)^p + b, \quad Q(Z) = \alpha^{1-q}(Z - b)^q + b.$$

In particular, P and Q commute. Even without the negative-support restriction, simultaneous nonconstant solutions force $P \circ Q = Q \circ P$.

Proof. Apply theorem 11.2; the representation is unique from its leading monomial and constant term. For the last assertion, commute the dilations and use the fact that they fix the coefficients: $(P \circ Q)(y) = (Q \circ P)(y)$. Transcendence of y turns equality at y into equality of rational functions. $\square$

12 A sharp infinitesimal-tail obstruction

The rational-map obstruction has a quantitative form. It identifies the first infinitesimal term that prevents an otherwise legitimate surreal solution from being an omnific integer.

Theorem 12.1 (First forbidden tail). *Let $P \in \mathbb{C}[Z]$ have degree $d \geq 2$, with leading coefficients p_d, p_{d-1} . Put*

$$b = -\frac{p_{d-1}}{dp_d}, \quad p_d \alpha^{d-1} = 1, \quad Q(X) = \alpha^{-1}(P(\alpha X + b) - b).$$

Then Q is monic and centered. If $Q \neq X^d$, write

$$Q(X) = X^d + cX^{d-r} + \text{lower powers}, \quad 2 \leq r \leq d, \quad c \neq 0.$$

For each positive scale γ , the infinite solution with leading coefficient α has expansion, with $T = t^{-\gamma}$,

$$y = \alpha T + b - \frac{\alpha c}{d} T^{1-r} + O(T^{-r}). \quad (12.1)$$

Its first positive-valuation term has valuation exactly $(r-1)\gamma$. In particular it does not belong to the negative-support ring.

Proof. The normalization eliminates the coefficient of X^{d-1} and makes the leading coefficient one. The infinity case of theorem 11.1 gives a unique normalized Laurent solution

$$F(T) = T + \sum_{n \geq 0} b_n T^{-n}, \quad F(T^d) = Q(F(T)),$$

with $y = \alpha F(T) + b$. Comparison at degree $d-1$ gives $b_0 = 0$. Successive comparison at degrees $d-2, d-3, \dots, d-r+1$ gives $b_1 = \dots = b_{r-2} = 0$. At degree $d-r \geq 0$, the left side has no term, and the right-side coefficient is $db_{r-1} + c$. Therefore $b_{r-1} = -c/d \neq 0$. This proves the expansion and the exact valuation. $\square$

Here $O(T^{-r})$ denotes an ordinary formal Laurent tail with powers at most $-r$; it is not an analytic estimate at all surreal magnitudes. After $T = t^{-\gamma}$, this notation has the exact support meaning required by the theorem.

12.1 Examples and exact coefficients

For $P(Z) = Z^2 + 1$, the unique infinite branch at each positive scale begins

$$y = T - \frac{1}{2}T^{-1} - \frac{3}{8}T^{-3} - \frac{3}{16}T^{-5} - \frac{45}{128}T^{-7} + \dots \quad (12.2)$$

It is a perfectly well-defined real surreal solution of $S_2y = y^2 + 1$ after $T = \omega^\gamma$, but is not omnific. There are no nonconstant omnific solutions to this equation.

For $P(Z) = Z^2 - 2$ there is an exact finite Laurent identity:

$$y = T + T^{-1}, \quad S_2y = T^2 + T^{-2} = y^2 - 2.$$

Again the infinitesimal term prevents an omnific solution. This example also explains why the full Hahn-field conclusion is Laurent-series rigidity, not polynomial rigidity.

For $P(Z) = Z^3 + Z$ one obtains

$$y = T - \frac{1}{3}T^{-1} - \frac{8}{81}T^{-5} - \frac{8}{243}T^{-7} + \dots$$

By contrast, $P(Z) = (Z - 3)^2 + 3$ has the omnific solutions $y = 3 + \omega^\gamma$, since its centered normalization is exactly Z^2 . The finite exact checks supplied with the article verify the displayed coefficients and the coordinate identities through the stated truncation orders; theorem 12.1 proves the infinite statement.

13 A strict hierarchy of autonomous algebraic orders

Definition 13.1. For a fixed dilation S_d , define the *autonomous algebraic order* of y over k to be the least $r \geq 0$ for which

$$y, S_dy, \dots, S_d^r y$$

are algebraically dependent over k . If no such r exists, set the order to $+\infty$. Constants have order zero. This definition concerns algebraic dependence of a transform tuple; it is not the usual order of an arbitrary variable-coefficient linear equation.

Theorem 13.2 (An infinite-support element of exact order two). *For each integer $d \geq 2$, the omnific integer*

$$u_d = \sum_{n \geq 0} \omega^{d^{-n}}$$

has exact autonomous algebraic order two over $\mathbb{C}$ for S_d . It satisfies

$$S_d^2 u_d - S_d u_d = (S_d u_d - u_d)^d. \quad (13.1)$$

Every two distinct forward transforms are nevertheless algebraically independent over $\mathbb{C}$.

Proof. Its decreasing normal-form exponents make the sum legitimate. Dilation is strong, and direct comparison of supports gives

$$S_d u_d = u_d + \omega^d.$$

Applying S_d once more gives $S_d^2 u_d - S_d u_d = \omega^{d^2} = (\omega^d)^d$, proving (13.1). The polynomial $Z - Y - (Y - X)^d$ is nonzero, so the order is at most two. By theorem 10.1, the infinite support makes $u_d, S_d u_d$ algebraically independent. Thus the order is exactly two. The last assertion is corollary 10.2. $\square$

Decreasing-denominator generalized series and their linear Mahler identities are already present in the literature; compare [10, Remark 2.18] and [9]. The series pattern is therefore not offered as a new construction. The point here is the rigorous *minimal autonomous order* and the resulting sharp boundary of the first-order classification in the omnific setting. [write] In the collection, the linear identity $S_2 u_2 - u_2 = \omega^2$ is the Conway form, after one application of S_2 , of the second series in the single-dilation report's example `dsup:ex:opposite` (section 1.5). [merge] Source 03 re-proves the theorem without theorem 10.1, by a character argument for the transcendence of u_d over $\mathbb{C}(\omega)$ (proposition 24.7), and it is the case $\rho = 1$ of source 02's infinite supports of rank ρ and exact order $\rho + 1$ (theorem 24.8).

Theorem 13.3 (Every finite order occurs). *Let $g_1, \dots, g_r$ be positive surreal exponents linearly independent over $\mathbb{Q}$, and let $a_1, \dots, a_r \in \mathbb{R}^\times$. Then*

$$x = \sum_{j=1}^r a_j \omega^{g_j}$$

has exact autonomous algebraic order r over $\mathbb{C}$ for every integer $d \geq 2$.

Proof. Put $T_j = \omega^{g_j}$. The T_j are algebraically independent over $\mathbb{C}$, since distinct monomials in them have distinct normal-form exponents. For $0 \leq i < r$, put

$$Y_i = \sum_{j=1}^r a_j T_j^{d^i}.$$

Their Jacobian determinant is

$$\left(\prod_{j=1}^r a_j \right) \left(\prod_{i=0}^{r-1} d^i \right) \det(T_j^{d^i-1})_{0 \leq i < r, 1 \leq j \leq r}. \quad (13.2)$$

The last determinant is nonzero: its permutation terms have distinct exponent vectors, since the integers $d^i - 1$ are distinct. In characteristic zero, a nonzero Jacobian implies algebraic independence. For completeness, a polynomial relation of least total degree would, on differentiation, yield a nontrivial linear relation between the differentials dY_i . At least one evaluated partial derivative is nonzero, because otherwise a nonzero partial derivative would be a relation of smaller degree. This contradicts the determinant.

Thus the first r transforms are independent. The first $r + 1$ lie in the field $\mathbb{C}(T_1, \dots, T_r)$ of transcendence degree r , so they are dependent. The least order is exactly r . $\square$

[merge] When $a_1 = \dots = a_r = 1$, the independence of $Y_0, \dots, Y_{r-1}$ is the algebraic independence of the power sums of degrees $1, d, \dots, d^{r-1}$ in r variables, a special case of Melánová–Sturmfels–Winter [18, Proposition 3] (any $m \leq n$ power sums of distinct positive degrees in n variables are independent, by a Jacobian computation with Vandermonde minors). The original manuscript did not cite it; no priority is claimed for that case. The theorem is also the independent-exponent case of the exact finite-support formula theorem 21.2.

Proposition 13.4 (Bounds from the rational rank of finite support). *Let x be a nonconstant finite-support omnific integer and let r be the dimension over $\mathbb{Q}$ of the span of its positive exponents. Its order is at most r . It is exactly one for $r = 1$ and exactly two for $r = 2$; for $r \geq 3$, the present results give the bounds $2 \leq \text{order} \leq r$.*

Proof. Choose a rational basis of the exponent span and clear the finitely many denominators. Then every transform of x lies in a rational function field generated by r algebraically independent monomials. This gives the upper bound. The first-order classification gives precisely the remaining assertions. $\square$

[merge] *Status (batch 33)*. The order is exactly r in every rank: corollary 21.4, proved independently by sources 02 and 03, gives equality for $r \geq 3$ as well, for every positive rational dilation other than 1, and for all finite-support surreal and surcomplex numbers. The bounds stated above remain true; they are no longer the best known.

14 Boundaries that the conclusions cannot cross

14.1 Positive characteristic

If $k = \mathbb{F}_p$ and the dilation is S_p , then

$$S_p y = y^p$$

for every Hahn series y : Frobenius acts coefficientwise on these strong sums. Hence completely arbitrary well-ordered supports can satisfy a degree-one autonomous relation in the dilated variable. The characteristic-zero hypothesis cannot be dropped. In the proof, both Newton–Puiseux and the nonvanishing of the coefficient d are substantive.

14.2 A named nonconstant coefficient

The infinite-support series u_d satisfies the linear equation

$$S_d u_d - u_d = \omega^d.$$

The right side is not in the ordinary constant field. Thus there is no extension of the support-collapse theorem to unrestricted nonconstant coefficients. The distinction from variable-coefficient Mahler algorithms [11, 10] is structural, not a matter of proof technique. [write] At $d = 2$ this equation is the Conway form of `dsup:ex:opposite` after one application of S_2 (section 1.5).

14.3 Identity dilation and arbitrary rational dilations

For $d = 1$, every series satisfies $S_1 y - y = 0$, so the theorem is false. For a nonintegral rational $q > 1$, theorem 11.1 rules out explicit rational-map solutions. That argument does not classify implicit relations $F(y, S_q y) = 0$. In the Puiseux step the leading order becomes qe , and taking an e th root need not produce an ordinary integer-order germ. The proof given here must not be quoted as covering that case. [write] Implicit relations of this kind do have nonconstant solutions: by the single-dilation report’s `dsup:lem:rationalrecognition`, for divisible Γ and $q = a/b$ the nonzero solutions of $(S_q x)^b = x^a$ are exactly the monomials $\mu_{|a-b|}(k) t^\Gamma$. They satisfy the cyclic-support conclusion trivially; the general implicit case remains open (section 1.5).

14.4 Full first-order theories

The decision procedure below is for one specified autonomous equation with algebraic-number coefficients. It says nothing about arbitrary quantifier alternations in an expanded surreal field, arbitrary systems of omnific Diophantine equations, or equality of arbitrary infinite normal-form descriptions. In particular, it does not contradict the undecidability phenomena discussed in the repository’s single-dilation report [2].

15 Effective finite shapes and a decision procedure

The sharp degree bound turns the omnific problem into a finite algebraic problem, even though a general omnific integer is specified by a potentially transfinite normal form.

Theorem 15.1 (Finitely many bounded polynomial shapes). *Let k be algebraically closed of characteristic zero, $0 \neq F \in k[X, Y]$, and $d \geq 2$. Put $D = \deg_Y F$. The set*

$$\mathcal{P}_{\leq D}(F, d) = \{P \in k[T] : 1 \leq \deg P \leq D, F(P(T), P(T^d)) = 0\} \quad (15.1)$$

is finite. Every nonconstant solution in a negative-support Hahn ring has the form $P(t^{-\delta})$ with P in this finite set and $\delta > 0$. The analogous statement holds elementwise for omnific and Gaussian omnific integers, retaining the appropriate coefficient and constant-term conditions.

Proof. Fix a degree M . For a polynomial P of that degree satisfying the identity, view $P(t^{-1})$ in $k((t^{\mathbb{Q}}))$. It has valuation $-M$. By theorem 7.1 it equals $f(t^{\delta})$ for one of finitely many infinity profiles f of pole order e . Valuation comparison forces $\delta = M/e$. Thus, for each fixed M and profile, there is at most one possible polynomial P . Finiteness follows on taking $1 \leq M \leq D$. Exhaustion of all negative-support solutions is precisely the degree-bounded assertion of theorem 8.1 and transcendence of the chosen monomial. Filtering the coefficients gives the omnific versions. $\square$

The bound on the polynomial degree is essential to this finiteness statement. For example, $S_d y = y^d$ is satisfied by $P(T) = T^n$ for arbitrarily large n . These do not create new shapes modulo the free monomial scale, since $(t^{-\delta})^n = t^{-n\delta}$.

Theorem 15.2 (Decidable nonconstant omnific solvability). *Let $F \in \overline{\mathbb{Q}}[X, Y] \setminus \{0\}$ be supplied by exact algebraic-number coefficients and let $d \geq 2$. There is an algorithm to decide whether $F(x, S_d x) = 0$ has a nonconstant solution in $\mathbf{Oz}$, or in $\mathbf{Oz}[i]$. It produces a finite list of polynomial shapes whose positive surreal monomial evaluations give all such solutions. Every coefficient of those shapes is an algebraic number.*

Proof. If $D = \deg_Y F = 0$, no nonconstant solution exists. Otherwise, for each $1 \leq M \leq D$, introduce unknowns $a_0, \dots, a_M$ and form

$$P(T) = a_0 + a_1 T + \dots + a_M T^M.$$

Set every coefficient in T of $F(P(T), P(T^d))$ equal to zero and impose $a_M \neq 0$, for instance by adding an auxiliary inverse variable. These are finite polynomial systems with algebraic-number coefficients. By theorem 15.1, each has only finitely many solutions in $\mathbb{C}$. A finite algebraic set over $\overline{\mathbb{Q}}$ has algebraic-number coordinates. Its points can be computed by standard zero-dimensional polynomial-system methods. Alternatively, separate real and imaginary parts and use exact real-algebraic point isolation [20]; a nonzero complex leading coefficient is imposed by inverting its squared modulus.

For $\mathbf{Oz}$, retain precisely the shapes with every coefficient real and $a_0 \in \mathbb{Z}$. For $\mathbf{Oz}[i]$, retain those with $a_0 \in \mathbb{Z}[i]$. Reality, equality, and integrality of exact algebraic numbers are decidable. The retained list is nonempty exactly when a nonconstant solution exists. Its evaluations at $T = \omega^{\delta}$ give all solutions by theorem 10.1. A witness may always be produced already at $T = \omega$. $\square$

The algorithm is a termination result, not a practical complexity claim. The coefficient systems can be large, and a production implementation would exploit Puiseux branches and the finite-profile structure before using general elimination. The supplied Python program implements exact checks of representative identities, not this complete elimination algorithm.

15.1 A finite computational specification

The equation expanded in the proof has degree at most $M(\deg_X F + d \deg_Y F)$ in T . Thus there is an explicit finite bound on the number of coefficient equations. Every returned candidate is certified simply by the exact polynomial identity $F(P(T), P(T^d)) = 0$, the leading-coefficient test, and the constant-term condition. No truncation of an unknown infinite surreal number is used as a certificate of support termination.

The Newton-weight test of corollary 7.4 is a different algorithm: it decides nonconstant solvability in the full surcomplex field, where infinite Laurent tails are allowed. The two procedures can therefore legitimately give different answers, as for $S_2x = x^2 + 1$.

16 Further research questions and priorities

The questions below are proposed research directions arising from this article. Unless explicitly stated otherwise, they are not claimed to be previously published named open problems. Their purpose is to specify concrete next theorems rather than to suggest that the present first-order classification covers a wider theory than it actually does.

Research question 1 (Exact order from finite support; high priority). For a finite-support omnific integer whose positive exponents have rational rank $r \geq 3$, is its autonomous order always r ? The answer is established here for $r \leq 2$ and for sums supported on a rationally independent set. The unresolved case allows several supported monomials in the same r -dimensional exponent span. A useful first target is rank three with four support terms. The bound in proposition 13.4 isolates the precise remaining gap.

[merge] *Status (batch 33): answered, yes, in every rank.* Sources 02 and 03 prove that the order of a finite-support omnific or Gaussian omnific integer is the rational rank of its positive exponents, with arbitrary dependent supports, for every integer $d \geq 2$ and every rational $d > 0$, $d \neq 1$ (corollary 21.4); the rank-three, six-term example of example 23.1 has order three.

Research question 2 (The support spectrum at order two; high priority). Which well-ordered Hahn supports can occur for omnific solutions of $F(x, S_d x, S_d^2 x) = 0$ over the ordinary constants? Infinite support of order type ω occurs by theorem 13.2. Determine whether fixed equation degree or fixed autonomous order bounds the possible support order types, denominator growth, or rational rank. The goal is a classification or a sharp no-bound theorem, not an extrapolation from the one telescoping example.

[merge] *Status (batch 33): the rational-rank component is answered.* A relation of autonomous order n , with any support, forces rational support rank at most n (corollary 20.11, both sources), and the bound is sharp. Source 02 exhibits infinite supports of every rank $\rho \geq 1$ with exact order $\rho + 1$ (theorem 24.8). Whether fixed degree or order bounds the support order type or the denominator growth remains open; theorem 13.2 shows unbounded denominators at order two.

Research question 3 (Formal verification of the core theorem; high priority). Formalize the chain from strong Hahn evaluation through the monomial recognizer, the formal Böttcher coordinate, and the evaluated Puiseux branch to theorem 6.1. The most substantial imported obligation is a usable Newton–Puiseux theorem with the *individual branch* degree bound. A staged formalization can state that theorem as an explicit parameter before replacing it with a verified construction; it should not silently add it as an axiom and call the entire result verified.

Research question 4 (An efficient shape solver). Implement theorem 15.2 for rational or algebraic coefficients, using the Newton-weight filter, Puiseux branches, and polynomial degree bounds before general elimination.

Prove bounds in terms of the input bidegree, coefficient height, and d . Can the finite candidate list be computed without constructing any nonpolynomial Böttcher profile? Can equivalent shapes under monomial rescaling be canonically minimized?

Research question 5 (Nonintegral rational dilation). Classify autonomous implicit equations $F(y, S_q y) = 0$ for $q \in \mathbb{Q}_{>1} \setminus \mathbb{Z}$. The explicit rational case has no nonconstant solutions, but a ramified correspondence can have the required rational Newton slope. Determine whether an appropriate generalized Böttcher coordinate still forces a controlled support subgroup, or permits unbounded denominators already at order one.

[write] *Status in the collection: open, with one partial datum.* The single-dilation report's `dsup:lem:rationalrecognition` solves the particular relation $(S_q x)^b = x^a$, $q = a/b$, exactly: for divisible Γ its nonzero solutions are $\mu_{|a-b|}(k) t^\Gamma$. So the class of implicit relations with nonintegral q is not empty of nonconstant solutions, and for this relation the solutions are single monomials. Nothing is known here about general F . [merge] *Status (batch 33): one further partial datum.* By remark 20.12, every solution of $F(y, S_q y) = 0$ over k , for any rational $q > 0$, $q \neq 1$, and any support, has rational support rank at most one. This says nothing about denominators; the question stays open.

Research question 6 (Tame positive-characteristic variants). When $\text{char } k = p > 0$ and $p \nmid d$, which support-collapse statements survive under restrictions on Puiseux ramification? Separate failure caused by wild algebraic branches from failure of the coordinate construction. The Frobenius example with $d = p$ shows that no unrestricted analogue is possible, but does not settle the tame cases.

Research question 7 (Systems and higher-dimensional correspondences). For a tuple $\mathbf{y}$ satisfying $\mathbf{S}_d \mathbf{y} = \mathbf{R}(\mathbf{y})$, what bounds the number of independent support scales? In one dimension, a superattracting germ is formally conjugate to a monomial. Higher-dimensional germs need not have such a coordinate. Identify hypotheses under which a finite-rank support module, rather than one cyclic group, is forced.

Research question 8 (Controlled nonconstant coefficients). Replace the ordinary constant field by a finitely generated monomial field $k(t^\Delta)$. A linear equation already admits unbounded denominators, so cyclic support is too strong. Find the correct substitute: perhaps a bound on the support module generated under multiplication and division by d , relative to Δ , together with an ordinal support bound. Distinguish finite-generation assertions from mere countability.

[merge] *Status (batch 33): the relative-rank part is answered.* A relation among m distinct positive rational dilates with coefficients in $k(t^\Delta)$ forces support rank at most $m - 1$ modulo $\mathbb{Q}\Delta$ (remark 20.13; stated by source 02, and implied by the relative theorem of both sources). The support module, denominator and ordinal parts remain open.

Research question 9 (Arithmetic and convergence of the profiles). For algebraic coefficients, all profile coefficients are algebraic by corollary 7.2. Determine effective denominator and height bounds, and compare their Archimedean and non-Archimedean analytic convergence. The underlying Böttcher arithmetic is already an established subject [6]; the new target is a uniform statement through the ramified correspondence and the finite-profile list.

Research question 10 (Simultaneous algebraic correspondences). Describe the intersection of the finite-profile solution classes for two multiplicatively independent dilations. The rational omnific case is completely rigid by corollary 11.3, but implicit correspondences and full Laurent tails allow more geometry. Determine which compatibility conditions replace commutation of two rational maps.

Research question 11 (Definability of bounded shape complexity). For each fixed bound M , describe the locus of omnific elements whose least first-order relation degree is at most M , in the language with one

named dilation and in weaker languages. The repository already recovers normal-form data from a dilation [2]; the question here is the complexity and uniformity of the *bounded-degree* stratification. Do not confuse a countable union of definable loci with one first-order definable set.

Research question 12 (Interaction with surreal exponential and derivation). The dilation in this article is defined by the normal-form exponent group, not by the surreal exponential. Study how the finite-profile classes behave under the surreal exponential and under specified surreal exponential-compatible derivations. Even a negative closure theorem would clarify which parts of difference algebra and differential algebra can be combined without changing the constant field or losing support control.

The recommended order is to formalize the support-collapse theorem, implement the finite-shape solver, and then attack the rank-three and order-two problems. Each has a precise success criterion and can be pursued without a general theory of analysis on the entire proper class. [merge] The rank-three problem is now solved (Research question 1), and the rank part of the order-two problem (Research question 2); the further questions of the additions are Research questions 13–30 in section 27.5.

17 Proof audit and concluding assessment

17.1 Dependency ledger

Input or conclusion	Status and dependency
Hahn field arithmetic and normal forms	Classical framework; used in set-sized workspaces.
Strong support lemma	Classical Neumann lemma; supplies all infinite formal evaluations.
Newton–Puiseux factorization	Classical characteristic-zero theorem; individual denominator bound proved here by the roots-of-unity orbit.
Formal Böttcher coordinate	Classical theorem; explicit logarithmic construction and uniqueness proof included.
Monomial recognizer	Already present for S_2 in the repository; general- d least-term proof included.
Support collapse and finite profiles	Proposed contribution; deductions from the preceding inputs, with no rank-one hypothesis.
Omnific criterion and sharp degree	Proposed consequences; follow by intersecting cyclic support with a bounded interval.
Finite-shape decision procedure	Uses the sharp degree bound, finite profiles, and standard exact finite algebraic-system solving.
Higher-order examples	Exact written identities and algebraic independence arguments; the generalized series pattern itself is prior.

The most delicate proof points have been made explicit: the selected Puiseux denominator is not a common splitting denominator; the root of the algebraic branch must be chosen to equal the actual $S_d z$; formal evaluation must respect composition and dilation; and the nondivisible-group descent uses a subgroup intersection containing the leading valuation. None of these is delegated to numerical evidence.

17.2 What has and has not been checked

The accompanying exact checks cover six inverse Böttcher expansions through degree 32 in the normalized power-series variable, twenty resultant examples, ten nonzero Jacobian instances, forty-five finite telescoping identities, four Frobenius examples, and six Newton-weight examples. All passed in the execution recorded

in `verification_results.json`. The calculations use exact rational or polynomial arithmetic. Their purpose is to detect sign, normalization, indexing, and small-example errors.

These checks do not certify the full support theorem, the correctness of every classical dependency, or the decision procedure’s implementation. No Lean build was performed and no new Lean theorem is supplied. The repository’s pre-existing formalization status is not inherited by this article. Independent mathematical review is still needed.

The source search covered the named primary literature and the specified repository reports and inventory. The Nishioka–Nishioka paper was located and its abstract-level scope checked; its full text was not available through the retrieval route used here, and no detailed theorem from it is invoked. This is a limitation on novelty assessment, not a missing premise in any proof. An absence of matching search results is not proof of priority. [write] The full text of [7] was read when the manuscript joined the collection; what it contains, and the priority caveat that follows, are in Appendix C.3 and section 18.

17.3 Conclusion

The main structural distinction is now exact. At autonomous order one, an algebraic relation with an integer exponent dilation collapses an arbitrary Hahn support to a cyclic Laurent scale. In the omnific ring, that collapse forces a finite polynomial in one monomial, and the primitive polynomial degree is recoverable as an algebraic extension degree. For a fixed equation there are only finitely many shapes modulo scale, giving an effective solvability reduction for algebraic input.

At order two the finite-support conclusion already fails. This sharp change is a useful organizing principle for further surreal difference algebra. It supplies both positive classifications and explicit negative results without relying on a problematic global topology or treating a proper class as an ordinary analytic field.

18 Consolidated ledger of non-claims

[write] The source states its limitations where they arise, and each is kept in place. This ledger collects them with their locations; “README” and “audit” refer to the delivered README (not shipped; its content is in the collection README) and to `SOURCE_AUDIT.md`. Items S1–S24 are the source’s; items W1–W7 were added when the report joined the collection. [merge] The limitations of sources 02 and 03 are listed as items R1–R23 in section 27.6.

- S1. Not refereed. No named published conjecture is claimed solved. Novelty is proposed, not certified; priority is not certified by an exhaustive literature review, and an absence of search results is not proof of priority (title page; section 17; README; audit).
- S2. The classical inputs are credited, not claimed: Hahn arithmetic and normal forms, the Neumann support lemma, Newton–Puisseux factorization, formal Böttcher coordinates and their arithmetic, the repository’s $d = 2$ recognizer, resultants and exact algebraic-system solving. Mahler equations, Böttcher coordinates and non-Puisseux solutions are not new topics (sections 1, 3, 5 and 17; audit).
- S3. The repository comparison covers selected reports and the inventory at the pin. It is not a line-by-line review of the repository and not an exhaustive repository-wide absence certificate; no clone or repository build was made and no repository file was changed (section 1; README; audit).
- S4. The headline equivalence is about polynomial relations with S_d over the ordinary constants; it does not say that an infinite omnific integer is algebraic over $\mathbb{R}$ (section 1).

- S5. This is not a factorization theorem and settles none of the remaining factorization conjectures for omnific integers. The primality of $\omega^{\sqrt{2}} + \omega + 1$ is cited, not proved or claimed; dilation independence is a different property (section 1, section 10; audit).
- S6. The encyclopedia page was used for orientation only, never in place of a primary source (section 1; audit).
- S7. The late-2025 preprints [12, 13] are cited from their abstracts for context, not as proof inputs (section 1; bibliography; audit).
- S8. The source compared Nishioka–Nishioka [7] at abstract level only; no result of it is a premise, and its full comparison was named as a priority task before any publication-level novelty claim (section 17; README; audit). See W2.
- S9. Membership in $k((t^\delta))$ is necessary, not sufficient, for a relation; the support conclusion does not turn strong evaluation into valuation convergence (the remark after theorem 6.1).
- S10. Coefficient descent is for a fixed coefficient field of the equation (after corollary 7.2).
- S11. The Newton-weight test decides existence, not whether an arbitrary encoded series satisfies the equation, and does not make arbitrary complex coefficients computable input; it decides solvability in the full field, so it can differ from omnific solvability, as for $S_2x = x^2 + 1$ (after corollary 7.4; section 15).
- S12. The polynomial-field identity corollary 9.3 is a consequence and consistency check, not a priority claim in polynomial decomposition theory; it fails for Laurent polynomials such as $T + T^{-1}$ (after corollary 9.3; audit).
- S13. Pairwise independence of dilates is not independence of a longer tuple (the example of u_d in section 10).
- S14. Over a field that is not algebraically closed, and for real solutions, the fixed point and leading-coefficient root in theorem 11.1 are genuine conditions, not automatic consequences of the complex theorem (after theorem 11.1).
- S15. In theorem 12.1, $O(T^{-r})$ is a formal Laurent tail, not an analytic estimate at all surreal magnitudes; the finite checks verify displayed coefficients through stated truncation orders only (after theorem 12.1; section 12).
- S16. The decreasing-denominator series pattern of u_d and its linear Mahler identity are prior [10, Remark 2.18], [9]; only the minimal autonomous order is claimed (after theorem 13.2; section 17; audit).
- S17. Characteristic zero cannot be dropped: in characteristic p with $d = p$, Frobenius gives arbitrary supports (section 14).
- S18. There is no extension to unrestricted nonconstant coefficients, by $S_d u_d - u_d = \omega^d$ (section 14).
- S19. For $d = 1$ the theorem is false; implicit relations $F(y, S_q y) = 0$ with nonintegral rational q are not covered by the proof (section 14). See W3.
- S20. The decision procedure concerns one specified equation with algebraic-number coefficients. It says nothing about full first-order theories, arbitrary systems of omnific Diophantine equations or equality of infinite normal-form descriptions, and does not contradict the single-dilation report’s undecidability (section 14).
- S21. The decision procedure is a termination result, not a complexity claim; it is not implemented, and `code/verify.py` checks representative identities only. The degree bound is essential to the finiteness of shapes (section 15; README; audit).

- S22. The exact autonomous order for finite support of rational rank $r \geq 3$ is open; only $2 \leq \text{order} \leq r$ is proved (proposition 13.4; section 16). [merge] *Status (batch 33)*: no longer open; the order is r (corollary 21.4).
- S23. The twelve research questions are proposed directions, not previously published open problems. A formalization should not add the Newton–Puiseux theorem as an axiom and call the result verified (section 16; audit).
- S24. No Lean formalization is supplied and no repository Lean build was performed; the collection’s formalization status is not inherited. The finite exact checks cover finite instances only; they do not certify the infinite theorems, the classical dependencies or the decision procedure. Independent mathematical review remains necessary. The script should not be run with Python optimization flags that disable assertions (title page; section 17; Appendix B; README; audit).
- W1. No `adr`: label has a Lean mapping in `FORMALIZATION.md` (section 1.5).
- W2. The full text of [7], read for this report, records Mahler’s 1983 criterion [8] for a convergent power-series solution of a constant-coefficient relation $P(f(z), f(z^d)) = 0$. For ordinary power series over a number field or $\mathbb{C}$, existence of nonconstant solutions of such relations is therefore prior to this report, and no priority is claimed for theorem 7.3 and corollary 7.4 or theorem 7.1 restricted to that rank-one setting. Mahler’s paper itself was not read; its content is taken from [7] (Appendix C.3).
- W3. The single-dilation report’s power-equation lemma gives nonconstant solutions of $(S_q x)^b = x^a$ for nonintegral q . This is a partial datum for the question on nonintegral rational dilation, which stays open (section 1.5, section 16).
- W4. None of the three questions of the single-dilation report (`dsup:sec:questions`) is answered; its question on definability from other exponent automorphisms is untouched beyond integer dilations. Its non-claim on variable-coefficient Mahler systems stands (section 1.5).
- W5. The exponent dilations here are not the argument dilations $f(z) \mapsto f(qz)$ of the holonomic-rigidity report, and nothing of that report is used or extended (section 1.5).
- W6. Nothing here bears on which value-group automorphisms admit exponential lifts (`q:image` of the exponential-automorphism report); S_d has no exponential lift by that report’s `thm:dilation`. None of the surcomplex-automorphisms report’s questions is addressed (section 1.5).
- W7. The source’s repository statements were checked at the placement commit and stand; they are kept as provenance of its pin (Appendix C.2).

19 Support rank and dilation orbits: two later manuscripts

[merge] Sections 19 to 27 print two manuscripts of batch 33 of the collection, written against this report. In the report’s local numbering (Appendix C.5) the original manuscript is source 01, and the additions are:

- *source 03*, archive `surreal_dilation_rank_research`, manuscript 04 of batch 33, *Support Rank and Algebraic Independence of Dilation Orbits*, pinned to `efc5446`. It is the *base* of this part: it works over an arbitrary ordered subfield $\mathcal{F}$ of the coefficient field, so that real dilations are allowed, relative to a full Hahn subfield $K_H(k)$, and it adds ordinal families, a uniform tail beyond any set of parameters, a prime-field formula and a polynomial-time certificate;

- *source 02*, archive `surreal_dilation_rank`, manuscript 02 of batch 33, *Support Rank and Algebraic Freeness under Surreal Dilations*, pinned to 958b5c4. It proves the same central theorems over $\mathbb{Q}$ by the same method, and adds a general finite denominator index, an infinite-support hierarchy of rank ρ and exact order $\rho + 1$ for every $\rho \geq 1$, set-sized families of free orbits over a prescribed base with equivariant embeddings of free difference rings into $\mathbf{Oz}$ and $\mathbf{Oz}[i]$, a coefficient-matrix algorithm for rational profiles, and the relative part of Research question 8.

Both sources compared themselves with this report at their pins. Its `article.tex` is the same file at both pins, at the placement commit aa9c891 and at the commit in which the report was written, so their descriptions of sections 13 and 16 are accurate. Neither source uses theorem 6.1, Newton–Puiseux factorization or the Böttcher coordinate: both state this, and the proofs below confirm it. A theorem proved by both sources is printed once, with both named; genuinely different proofs are kept as marked second routes. Text added in the merge is marked [merge]; a bracket such as [03, 02] after a heading names the sources that prove the statement.

19.1 What the additions settle in this report

1. *Research question 1* (exact order from finite support) is answered affirmatively in every rank by both sources: the autonomous order of a finite-support omnific or Gaussian omnific integer is the rational rank of its positive exponents (corollary 21.4), for every integer $d \geq 2$ and indeed every rational $d > 0$, $d \neq 1$. So the bound $2 \leq \text{order} \leq r$ of proposition 13.4 is an equality for $r \geq 3$ as well.
2. *Research question 2*, rank component: an autonomous relation of order n , with any support, forces rational support rank at most n (corollary 20.11), again by both sources. The order-type and denominator components stay open. Source 02 adds the data of theorem 24.8: infinite supports of every rank $\rho \geq 1$ with exact order $\rho + 1$.
3. [merge] *Research question 5*, a partial datum that neither source states: with $m = 2$, corollary 20.11 shows that every solution of $F(y, S_q y) = 0$ over k , for rational $q > 0$, $q \neq 1$, and with any support, has rational support rank at most one (remark 20.12).
4. *Research question 8*, relative part (source 02 states it; the relative theorem of both sources gives it): a relation in m distinct dilates with coefficients in $k(t^\Delta)$ forces rank at most $m - 1$ modulo the rational span of Δ (remark 20.13).
5. theorem 13.3 is the independent-exponent case of theorem 21.2; with all $a_j = 1$ its independence statement is a special case of Melánová–Sturmfels–Winter [18, Proposition 3], credited after theorem 13.3. Theorem 13.2 is re-proved by source 03 without theorem 10.1 (proposition 24.7) and is the case $\rho = 1$ of source 02’s theorem 24.8.

Not settled: Research questions 3, 4, 6, 7 and 9–12, the order-type and denominator parts of Research question 2, and the general case of Research question 5. [merge] Source 03’s question on the exact order of $u_{d,\ell}$ (Research question 13) is *not* answered by source 02: its hierarchy raises the rank with a gap of one, whereas $u_{d,\ell}$ has rank one and a candidate gap of ℓ (remark 24.10).

19.2 Conventions and renamed symbols

[merge] Both sources write Hahn series with *growth* support, $\sum c_g X^g$ (source 03) or $\sum a_\gamma t^\gamma$ (source 02), with reverse well-ordered supports, largest exponents leading and $X^g = t^g = t^{-g}$. This report uses the *valuation*

convention of section 2: $K_\Gamma(k) = k((t^\Gamma))$ with well-ordered supports and $t^\gamma \mapsto \omega^{-\gamma}$ (2.4). The two sources' Hahn-field statements are converted: every support is negated, “reverse well ordered” becomes “well ordered”, every greedy maximum becomes a minimum, and every “largest exponent” of a determinant becomes its valuation. Support rank is unchanged, since negation does not change a linear span. Statements about surreal numbers are written in Conway's ω -notation, where both conventions agree. The tempting false reading is that the “leading” term of a determinant is its largest t -exponent: in this report it is the term of least valuation, the largest ω -exponent.

The following symbols of the sources are renamed; everything else is as in the sources.

Here	Source 03 / source 02	Reason
$\mathcal{F}$, ordered subfield of k	F / (only $\mathbb{Q}$)	F is a relation polynomial in this report
Γ , ordered $\mathcal{F}$ -vector space	G / V	section 2
$K_\Gamma(k) = k((t^\Gamma))$, $K_H(k) = k((t^H))$	K_G, K_H / $\mathcal{K}_k(V), \mathcal{K}_k(H)$	valuation convention
$\rho_{\mathcal{F}, H}(y), \rho(y) = \rho_{\mathbb{Q}, 0}(y)$	$\rho_{F, H}, r(x) / r_H(y), r(y)$	r is an order in section 13
$\rho_{\text{cm}}(f)$, coefficient-matrix rank	$— / \rho(f)$	ρ is the support rank
$\text{aord}_{d, \mathfrak{B}}(y)$	$\text{aord} / \text{aord}_{S_d, B}$	base field $\mathfrak{B}$
∂_λ , Euler derivation; $\partial_1, \dots, \partial_m$	D_λ, D_i / D_ℓ, D_j	D never denotes a derivation here
$\lambda, \lambda_i, \vec{\lambda}(\gamma)$	$\lambda, \lambda_i, \Lambda(g) / \ell, \ell_j$	Λ is a lattice below
$b_1, \dots, b_m$, greedy basis	b_j / g_j	g_j are exponents in theorem 13.3
Σ , a support	A / E	$A_\Gamma(k)$ is the negative-support ring
η , leading exponent	E / η	
$q_1 > \dots > q_m$	descending / ascending	one ordering; see remark 20.9
Q , a finite set of dilations	Q / $—$	Q is a denominator
F , a relation polynomial	P / P	as in section 6
Λ , a finitely generated group or lattice	L / L	L is a field in section 7
$\Sigma_{P, Q} = \text{supp } P \cup \text{supp } Q; \tau$	$B; T$ / $—$	T is a monomial variable
T_1, T_2, T_3 , independent monomials	X, Y, Z / X, Y, Z	X, Y, Z are polynomial variables
$\Xi = \omega$	$— / Y$	Y is a variable
$\Xi_\alpha, g_{\alpha, n}$	$Z_\alpha, b_{\alpha n}$ / $—$	Z, b_j taken
$Y_\alpha, h_{\alpha, n}, v_0, \mathcal{I}$	$— / Y_\alpha, \gamma_{\alpha, n}, B, T$	
$\mathfrak{B}$, a set-sized base field; $\mathfrak{P}$, a parameter set	$E; P$ / F, B	F, P, B taken
$\mathcal{Z}_\rho, w_\rho = (S_d - 1)\mathcal{Z}_\rho$	$— / Z_r, P$	
$\mathcal{D}_y$, denominator group; N, N'	$— / A_y; N, M$	$A_\Gamma(k), M$ taken
$u_d^{[s]}$	$— / V_s$	V was the exponent space
τ_χ, τ_j , character twists	T_j / σ_χ	
$\sigma_1, \sigma_2, \sigma_3$, elementary symmetric	e_1, e_2, e_3 / $—$	e is an integer here
N_0, B_0, c_* (certificate)	M, B, C / $—$	
φ , a coefficient automorphism	$— / \rho$	
$\mathfrak{B} \subseteq \mathfrak{L}$ in lemma 20.2	$E \subseteq L$ / $B \subseteq K$	E, L, K taken
e_β , coefficients in lemma 20.1	d_u / b_β	d is the dilation
w_1, w_2, w_3 in example 24.9	$— / p, q, r_0$	q is a dilation
ℓ , observations in theorem 26.1	t / $—$	t is the Hahn variable

The letter ρ also names a positive valuation in the proof of lemma 3.3; that use is local. Kept from source 02 as local names: the matrix M of theorem 20.8 (elsewhere M is a degree), the abstract variables $X_1, \dots, X_s$ of section 22 and the monomials A, B of example 24.9 (not the ring $A_\Gamma(k)$). H is a subspace of exponents, not the Böttcher series H_a of proposition 5.2. The integer $d \geq 2$, the rational q , and S_q keep their meanings; S_a for $a \in \mathcal{F}_{>0}$ extends them. “Rank” always means rational (or $\mathcal{F}$ -linear) dimension of a support span, never a matrix rank unless a matrix is named. “Order” always means the autonomous algebraic order.

19.3 The setting

Definition 19.1 (Ordered scalar fields and exponent spaces [03, 02]). Let $\mathcal{F} \subseteq k$ be an ordered subfield of a field k ; in particular $\text{char } k = 0$. An *ordered $\mathcal{F}$ -vector space* is an $\mathcal{F}$ -vector space Γ whose additive group is totally ordered, with multiplication by every positive element of $\mathcal{F}$ preserving strict inequalities. For a set-sized such Γ put $K_\Gamma(k) = k((t^\Gamma))$, the full Hahn field with well-ordered supports, and for an $\mathcal{F}$ -subspace $H \subseteq \Gamma$ let $K_H(k) = k((t^H)) \subseteq K_\Gamma(k)$. Define the *relative support rank*

$$\rho_{\mathcal{F},H}(y) = \dim_{\mathcal{F}} \frac{H + \text{span}_{\mathcal{F}} \text{supp } y}{H}, \quad \rho(y) = \rho_{\mathbb{Q},0}(y). \quad (19.1)$$

The zero exponent contributes nothing. Source 02 works throughout with $\mathcal{F} = \mathbb{Q}$; source 03 allows any $\mathcal{F}$. The case $\mathcal{F} = \mathbb{Q}$ suffices for every exact finite-support statement, and $\mathcal{F} = \mathbb{R}$, $k = \mathbb{C}$ is used for the stronger independence examples.

The Hahn support facts used are the standard ones [4]: a finite sum of well-ordered subsets of Γ is well ordered, and a fixed exponent has only finitely many representations as a sum of one element from each of finitely many well-ordered sets. Source 03 notes the two-set case directly: infinitely many decompositions $u = g + h$ would, after selection, give a strictly increasing sequence of g 's and a strictly decreasing sequence of h 's, contradicting well-ordering; the finite case follows by induction. A family of series is strongly summable when the union of its supports is well ordered and each exponent lies in only finitely many supports; its sum is coefficientwise. Source 02 stresses that strong summation must not be confused with convergence of partial sums in a valuation topology: bounded increasing valuations often give strong sums without that convergence (compare section 2). No order on k is needed; k may be $\mathbb{C}$.

For $a \in \mathcal{F}_{>0}$, multiplication by a is an order automorphism of Γ , so

$$S_a \left(\sum_{\gamma} c_{\gamma} t^{\gamma} \right) = \sum_{\gamma} c_{\gamma} t^{a\gamma} \quad (19.2)$$

is a strong k -automorphism of $K_\Gamma(k)$, additive and multiplicative because $a(g + h) = ag + ah$, with

$$S_a S_b = S_{ab}, \quad S_1 = \text{id}, \quad S_a^{-1} = S_{1/a}, \quad S_a(K_H(k)) = K_H(k). \quad (19.3)$$

It agrees with (2.1) for $a = d$ and with the S_q of section 11 for $a = q$. It is not $y \mapsto y^a$, not argument dilation of an analytic function, and not the surreal exponential; the single-dilation report studies these exponent dilations and their definability properties [2]. “Autonomous” means that a relation has coefficients in the stated base, usually k ; in the relative version the base is $K_H(k)$. A relation with coefficients involving further monomials is not a relation over k , and passing to a larger core changes the relevant rank.

Definition 19.2 (Autonomous algebraic order over a base [02, 03]). For $d \in \mathbb{Q}_{>0} \setminus \{1\}$ and a subfield $\mathfrak{B}$ of the ambient field, put

$$\text{aord}_{d,\mathfrak{B}}(y) = \min\{n \geq 0 : y, S_d y, \dots, S_d^n y \text{ are algebraically dependent over } \mathfrak{B}\},$$

with value ∞ when no such n exists; elements of $\mathfrak{B}$ have order zero. For $\mathfrak{B} = k$ and integer $d \geq 2$ this is the autonomous algebraic order of section 13. An infinite family is algebraically independent when each finite subfamily is; for the orbit $\{S_d^n y\}$ this is *difference transcendence* over $\mathfrak{B}$. It is not differential transcendence under any chosen derivation.

Surreal transfer. Every surreal has a unique Conway normal form $x = \sum_g r_g \omega^g$ with set-sized reverse well-ordered support of exponents g and real $r_g \neq 0$, and every such series is a surreal [15, Chapter 5]; a surcomplex number has the analogous complex expansion, obtained from the union of its two real supports. Under (2.4), a set-sized ordered $\mathcal{F}$ -subspace $\Gamma \subseteq \mathbf{No}$ gives the embedded fields $\mathbb{R}((t^\Gamma)) \subseteq \mathbf{No}$ and $\mathbb{C}((t^\Gamma)) \subseteq \mathbf{No}[i]$. Finitely many given numbers, or the members of a given set, have a set-sized union of supports, so every finite algebraic relation among them lives in one set-sized Hahn field. In ω -notation $S_a(\sum r_g \omega^g) = \sum r_g \omega^{ag}$, and (2.5) shows that positive real and rational dilations preserve $\mathbf{Oz}$ and $\mathbf{Oz}[i]$; neither positivity nor integrality of the nonconstant coefficients is required. All powers ω^g are Conway's canonical monomials, not values of Gonshor's exponential.

Set theory. The Hahn-field theorems are set-sized algebra in ZFC. Linear functionals are extended by the basis theorem on a set-sized space only. For proper-class surreal conclusions each proposed finite relation is placed in one set-sized workspace; no global basis of $\mathbf{No}$ over $\mathbb{Q}$ or $\mathbb{R}$, no proper-class sum and no proper-class determinant is used. The ordinal-indexed families below are given by explicit formulas, and their algebraic independence means that every finite subfamily is independent: a class statement with set-sized proof instances, not a class-sized polynomial ring.

20 The greedy leading Jacobian term and the rank obstruction

20.1 Euler derivations and the Jacobian criterion

Lemma 20.1 (Support-preserving Euler derivations [03, 02]). *Let $\lambda : \Gamma \rightarrow k$ be $\mathcal{F}$ -linear. Then*

$$\partial_\lambda \left(\sum_\gamma c_\gamma t^\gamma \right) = \sum_\gamma c_\gamma \lambda(\gamma) t^\gamma \quad (20.1)$$

is a k -linear derivation of $K_\Gamma(k)$ that commutes with strong sums, and

$$\partial_\lambda S_a = a S_a \partial_\lambda \quad (a \in \mathcal{F}_{>0}). \quad (20.2)$$

If $\lambda(H) = 0$, then ∂_λ vanishes on $K_H(k)$.

Proof. The support of the right side of (20.1) is contained in the original support, so the series exists; additivity and k -linearity are coefficientwise. For $f = \sum c_\gamma t^\gamma$ and $h = \sum e_\beta t^\beta$ the coefficient of t^δ in $\partial_\lambda(fh)$ is the finite sum

$$\sum_{\gamma+\beta=\delta} c_\gamma e_\beta \lambda(\gamma + \beta) = \sum_{\gamma+\beta=\delta} c_\gamma e_\beta (\lambda(\gamma) + \lambda(\beta)),$$

the coefficient of $(\partial_\lambda f)h + f \partial_\lambda h$. The same support-subset property and finite coefficient sums give commutation with strong sums, and $\lambda(a\gamma) = a\lambda(\gamma)$ gives (20.2). An element of $K_H(k)$ has all its exponents in H , so its image is zero when $\lambda(H) = 0$. $\square$

Source 03 takes λ with values in $\mathcal{F}$ and source 02 takes $\mathbb{Q}$ -linear λ with values in k ; the statement above covers both. These are deliberately chosen coordinate derivations. They need not be Hardy-type derivations and are not the Berarducci–Mantova derivation or an analytic derivative of surreal functions, nor are they asserted to be exponential-compatible. Their role is algebraic: they separate independent exponent directions while annihilating the chosen coefficient core. General series derivations have an established theory [17]; this simple construction is proved so that no further differential-algebra theorem is a premise.

Lemma 20.2 (An invertible derivative matrix detects independence [03, 02]). *Let $\mathfrak{B} \subseteq \mathfrak{L}$ be fields of characteristic zero, let $\partial_1, \dots, \partial_m$ be derivations of $\mathfrak{L}$ vanishing on $\mathfrak{B}$, and let $z_1, \dots, z_m \in \mathfrak{L}$. If $\det[\partial_i(z_j)]_{j,i} \neq 0$, then $z_1, \dots, z_m$ are algebraically independent over $\mathfrak{B}$.*

Proof. Otherwise choose a nonzero $F \in \mathfrak{B}[Y_1, \dots, Y_m]$ of least total degree with $F(z_1, \dots, z_m) = 0$; its degree is positive. Applying ∂_i gives

$$\sum_{j=1}^m \frac{\partial F}{\partial Y_j}(z_1, \dots, z_m) \partial_i(z_j) = 0 \quad (1 \leq i \leq m).$$

Invertibility forces every evaluated partial derivative to vanish. In characteristic zero some formal partial derivative of the nonconstant F is a nonzero polynomial of smaller total degree, a smaller relation. This is a contradiction. $\square$

The proof shows the two hypotheses at work: the derivations must kill the *whole* coefficient field of the proposed relation, and characteristic zero prevents all partial derivatives of a minimal relation from vanishing. The relative core $K_H(k)$ is chosen to meet the first; the characteristic- p argument of section 26 replaces the second. The same criterion closes the proof of theorem 13.3, and source 02 compares it with the classical form [19, Theorem 6].

20.2 Greedy selection and strict rearrangement

Let $\Sigma \subseteq \Gamma$ be well ordered, and assume that its image in Γ/H spans a space of dimension at least m . Define successively

$$b_j = \min(\Sigma \setminus (H + \text{span}_{\mathcal{F}}\{b_1, \dots, b_{j-1}\})), \quad 1 \leq j \leq m. \quad (20.3)$$

Each set is nonempty by the rank hypothesis and has a least element by well-ordering. The images of $b_1, \dots, b_m$ in Γ/H are independent, and $b_1 < b_2 < \dots < b_m$. Elements of $\Sigma \cap H$, however small, are skipped: they cannot contribute to derivatives annihilating $K_H(k)$. Extending these images to a basis of the set-sized space Γ/H gives $\mathcal{F}$ -linear maps $\lambda_1, \dots, \lambda_m : \Gamma \rightarrow \mathcal{F}$ with

$$\lambda_i(H) = 0, \quad \lambda_i(b_j) = \delta_{ij}. \quad (20.4)$$

Write $\vec{\lambda}(\gamma) = (\lambda_1(\gamma), \dots, \lambda_m(\gamma)) \in \mathcal{F}^m$. The remaining basis coordinates are discarded; a term with zero projected determinant simply disappears from the Jacobian expansion.

Lemma 20.3 (Coordinatewise dominance of the greedy basis [03, 02]). *If $a_1 < \dots < a_m$ are elements of Σ with $\vec{\lambda}(a_1), \dots, \vec{\lambda}(a_m)$ linearly independent over $\mathcal{F}$, then $a_j \geq b_j$ for $1 \leq j \leq m$.*

Proof. If $a_j < b_j$, then $a_1, \dots, a_j$ are all below b_j . By the minimality defining b_j , every element of Σ below b_j lies in $H + \text{span}_{\mathcal{F}}\{b_1, \dots, b_{j-1}\}$, so its $\vec{\lambda}$ -image lies in the span of the first $j-1$ coordinate vectors, and the j images $\vec{\lambda}(a_1), \dots, \vec{\lambda}(a_j)$ would be dependent. $\square$

This is the part of the greedy principle for linear matroids [16] that is needed. It compares the j th selected element with the j th element of *every* competing independent selection, which is stronger than optimizing one unweighted sum. The proof is finite dimensional even when Σ has uncountable order type, and no traversal of an infinite support is claimed as an algorithm.

Lemma 20.4 (Strict rearrangement [03, 02]). *Let $a_1 < \dots < a_m$ lie in an ordered $\mathcal{F}$ -vector space and let $q_1 > \dots > q_m > 0$ lie in $\mathcal{F}$. For every permutation π ,*

$$\sum_{j=1}^m q_j a_{\pi(j)} \geq \sum_{j=1}^m q_j a_j,$$

with equality only for the identity.

Proof. If $i < j$ but $a_{\pi(i)} > a_{\pi(j)}$, exchanging the two assigned elements lowers the sum by $(q_i - q_j)(a_{\pi(i)} - a_{\pi(j)}) > 0$, and sorting a nonidentity permutation needs at least one such exchange. Only the ordered-vector-space axioms are used: no Archimedean hypothesis and no real-valued weight. $\square$

Proposition 20.5 (Unique minimizing support tuple [03, 02]). *Let $q_1 > \dots > q_m > 0$ in $\mathcal{F}$. Among all ordered tuples $(g_1, \dots, g_m) \in \Sigma^m$ with $\det[\lambda_i(g_j)]_{j,i} \neq 0$, the sum $\sum_j q_j g_j$ has the unique minimum*

$$\eta = q_1 b_1 + \dots + q_m b_m, \quad (20.5)$$

attained only at $(g_1, \dots, g_m) = (b_1, \dots, b_m)$.

Proof. A nonzero determinant makes the g_j distinct; sort them as $a_1 < \dots < a_m$. Lemma 20.4 gives $\sum_j q_j g_j \geq \sum_j q_j a_j$, and lemma 20.3 with $q_j > 0$ gives $\sum_j q_j a_j \geq \sum_j q_j b_j$. The first inequality is strict unless the tuple is already increasing; the second is strict if some $a_j > b_j$, because a sum of nonnegative elements with a positive summand is positive in an ordered group. So equality forces $g_j = b_j$ for all j , and the greedy tuple has identity projected matrix by (20.4). $\square$

The proposition excludes the dangerous case in which several extreme terms have the same exponent and cancel: it identifies the extremal *ordered tuple*, not merely an extremal value of a weight on possibly cancelling terms.

20.3 The leading Jacobian term

Theorem 20.6 (The leading Jacobian term [03, 02]). *Let $y = \sum_{\gamma \in \Sigma} c_\gamma t^\gamma \in K_\Gamma(k)$, choose the greedy exponents and dual functionals as above, and let $q_1 > \dots > q_m > 0$ in $\mathcal{F}$. Put $J_{ji} = \partial_{\lambda_i}(S_{q_j} y)$. Then $v(\det J) = \eta$, and*

$$[t^\eta] \det J = \left(\prod_{j=1}^m q_j \right) \left(\prod_{j=1}^m c_{b_j} \right) \neq 0. \quad (20.6)$$

Proof. By lemma 20.1, $J_{ji} = q_j \sum_{\gamma \in \Sigma} c_\gamma \lambda_i(\gamma) t^{q_j \gamma}$. Expanding the determinant and collecting one support exponent from each row, an ordered tuple $(g_1, \dots, g_m)$ contributes

$$\left(\prod_j q_j \right) \left(\prod_j c_{g_j} \right) \det[\lambda_i(g_j)]_{j,i} t^{\sum_j q_j g_j}. \quad (20.7)$$

The regrouping is legitimate: there are finitely many permutations, each product has support in the finite sum of the well-ordered sets $q_j \Sigma$, and each coefficient receives finitely many contributions; no conditionally convergent series is rearranged. Tuples with zero projected determinant contribute zero. By proposition 20.5 the unique least exponent among the others is η , from $(b_1, \dots, b_m)$, whose projected matrix is the identity. Each c_{b_j} is nonzero by definition of the support, and each q_j is nonzero in the characteristic-zero field k . $\square$

Theorem 20.7 (Support-rank obstruction [03, 02]). *Let $y \in K_\Gamma(k)$ and let $a_1, \dots, a_m$ be distinct elements of $\mathcal{F}_{>0}$, $m \geq 1$. If $\rho_{\mathcal{F},H}(y) \geq m$, then $S_{a_1}y, \dots, S_{a_m}y$ are algebraically independent over $K_H(k)$. Consequently, for every finite list of distinct positive dilations,*

$$\text{trdeg}_{K_H(k)} K_H(k)(S_{a_1}y, \dots, S_{a_m}y) \geq \min\{m, \rho_{\mathcal{F},H}(y)\}, \quad (20.8)$$

where the minimum is m when the rank is infinite.

Proof. Choose $b_1, \dots, b_m$ by the rank hypothesis. The derivations ∂_{λ_i} vanish on $K_H(k)$ by (20.4) and lemma 20.1, and their matrix on the dilates, with the dilations listed in decreasing order, is invertible by theorem 20.6. Lemma 20.2 applies, and reordering a tuple does not affect independence. For rank smaller than m apply this to a subset of that many dilations; for rank zero the bound is zero. $\square$

Source 03 proves this for every ordered $\mathcal{F}$; source 02 proves it for $\mathcal{F} = \mathbb{Q}$ (its relative support-rank obstruction), with a different order of choices, recorded as a second route.

Theorem 20.8 (Vector-valued greedy determinant [02]). *Let $\Sigma \subseteq \Gamma$ be well ordered and let $c_\gamma \in k^m$, $\gamma \in \Sigma$, span k^m ; ignore zero vectors. Choose*

$$g_1 = \min\{\gamma : c_\gamma \neq 0\}, \quad g_j = \min\{\gamma : c_\gamma \notin \text{span}_k(c_{g_1}, \dots, c_{g_{j-1}})\} \quad (2 \leq j \leq m).$$

Then $g_1 < \dots < g_m$, and if $h_1 < \dots < h_m$ index independent vectors then $h_j \geq g_j$ for each j . For $q_1 > \dots > q_m > 0$ in $\mathbb{Q}$, the matrix $M_{ij} = \sum_{\gamma \in \Sigma} c_{\gamma,j} t^{q_i \gamma}$ has nonzero determinant, with

$$v(\det M) = \sum_{j=1}^m q_j g_j, \quad [t^{v(\det M)}] \det M = \det(c_{g_i,j})_{1 \leq i,j \leq m} \neq 0.$$

Multiplying row i by q_i multiplies the coefficient by $\prod_i q_i$ and leaves the valuation unchanged.

Proof. The choices exist because the family has rank m . Every vector with exponent below g_j lies in the span of the first $j-1$ selected vectors, since otherwise it would be eligible in place of g_j . If $h_j < g_j$, the j independent vectors indexed by $h_1, \dots, h_j$ would lie in that $(j-1)$ -dimensional span. Strict ascent holds because exponents below an earlier selected exponent were already ineligible and the selected exponent itself lies in the enlarged span.

A choice of exponents $\gamma_1, \dots, \gamma_m$, one per row, contributes $\det(c_{\gamma_i,j})_{i,j} t^{q_1 \gamma_1 + \dots + q_m \gamma_m}$; the union of possible exponents lies in $q_1 \Sigma + \dots + q_m \Sigma$ and each coefficient has finitely many contributions. Dependent choices contribute zero. For an independent choice sorted as $h_1 < \dots < h_m$, lemma 20.4 and the dominance just proved give $\sum_i q_i \gamma_i \geq \sum_j q_j h_j \geq \sum_j q_j g_j$, with the first equality only for the increasing assignment and the second only when every $h_j = g_j$, by positivity of the weights. Exactly one independent row choice attains the minimum, with rows $c_{g_1}, \dots, c_{g_m}$. $\square$

Remark 20.9 (Source 02's ordering, and its route to theorem 20.7). [merge] Source 02 lists the parameters increasingly, $0 < q_1 < \dots < q_m$, and states its leading exponent as $\sum_j q_{m+1-j} g_j$ with coefficient $(-1)^{m(m-1)/2} \det(c_{g_i,j})$: reversing the rows of M gives the sign. This is the same statement. For finite Σ the greedy basis is the usual greedy basis of a linear matroid [16], and the proof shows why an infinite well-ordered ground set adds no difficulty when the requested rank is finite. *Second route to theorem 20.7* (source 02, for $\mathcal{F} = \mathbb{Q}$). Choose m supported exponents independent modulo H , take coordinate functionals to $\mathbb{Q}$ on their span modulo H , and extend them to Γ/H by extending a rational basis; composing with $\Gamma \rightarrow \Gamma/H$ gives $\lambda_1, \dots, \lambda_m : \Gamma \rightarrow \mathbb{Q} \subseteq k$ vanishing on H , of rank m on the selected exponents. With $y = \sum c_\gamma t^\gamma$ put

$\vec{c}_\gamma = c_\gamma \vec{\lambda}(\gamma) \in k^m$; these span k^m , since the selected square minor is a nonzero rational determinant times nonzero coefficients. The Euler matrix is $(\partial_{\lambda_j} S_{q_i} y)_{i,j} = (q_i \sum_\gamma \vec{c}_{\gamma,j} t^{q_i \gamma})_{i,j}$, whose determinant is nonzero by theorem 20.8; conclude with lemma 20.2. Here the functionals are fixed first and the greedy basis is chosen among their images, whereas source 03 chooses the greedy basis first and makes the functionals dual to it. Both routes need a rank- m family of coefficient vectors, not merely m nonzero coefficients, and discard zero Euler vectors before the greedy choice.

Remark 20.10 (Why the argument handles difficult supports [03, 02]). The least exponent of y alone may not suffice: it may lie in the core, or several of the smallest exponents may span one direction. Greedy selection skips exactly those, and an arbitrary independent set would not do, because terms outside it could dominate and cancel; coordinatewise dominance settles both issues. Different independent subsets can produce the same nonleading exponent and cancel; the theorem does not claim that all determinant terms are distinct, only that one extremal nonzero contribution is isolated. Coefficient signs and phases never enter the selection: they multiply one nonzero leading coefficient after cancellation has been excluded, so arbitrary real or complex coefficients are allowed with no positivity or genericity assumption. Infinite order types, accumulation toward a cut, and the absence of an Archimedean embedding of Γ create no extra case.

Corollary 20.11 (Every relation bounds support rank [03, 02]). *If a nonzero $F \in K_H(k)[Y_1, \dots, Y_m]$ satisfies $F(S_{a_1} y, \dots, S_{a_m} y) = 0$ for distinct $a_j \in \mathcal{F}_{>0}$, then $\rho_{\mathcal{F},H}(y) \leq m - 1$. In particular, for rational $d > 0$, $d \neq 1$, a nonzero relation $F(y, S_d y, \dots, S_d^n y) = 0$ over $K_H(k)$ forces $\rho_{\mathbb{Q},H}(y) \leq n$, and*

$$\text{aord}_{d,k}(y) = n < \infty \implies \rho(y) \leq n.$$

If $\rho_{\mathcal{F},H}(y)$ is infinite, all distinct positive $\mathcal{F}$ -dilates of y are jointly algebraically independent over $K_H(k)$.

Proof. Rank at least m contradicts theorem 20.7; the parameters $1, d, \dots, d^n$ are distinct because $d > 0$ and $d \neq 1$. An infinite rank satisfies the hypothesis of theorem 20.7 for every finite m . $\square$

In the absolute case this is a necessary condition on the normal form of every solution, not a statement about chosen examples. It is sharp: sums of n monomials with independent exponents have order exactly n (theorem 13.3). It does not say that rank at most n suffices for a relation of order n , and it bounds neither the degree of a relation nor the support's order type or denominators.

Remark 20.12 (Research question 5: a partial datum). [merge] Neither source states the following. Let Γ be any ordered abelian group in which S_q is defined, $q \in \mathbb{Q}_{>0} \setminus \{1\}$, and let $y \in K_\Gamma(k) \setminus k$ satisfy $F(y, S_q y) = 0$ for a nonzero $F \in k[X, Y]$, with $\text{char } k = 0$. Embed $K_\Gamma(k)$ coefficientwise into $k((t^{\Gamma \otimes \mathbb{Q}}))$; this commutes with S_q and keeps the relation. Corollary 20.11 with $m = 2$, $\mathcal{F} = \mathbb{Q}$, $H = 0$ gives $\rho(y) \leq 1$: the support of y spans at most a one-dimensional rational space. For integer $q = d \geq 2$ this is weaker than theorem 6.1. For nonintegral q it is new information on Research question 5: no nonconstant solution can have two rationally independent support directions, whatever the generalized Böttcher theory turns out to be. It says nothing about denominators, which is the other half of that question, and is consistent with the monomial solutions recorded in section 1.5. Research question 5 stays open.

Remark 20.13 (Research question 8: the relative part). Source 02 records that its relative theorem answers a part of Research question 8. If the coefficients of a relation lie in the monomial field $k(t^\Delta)$ of a subgroup $\Delta \subseteq \Gamma$, they lie in $K_H(k)$ for $H = \mathbb{Q}\Delta$, so a nonzero relation among m distinct positive rational dilates forces rank at most $m - 1$ modulo the rational span of Δ . [merge] Source 03's theorem 20.7 gives the same, over any $\mathcal{F}$. Neither conclusion bounds the support order type, the denominators, or the support module generated under multiplication and division by d relative to Δ , which that question also asks for; those parts remain open. A relation with such coefficients can exist where none exists over k : $S_d u_d - u_d = \omega^d$ (section 14).

21 Finite support: the exact order in every rank

Lemma 21.1 (Independent monomials over a Hahn core [03, 02]). *If $\beta_1, \dots, \beta_\rho \in \Gamma$ are $\mathbb{Q}$ -linearly independent modulo a $\mathbb{Q}$ -subspace H , then $t^{\beta_1}, \dots, t^{\beta_\rho}$ are algebraically independent over $K_H(k)$.*

Proof. For distinct integer multi-indices $\nu \neq \mu$ the cosets $H + \sum_i \nu_i \beta_i$ and $H + \sum_i \mu_i \beta_i$ are disjoint. A nonzero coefficient in $K_H(k)$ times the corresponding monomial has nonempty support in its own coset, so a nonzero polynomial relation would be a sum of nonzero series with pairwise disjoint supports, which cannot vanish. $\square$

Theorem 21.2 (Exact finite-support rank [03, 02]). *Let $\mathcal{F} = \mathbb{Q}$, and let $y \in K_\Gamma(k)$ have finite support with $\rho = \rho_{\mathbb{Q}, H}(y)$. For every finite set $Q \subseteq \mathbb{Q}_{>0}$,*

$$\text{trdeg}_{K_H(k)} K_H(k)(S_Q y : q \in Q) = \min\{\rho, |Q|\}. \quad (21.1)$$

The field generated over $K_H(k)$ by all positive rational dilates of y has transcendence degree ρ . For every rational $d > 0$, $d \neq 1$, $\text{aord}_{d, K_H(k)}(y) = \rho$; in particular $\text{aord}_{d, k}(y) = \rho(y)$.

Proof. The lower bound is theorem 20.7. For the upper bound choose $\beta_1, \dots, \beta_\rho$ spanning the image of the finite support Σ modulo H and write $\gamma = h_\gamma + \sum_i u_{\gamma i} \beta_i$ with $h_\gamma \in H$, $u_{\gamma i} \in \mathbb{Q}$, for $\gamma \in \Sigma$. Since Q and Σ are finite there is an integer $N \geq 1$ with $Nqu_{\gamma i} \in \mathbb{Z}$ for all occurring q, γ, i . With $z_i = t^{\beta_i/N}$,

$$t^{q\gamma} = t^{qh_\gamma} \prod_i z_i^{Nqu_{\gamma i}}, \quad t^{qh_\gamma} \in K_H(k),$$

so every $S_Q y$ lies in $K_H(k)(z_1, \dots, z_\rho)$, a rational function field of transcendence degree ρ by lemma 21.1; the generated field has transcendence degree at most ρ and at most $|Q|$. Every monomial t^γ with γ in the rational span of Σ is algebraic over $K_H(k)(t^{\beta_1}, \dots, t^{\beta_\rho})$, so the field of all positive rational dilates lies in an algebraic extension of that field and has transcendence degree exactly ρ . Finally, the first ρ consecutive transforms are independent and the first $\rho + 1$ dependent; for $\rho = 0$, $y \in K_H(k)$ and the order is zero. $\square$

Source 03 proves the relative statement (21.1); source 02 proves the absolute one ($H = 0$) with the statement on all dilates, choosing a rational basis $e_1, \dots, e_\rho$ of the support span and writing a finite list of dilates as Laurent polynomials in $t^{e_j/N}$. The two proofs are the same. The substance is the lower bound, which must hold for *every* nonzero coefficient pattern: the proof exhibits one noncancelling determinant term rather than invoking genericity. Cancellation among support terms and rational dependencies among exponents are allowed, and the coefficients may satisfy arbitrary algebraic relations.

Corollary 21.3 (The uniform matroid of the rational orbit [03, 02]). *For a finite-support y with $\rho = \rho(y) > 0$, a set of distinct positive rational dilates of y is algebraically independent over k if and only if it has at most ρ elements. So the algebraic matroid on $\{S_Q y : q \in \mathbb{Q}_{>0}\}$ is the uniform matroid of rank ρ . The same holds over $K_H(k)$ with $\rho = \rho_{\mathbb{Q}, H}(y)$.*

There are no exceptional finite subsets of distinct positive rational parameters. The statement says which subsets are independent; it computes no minimal equation and no extension degree, and it does not say that ρ dilates recover every support monomial by rational functions. An infinite collection of rational dilations still generates a field of transcendence degree ρ , but that field can be an infinite algebraic extension of the field of ρ chosen observations: the denominators of the dilations need not have a common bound.

Corollary 21.4 (Omnific autonomous order; Research question 1 [03, 02]). *Let $x \in \mathbf{Oz}$ or $x \in \mathbf{Oz}[i]$ have finite support, and let ρ be the $\mathbb{Q}$ -dimension of the span of its nonzero exponents. For every integer $d \geq 2$, and every rational $d > 0$ with $d \neq 1$, the autonomous algebraic order of x over $\mathbb{C}$ is ρ ; for real x its order over $\mathbb{R}$ is also ρ . Any m distinct positive rational dilates of x generate a field of transcendence degree $\min\{m, \rho\}$ over $\mathbb{C}$. The same holds for every finite-support surreal or surcomplex number, with negative exponents allowed.*

Proof. All supports involved lie in the set-sized $\mathbb{Q}$ -span of the finite original support. Apply theorem 21.2 with $k = \mathbb{C}$ and $H = 0$, and with $k = \mathbb{R}$ for the real statement. Source 02 adds that a determinant with nonzero real leading coefficient stays nonzero after scalar extension to $\mathbb{C}$, so the real and complex statements agree. $\square$

This is the equality asked for in Research question 1, including its unresolved case of several supported monomials in one ρ -dimensional exponent span, and it makes proposition 13.4 an equality in every rank. The restriction to omnific integers is unnecessary. For $\rho = 1$ and $\rho = 2$ it recovers proposition 13.4 and, with $m = 2$, the finite-support case of corollary 8.2, by a proof that uses none of sections 4 to 6.

21.1 Finite determination is not reconstruction

For ρ finite-support observations generating a field of transcendence degree ρ , the ambient monomial field is finitely generated and algebraic, hence a finite extension, over the observation field. The monomial coordinates are therefore determined generically up to finitely many algebraic possibilities. This is weaker than a birational reconstruction theorem, and much weaker than finiteness or uniqueness of every numerical fibre; special fibres can be exceptional even for power sums [18]. A polynomial invariant under a nontrivial permutation of its monomial coordinates keeps that symmetry after every dilation, so its observations cannot distinguish the permuted coordinates: exact rank does not remove finite symmetries (source 03). Uniform algebraic independence alone does not give birational recovery, and the finite extension degrees are additional information (source 02).

22 Rational functions of finitely many monomials

A series can have infinite support because a rational expression was expanded as a Hahn series. Such series still have exact order equal to rank. The two sources prove this by different routes, both printed: source 03 descends the function to its support subgroup and computes the rank from a reduced quotient; source 02 computes it from a coefficient matrix that needs no reduction and bounds the order by a Jacobian rank.

Throughout, $\Lambda \subseteq \Gamma$ is a finitely generated, hence free, subgroup, $k(\Lambda) = \text{Frac}(k[t^\Lambda]) \subseteq K_\Gamma(k)$, and $\mathbb{Q}\Lambda$ is its rational span. A rational function in $k(\Lambda)$ has Hahn support in Λ : dividing by the least monomial of the denominator and expanding the geometric series uses only sums and differences of elements of Λ (the Neumann expansion of lemma 3.1).

22.1 First route: rational support descent [03]

Lemma 22.1 (Rational support descent [03]). *Let W be a $\mathbb{Q}$ -subspace of $\mathbb{Q}\Lambda$. In $K_\Gamma(k)$,*

$$k(\Lambda) \cap k((t^W)) = k(\Lambda \cap W). \quad (22.1)$$

Proof. Projection onto the exponents in W is an additive map π_W of the Hahn field, and $\pi_W(hv) = h\pi_W(v)$ whenever h is supported in W : a summand of the coefficient at $w \in W$ uses an exponent $s \in W$ of h , so the exponent $w - s$ from v also lies in W ; this is checked coefficientwise on finite convolutions.

Let $f = P/Q \in k(\Lambda)$ be supported in W , pick $u \in \text{supp } Q$, and put $P' = t^{-u}P$, $Q' = t^{-u}Q$. The Laurent polynomial Q' has a nonzero constant coefficient, so $\pi_W(Q') \neq 0$, and projecting $fQ' = P'$ gives $f = \pi_W(P')/\pi_W(Q')$. Both projections are Laurent polynomials supported in $\Lambda \cap W$. The reverse inclusion holds because the series supported in W form a field. $\square$

The projection must not be mistaken for a ring retraction of the whole Hahn field: projection onto exponent zero kills t^g and t^{-g} for $g \neq 0$ but not their product. Only linearity when one factor is supported in the subgroup is used.

Theorem 22.2 (Exact order for rational monomial functions [03, 02]). *Let $f \in k(\Lambda)$, viewed in $K_\Gamma(k)$; its support may be infinite. Put $\rho = \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}} \text{supp } f$. For distinct positive rationals $q_1, \dots, q_m$,*

$$\text{trdeg}_k k(S_{q_1}f, \dots, S_{q_m}f) = \min\{\rho, m\}.$$

In particular $\text{aord}_{d,k}(f) = \rho$ for every rational $d > 0$, $d \neq 1$.

Proof [03]. Put $W = \text{span}_{\mathbb{Q}} \text{supp } f$. The support of f lies in Λ , so lemma 22.1 gives $f \in k(\Lambda \cap W)$. The subgroup $\Lambda \cap W$ is finitely generated, as a subgroup of the free group Λ of finite rank, and has rank ρ , because it contains the support spanning W and lies in the ρ -dimensional space W . For finitely many rational dilations choose a common denominator N ; all dilates lie in the monomial rational function field of the rank- ρ group $N^{-1}(\Lambda \cap W)$, which gives the upper bound ρ . The lower bound is theorem 20.7, applied to the possibly infinite support of f . The order statement follows as in theorem 21.2. $\square$

Theorem 22.3 (A finite rank formula from a reduced quotient [03]). *Write a nonzero $f \in k(\Lambda)$ as P/Q with coprime nonzero Laurent polynomials $P, Q \in k[t^\Lambda]$, and put $\Sigma_{P,Q} = \text{supp } P \cup \text{supp } Q$. Then*

$$\text{span}_{\mathbb{Q}} \text{supp } f = \text{span}_{\mathbb{Q}} \{a - b : a, b \in \Sigma_{P,Q}\}. \quad (22.2)$$

So the dimension of the difference span is the exact order in theorem 22.2.

Proof. Let W and V be the left and right sides. For $b \in \Sigma_{P,Q}$, both $t^{-b}P$ and $t^{-b}Q$ are supported in V , so their quotient lies in the Hahn field over V , and $W \subseteq V$.

Conversely, let $\lambda : \Lambda \rightarrow \mathbb{Z}$ be an integral linear functional vanishing on $W \cap \Lambda$. With an independent indeterminate τ , the substitution $t^a \mapsto \tau^{\lambda(a)}t^a$ is an automorphism of $k(\tau)[t^\Lambda]$ and a support-preserving coefficient twist of the associated Hahn field. Every exponent of f lies in W , so the twist fixes f ; writing P_τ, Q_τ for the twisted numerator and denominator, $P_\tau/Q_\tau = P/Q$. The Laurent polynomial ring $k(\tau)[t^\Lambda]$ is a unique factorization domain, coprimality of P, Q persists under the purely transcendental scalar extension, and the substitution preserves it. Two coprime pairs representing one fraction differ by a unit, so $P_\tau = c(\tau)t^u P$ and $Q_\tau = c(\tau)t^u Q$ with $c(\tau) \in k(\tau)^\times$ and $u \in \Lambda$. The twist does not change supports, so $u + \text{supp } P = \text{supp } P$, and a nonempty finite subset of a torsion-free group is not invariant under a nonzero translation; hence $u = 0$. Comparing nonzero coefficients gives $\tau^{\lambda(a)} = c(\tau)$ for every $a \in \Sigma_{P,Q}$, so $\lambda(a - b) = 0$ for all $a, b \in \Sigma_{P,Q}$. Finally W is a rational subspace of $\mathbb{Q}\Lambda$: a vector of $\mathbb{Q}\Lambda$ outside W is separated from W by a rational functional vanishing on W , which becomes integral on Λ after clearing denominators. So every difference $a - b$ lies in W , and $V \subseteq W$. $\square$

The factorization facts used are elementary: a Laurent polynomial ring is a localization of a polynomial unique factorization domain, its units are nonzero scalars times monomials, and an irreducible coefficient polynomial stays prime after adjoining and inverting a purely transcendental parameter, since its quotient ring stays a domain. No geometric assumption on k is needed. Reduction is essential: the unreduced $(T_1 + T_2)/(T_1 + T_2)$ has nonzero difference rank, whereas the function is constant with order zero. For a polynomial take $Q = 1$; the extra exponent zero makes the difference span the linear span of the support, recovering theorem 21.2.

22.2 Second route: the coefficient matrix [02]

Let $X_1, \dots, X_s$ be algebraically independent and let $f = P/Q$ with Laurent polynomials $P, Q \in k[X_1^{\pm 1}, \dots, X_s^{\pm 1}]$, $Q \neq 0$; a common factor need *not* have been cancelled. Define the vector Laurent polynomial

$$B_j = Q X_j \frac{\partial P}{\partial X_j} - P X_j \frac{\partial Q}{\partial X_j}, \quad B = (B_1, \dots, B_s) = \sum_u b_u X^u, \quad b_u \in k^s, \quad (22.3)$$

and put

$$\rho_{\text{cm}}(f) = \dim_k \text{span}_k \{b_u : u \in \mathbb{Z}^s\}; \quad (22.4)$$

only finitely many b_u are nonzero.

Lemma 22.4 (Coefficient matrix equals support rank [02]). *Let $e_1, \dots, e_s \in \Gamma$ be rationally independent and $y = f(t^{e_1}, \dots, t^{e_s})$. Then $\rho(y) = \rho_{\text{cm}}(f)$. In particular the rank does not depend on the Hahn ordering of the abstract exponent lattice or on the chosen fraction P/Q .*

Proof. With $\Lambda = \mathbb{Z}e_1 + \dots + \mathbb{Z}e_s$, the support of y lies in Λ . For $\vec{v} \in k^s$ let $\lambda_{\vec{v}}$ be the $\mathbb{Q}$ -linear functional with $\lambda_{\vec{v}}(e_j) = v_j$, extended to Γ . On the rational function field, $\partial_{\lambda_{\vec{v}}}$ acts as $\sum_j v_j X_j \partial_{X_j}$, so

$$\partial_{\lambda_{\vec{v}}} y = 0 \iff \sum_j v_j B_j = 0 \iff \vec{v} \cdot b_u = 0 \text{ for every } u.$$

By uniqueness of Hahn coefficients, $\partial_{\lambda_{\vec{v}}} y = 0$ if and only if $\vec{v} \cdot \vec{n} = 0$ for every supported lattice coordinate vector $\vec{n} \in \mathbb{Z}^s$. So the coefficient vectors b_u and the support coordinate vectors have the same annihilator in k^s , hence spans of the same dimension; the rank of a family of rational vectors does not change on extension from $\mathbb{Q}$ to a field of characteristic zero, because every minor is rational. Both invariances follow from these descriptions. $\square$

Lemma 22.5 (Jacobian rank for rational functions [02]). *For $f_1, \dots, f_m \in k(T_1, \dots, T_s)$ in characteristic zero, the rank of their ordinary, or logarithmic, Jacobian over $k(T_1, \dots, T_s)$ equals $\text{trdeg}_k k(f_1, \dots, f_m)$.*

Proof. A transcendence basis among the f_i has independent differentials: extend it to a transcendence basis of $k(T_1, \dots, T_s)/k$; the coordinate derivations of the purely transcendental subfield extend uniquely across the finite separable algebraic extension, since differentiating a minimal polynomial determines the derivative, and applying them proves independence. Every remaining f_i is separable algebraic over the field of that basis, and differentiating its minimal polynomial puts its differential in their span. The dT_j form a basis of the module of differentials, and multiplying column j by the nonzero T_j does not change rank. Compare the polynomial version [19, Theorem 6]. $\square$

Second proof of theorem 22.2 [02]. Write f as P/Q in rationally independent monomials $t^{e_1}, \dots, t^{e_s}$ as above, so that $\rho(y) = \rho_{\text{cm}}(f)$ by lemma 22.4. The lower bound follows from theorem 20.7. For the upper bound choose N with every Nq_i a positive integer and put $T_j = t^{e_j/N}$, so that $S_{q_i} y = f(T_1^{Nq_i}, \dots, T_s^{Nq_i})$. The i th logarithmic Jacobian row is

$$\frac{Nq_i}{Q(T^{Nq_i})^2} B(T^{Nq_i}),$$

which lies in the $k(T_1, \dots, T_s)$ -span of the fixed k -space spanned by the b_u , of dimension $\rho_{\text{cm}}(f)$. So the Jacobian has rank at most $\rho_{\text{cm}}(f)$, and lemma 22.5 gives the upper bound. The order statement follows as in theorem 21.2. $\square$

Remark 22.6 (Why the second upper bound is restricted [02]). For an arbitrary Hahn series, Euler derivations need not span all algebraically relevant derivations of the field generated by its dilates: a rank-one support can have order two (theorem 13.2). The finitely generated rational function field of lemma 22.5 is essential to the upper bound, and it is not applied to an arbitrary infinite Hahn expansion.

An exact algorithm and its input contract [02]. The invariant is computable by finite exact algebra.

1. Supply a finite rationally independent exponent basis and a rational Laurent expression P/Q over an effective field of characteristic zero.
2. Compute the s Laurent polynomials (22.3) and collect equal exponent vectors.
3. Form the matrix with rows b_u and compute its exact rank.
4. Return that rank as the autonomous order, and use theorem 22.2 for every requested finite tuple of dilations.

This terminates for the specified representation. It makes no claim that arbitrary surreal expressions have decidable rational linear relations, or that equality in an arbitrary coefficient field is effective; rational and exactly represented algebraic-number coefficients meet the input contract. A noninjective exponent parametrization must first be reduced to a certified rationally independent basis. For finite support one may skip differentiation and take the rank of the support coordinate matrix. For a rational profile the coefficient matrix matters: the union of the supports of an unreduced numerator and denominator can give the wrong answer. The supplied verifier checks examples; it is not a surreal computer algebra system.

Example 22.7 (Examples of both routes [03, 02]). (a) $f = T_1/T_2$ has reduced union support $\{(1, 0), (0, 1)\}$ and a one-dimensional difference span, so its order is one (source 03).

(b) $f = T_1/(T_1 + T_2)$: $(B_1, B_2) = (T_1T_2, -T_1T_2)$ and $\rho_{\text{cm}}(f) = 1$. The displayed quotient involves two independent monomials, but $f = 1/(1 + T_2/T_1)$ has one intrinsic scale; where T_2/T_1 is infinitesimal its expansion is $\sum_{n \geq 0} (-1)^n (T_2/T_1)^n$, of infinite support, rank one and order one, and the opposite ordering changes the expansion but not the rank (source 02).

(c) $f = (T_1 + T_2)/(1 + T_1T_2)$: $B_1 = T_1 - T_1T_2^2$ and $B_2 = T_2 - T_1^2T_2$, whose coefficient vectors span k^2 (source 02); its reduced difference rank is two (source 03). Its expansion is generally infinite; any two distinct positive rational dilates are independent, any three dependent, and the order is two. Source 02's verifier checks that multiplying numerator and denominator by $1 + T_1 + 2T_2$ changes neither example's invariant, and its recorded run also lists $(T_1^2T_2^3 + 1)/(T_1^2T_2^3 - 2)$, of rank one, and the constant $5/2$, of rank zero.

23 Worked examples and finite certificates

In this section T_1, T_2, T_3 are independent monomials, abstract Laurent variables or $T_j = \omega^{e_j}$ for rationally independent positive surreal e_j . Exponent vectors refer to T -monomials, and an auxiliary integer weight w orders them; “greatest w -weight” corresponds to least valuation when T^a is given valuation $-w \cdot a$. The two sources' rank-three certificates are different and both are printed.

23.1 Two rank-three certificates

Example 23.1 (Source 03's certificate). Let

$$f = T_1 + T_2 + T_3 + T_1T_2 - 2T_2T_3 + 3T_1^2T_3. \quad (23.1)$$

Its six supported exponent vectors span $\mathbb{Q}^3$ but are far from independent. Order them by $w = (1, 6, 36)$. Greedy selection by decreasing weight gives $b_1 = (0, 1, 1)$, $b_2 = (2, 0, 1)$, $b_3 = (0, 0, 1)$, with coefficient product $(-2)(3)(1) = -6$ and

$$\det \begin{pmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = -2.$$

With the ordinary logarithmic coordinate derivations $T_1\partial_{T_1}, T_2\partial_{T_2}, T_3\partial_{T_3}$ and the dilations in the order 4, 2, 1, the leading Jacobian exponent is $4b_1 + 2b_2 + b_3 = (4, 4, 7)$, and the coefficient of $T_1^4T_2^4T_3^7$ is

$$(4 \cdot 2 \cdot 1)(-6)(-2) = 96. \quad (23.2)$$

The extra determinant factor, compared with (20.6), comes from using coordinate derivations instead of the dual derivations normalized on the greedy basis; the unique-extremum argument is unchanged. So f, S_2f, S_4f are independent and f, S_2f, S_4f, S_8f are dependent. The exact expansion in source 03's program has 97 nonzero terms; only the one term (23.2) is needed for the proof. For an omnific instance put $T_1 = \omega$, $T_2 = \omega^\omega$, $T_3 = \omega^{\omega^2}$; the directions $1, \omega, \omega^2$ are rationally independent by uniqueness of their own normal forms, and in decreasing normal-form order

$$x = -2\omega^{\omega^2+\omega} + 3\omega^{\omega^2+2} + \omega^{\omega^2} + \omega^{\omega+1} + \omega^\omega + \omega. \quad (23.3)$$

All exponents are positive and the constant coefficient is zero, so $x \in \mathbf{Oz}$. It has autonomous order three for every integer $d \geq 2$. This is exactly the kind of dependent support that Research question 1 left open.

Example 23.2 (Source 02's certificate). Let

$$f = T_1^2T_3 + 2T_1T_2 - 3T_3 + 5T_2 + 7T_1 + 11. \quad (23.4)$$

Its five nonconstant supported exponents, $e_1, e_2, e_3, e_1 + e_2, 2e_1 + e_3$ for $T_j = \omega^{e_j}$, span dimension three but are not independent; theorem 21.2 shows that f, S_2f, S_4f are independent, that adding S_8f creates a relation, and that the order is exactly three. With the weight $(1, 4, 16)$, refined lexicographically to break ties, the greedy vectors are $b_1 = (2, 0, 1)$, $b_2 = (0, 0, 1)$, $b_3 = (1, 1, 0)$, with coefficients 1, -3, 2. The logarithmic Jacobian with rows $(T_j\partial_{T_j}f(T_1^q, T_2^q, T_3^q))_j$ for $q = 1, 2, 4$, in this order, has the unique leading monomial

$$-96T_1^9T_2T_3^6, \quad (9, 1, 6) = 4b_1 + 2b_2 + b_3. \quad (23.5)$$

Source 02's verifier expands the determinant independently and finds 45 nonzero terms, with exactly this leading term. [merge] The sign differs from (23.2) only because the rows are listed with increasing q (remark 20.9); in the order 4, 2, 1 the coefficient is $(4 \cdot 2 \cdot 1)(1 \cdot (-3) \cdot 2)\det(b_i) = 8 \cdot (-6) \cdot (-2) = 96$. The auxiliary order need not be the actual order of the surreal e_j : the certificate proves that the abstract Laurent polynomial is nonzero, and evaluation at rationally independent monomials is injective. So once a rationally independent exponent basis is certified, a finite certificate need not decide the order of arbitrary surreal expressions.

23.2 Further examples [03]

Example 23.3 (The linear span, not the affine span). For $f = T_1T_2 + T_1T_2^2$ the support $\{(1, 1), (1, 2)\}$ has affine span of dimension one but linear span of dimension two, so any two distinct positive rational dilates are independent and the order is two. Factoring out T_1T_2 does not lower the order: the common monomial factor is itself dilated and carries another exponent direction. The theorem concerns the linear span of the support, not the affine dimension of a Newton polytope or its number of edges. A constant shift is different: adding a constant changes only the zero exponent, which contributes nothing to rank, and translates every dilate by the same constant, so it neither creates nor destroys independence.

Example 23.4 (A relative example). Let Γ contain rationally independent h, β, γ , let $H = \mathbb{Q}h$, and let $f = t^h + t^\beta + t^\gamma + 2t^{h+\beta+\gamma}$. Its absolute rank is three and its rank modulo H is two. Three distinct positive rational dilates are independent over k , while over $K_H(k)$ any two are independent and any three dependent. The core knows the whole h -direction, including arbitrary Hahn series supported on it; this is what relative rank measures.

Example 23.5 (Power sums: a known case with explicit recovery). For $f = T_1 + T_2 + T_3$ let $p_j = T_1^j + T_2^j + T_3^j$ and let $\sigma_1, \sigma_2, \sigma_3$ be the elementary symmetric polynomials. Newton's identities give

$$\sigma_1 = p_1, \quad \sigma_2 = \frac{p_1^2 - p_2}{2}, \quad (23.6)$$

$$\sigma_3 = \frac{p_4 - \sigma_1^4 + 4\sigma_1^2\sigma_2 - 2\sigma_2^2}{4\sigma_1}, \quad (23.7)$$

a rational identity, checked from $p_3 = \sigma_1^3 - 3\sigma_1\sigma_2 + 3\sigma_3$ and $p_4 = \sigma_1p_3 - \sigma_2p_2 + \sigma_3p_1$. So $k(p_1, p_2, p_4) = k(\sigma_1, \sigma_2, \sigma_3)$. Given these three observations generically, T_1, T_2, T_3 are the roots of $Z^3 - \sigma_1Z^2 + \sigma_2Z - \sigma_3$, determined up to their six permutations, and all further power sums follow from $p_n = \sigma_1p_{n-1} - \sigma_2p_{n-2} + \sigma_3p_{n-3}$ ($n \geq 4$); in particular $S_8f = p_8$ is rational in f, S_2f, S_4f . These identities are classical and illustrate the difference between an order theorem and a reconstruction theorem. The independence of power sums is [18, Proposition 3], not a result of either source.

23.3 A certificate that avoids full expansion [03]

Let $f = \sum_{a \in \Sigma} c_a T^a$ with exact rational coefficients and $\Sigma \subseteq \mathbb{Z}^\rho$, $\text{span}_{\mathbb{Q}} \Sigma = \mathbb{Q}^\rho$ (rational exponent coordinates are first cleared by one common denominator), and let $q_1 > \dots > q_\rho > 0$ be rationals. A short independence certificate consists of a separating integer weight, a greedy basis, its determinant, and one exponent and coefficient. Let $N_0 = \max_{a \in \Sigma, i} |a_i|$ and

$$B_0 = 2N_0 + 2, \quad w = (1, B_0, B_0^2, \dots, B_0^{\rho-1}). \quad (23.8)$$

The values $w \cdot a$ are distinct on Σ : for two different vectors, at the largest index j where they differ the weighted contribution has absolute value at least B_0^j , while all lower contributions total at most $2N_0(B_0^j - 1)/(B_0 - 1) < B_0^j$. Sort Σ by decreasing $w \cdot a$ and keep a vector exactly when it raises the rational rank; this gives a basis $b_1, \dots, b_\rho$. For the ordinary logarithmic coordinate derivations the certified term is

$$\eta = \sum_{j=1}^{\rho} q_j b_j, \quad (23.9)$$

$$c_* = \left(\prod_j q_j \right) \left(\prod_j c_{b_j} \right) \det[b_{ji}]_{j,i}, \quad (23.10)$$

the unique term of greatest w -weight with nonzero determinant contribution. The proof of proposition 20.5 is applied with the scalar weights $w \cdot a$ in this finite setting; no claim that w preserves every comparison of an ambient surreal group is needed. Since $c_* \neq 0$ the Jacobian is not the zero Laurent polynomial, and an embedding into any Hahn field with independent monomial coordinates preserves this. For fewer than ρ observations, take the full greedy basis, invert its matrix, and use the first m dual functionals; the coefficient is then $\prod_j q_j \prod_j c_{b_j}$, as in theorem 20.6. Source 03's program checks both forms.

What is effective [03]. With finite rational-coordinate input, the certificate uses a sort, at most $|\Sigma|$ exact rank-increase tests, and one determinant or matrix inverse; standard exact linear algebra makes this polynomial time in the encoded input size, with separating weights of bit length $O(\rho \log(N_0 + 1))$. It certifies independence; it does not expand a possibly huge minimal relation, and the full expansions in the program are deliberately more expensive and are not the proposed method. For a surreal expression whose exponent dependencies are unresolved, an integer-coordinate presentation is additional input: neither source decides arbitrary equality or rational dependence of symbolic surreal exponents, and the greatest-element choices in the infinite-support proof are existence arguments, not a termination claim for encodings of infinite normal forms.

24 Infinite supports

24.1 Difference transcendence from infinite rank

Corollary 24.1 (Infinite-rank orbits [03, 02]). *If $\rho_{\mathcal{F}, H}(y)$ is infinite, then $\{S_{ay} : a \in \mathcal{F}_{>0}\}$ is algebraically independent over $K_H(k)$. In particular, for $\mathcal{F} = \mathbb{Q}$, y satisfies no nonzero autonomous relation of any finite order over $K_H(k)$ for any rational $d > 0$, $d \neq 1$.*

Proof. A relation involves finitely many members; apply theorem 20.7, or corollary 20.11. $\square$

This difference transcendence is over the stated base only. It is not differential transcendence under a chosen derivation, and it claims no independence from further surreal parameters, which need the relative version (section 25).

Theorem 24.2 (A countable omnific support with an independent real orbit [03, 02]). *The omnific integer*

$$\Xi = \sum_{n \geq 0} \omega^{\omega^{-n}} = \omega + \omega^{\omega^{-1}} + \omega^{\omega^{-2}} + \dots \quad (24.1)$$

has $\{S_a \Xi : a \in \mathbb{R}_{>0}\}$ algebraically independent over $\mathbb{C}$ (source 03). The same holds for $\Xi - \omega = \sum_{n \geq 1} \omega^{\omega^{-n}}$ with rational dilations, $\{S_q(\Xi - \omega) : q \in \mathbb{Q}_{>0}\}$ (source 02). Both elements have support of order type ω and are difference transcendental for every rational $d > 0$, $d \neq 1$.

Proof. The exponents ω^{-n} form a strictly decreasing sequence of positive surreals, so the support is legitimate, and the element is omnific by (2.5). The exponents are linearly independent over $\mathbb{R}$: a nontrivial finite real combination of the distinct monomials ω^{-n} is a normal form with distinct inner exponents $-n$ and cannot vanish. Let Γ be their set-sized $\mathbb{R}$ -span, put $\mathcal{F} = \mathbb{R}$, $k = \mathbb{C}$, $H = 0$, and apply corollary 24.1; for rational dilations one may use $\mathcal{F} = \mathbb{Q}$ and the rational span, as source 02 does. $\square$

The two appearances of ω have different roles: the inner monomials are independent *exponents*, the outer monomials are the terms of the omnific integer. The support is countable, but its exponent directions have infinite dimension even over $\mathbb{R}$, which is why continuum many positive real dilations can be independent. This is not a statement about classical transcendence of real exponentials.

Corollary 24.3 (Forward, backward and several dilations [02]). *For every $d \in \mathbb{Q}_{>0} \setminus \{1\}$ the elements $S_d^n(\Xi - \omega)$, $n \in \mathbb{Z}$, are algebraically independent over $\mathbb{C}$. For multiplicatively independent positive rationals $d_1, \dots, d_s$ the family $\{S_{d_1}^{n_1} \cdots S_{d_s}^{n_s}(\Xi - \omega) : (n_1, \dots, n_s) \in \mathbb{Z}^s\}$ is algebraically independent over $\mathbb{C}$.*

Proof. These are subfamilies of the positive rational orbit, with distinct parameters under the stated hypotheses. $\square$

24.2 Bounded denominators: two character arguments

Lemma 24.4 (Character extension [02]). *Let k be algebraically closed of characteristic zero. Every homomorphism from a subgroup A of an abelian group G to $k^\times$ extends to G , and a finite abelian group A has exactly $|A|$ such characters.*

Proof. Take a maximal extension by Zorn's lemma. If x is outside its domain and no positive multiple nx lies in the domain, give x any nonzero value; otherwise take the least such n and give x an n th root of the value at nx . This extends the character to the domain plus $\mathbb{Z}x$, contradicting maximality. The count follows from a decomposition into cyclic groups, since a cyclic group of order n has exactly n characters: k contains n distinct n th roots of unity. $\square$

Theorem 24.5 (Finite denominator index for algebraic Hahn elements [02]). *Let $\Lambda \subseteq \Gamma$ be free abelian of finite rank and $y \in k((t^{\mathbb{Q}^\Lambda}))$ algebraic of degree at most N over $k(t^\Lambda)$. Then*

$$\mathcal{D}_y = (\Lambda + \langle \text{supp } y \rangle_{\mathbb{Z}}) / \Lambda$$

is finite of order at most N . Consequently $\text{supp } y \subseteq N'^{-1}\Lambda$ for some integer $1 \leq N' \leq N$.

Proof. First let k be algebraically closed. Each character $\chi : \mathbb{Q}\Lambda/\Lambda \rightarrow k^\times$ defines the strong Hahn automorphism $\tau_\chi(\sum c_\gamma t^\gamma) = \sum c_\gamma \chi(\gamma + \Lambda) t^\gamma$, fixing $k(t^\Lambda)$. All $\tau_\chi(y)$ are roots of one nonzero polynomial of degree at most N over that field, so the character orbit of y has at most N elements. Two characters act equally on y exactly when their restrictions to $\mathcal{D}_y$ agree: equality is detected at every nonzero coefficient, and those support classes generate $\mathcal{D}_y$. Every character of $\mathcal{D}_y$ extends to $\mathbb{Q}\Lambda/\Lambda$ by lemma 24.4, so if $\mathcal{D}_y$ is finite its $|\mathcal{D}_y|$ characters give that many distinct conjugates, and $|\mathcal{D}_y| \leq N$. If $\mathcal{D}_y$ were infinite, its elements would have unbounded orders: for fixed e the e -torsion of $\mathbb{Q}\Lambda/\Lambda \cong \mathbb{Q}^p/\mathbb{Z}^p$ has only e^p elements, so a uniform bound would put $\mathcal{D}_y$ in a finite group. An element of order $e > N$ has e characters on its cyclic subgroup, which extend and give more than N conjugates, a contradiction. Taking $N' = |\mathcal{D}_y|$ kills every support class. For general k , embed coefficientwise in an algebraic closure: the polynomial keeps degree at most N , and the support and index are unchanged. $\square$

Lemma 24.6 (A rank-one character proof [03]). *If $f \in \mathbb{C}((t^{\mathbb{Q}}))$ is algebraic over $\mathbb{C}(t)$, then $\text{supp } f \subseteq \frac{1}{N}\mathbb{Z}$ for some integer $N \geq 1$.*

Proof. For each integer j , $\chi_j(g) = \exp(2\pi i j g)$ is a character of $(\mathbb{Q}, +)$, and $\tau_j(\sum c_g t^g) = \sum c_g \chi_j(g) t^g$ is a support-preserving field automorphism fixing $\mathbb{C}(t)$, since $\chi_j(1) = 1$. If $F(f) = 0$ for a nonzero $F \in \mathbb{C}(t)[Y]$, all $\tau_j(f)$ are among its finitely many roots, so $\tau_a(f) = \tau_b(f)$ for some integers $a > b$ and $\tau_{a-b}(f) = f$. At a nonzero coefficient at g this gives $\exp(2\pi i(a-b)g) = 1$, so $(a-b)g \in \mathbb{Z}$; take $N = a-b$. $\square$

Lemma 24.6 is the case $\Lambda = \mathbb{Z}$, $k = \mathbb{C}$ of theorem 24.5, without the degree bound; both are printed because their proofs differ (explicit exponential characters against extended abstract characters). Both conditions are necessary, not sufficient: a series with bounded denominators, or on a fixed lattice, can be transcendental

over the monomial field. They are character-orbit versions of a familiar ramification obstruction, and neither source claims the character method as new.

24.3 Rank one, order two

Proposition 24.7 (The rank-one order-two example [03]). *For $d \geq 2$ the omnific integer $u_d = \sum_{n \geq 0} \omega^{d^{-n}}$ of theorem 13.2 has rational support rank one and autonomous order two under S_d .*

Proof [03]; *a second route to theorem 13.2.* The support is decreasing, positive and in $\mathbb{Q}$, with one-dimensional rational span. Its denominators are unbounded, so lemma 24.6, read in $\mathbb{C}((t^{\mathbb{Q}}))$ through $\omega = t^{-1}$, shows that u_d is transcendental over $\mathbb{C}(\omega)$. Shifting the Hahn sum gives $S_d u_d - u_d = \omega^d$, so $\mathbb{C}(u_d, S_d u_d) = \mathbb{C}(u_d, \omega^d)$, of transcendence degree two: ω^d is transcendental over $\mathbb{C}$, and u_d is transcendental over $\mathbb{C}(\omega)$, hence over $\mathbb{C}(\omega^d)$. So the first two transforms are independent, while

$$S_d^2 u_d - S_d u_d = (S_d u_d - u_d)^d \quad (24.2)$$

makes the first three dependent. $\square$

This proof uses neither theorem 10.1 nor the support-collapse theorem behind it. The example does not contradict corollary 20.11, since order two allows rank one. It refutes the unwarranted extension of the finite-support *upper* bound to infinite supports, which is why the infinite-support theorem is a lower bound, not an equality. Infinite support alone does not force difference transcendence either: example 22.7(b) has infinite support and order one (source 02). Infinite support, ordinal support type, unbounded denominators (an infinite index modulo a lattice) and infinite rational rank are four different properties, and only the last forces independence of all dilates.

24.4 Rank ρ , order $\rho + 1$: multi-scale telescopes

Fix an integer $d \geq 2$, rationally independent $\gamma_1, \dots, \gamma_\rho \in \Gamma_{>0}$ and nonzero $a_1, \dots, a_\rho \in k$, and put

$$U_j = \sum_{n \geq 0} t^{-\gamma_j d^{-n}}, \quad \mathcal{Z}_\rho = \sum_{j=1}^{\rho} a_j U_j. \quad (24.3)$$

Each support $\{-\gamma_j d^{-n}\}$ is strictly increasing, a finite union of well-ordered sets is well ordered, and no two terms from different j share an exponent, by rational independence. For $\Gamma \subseteq \mathbf{No}$ and real a_j , (2.4) gives $\mathcal{Z}_\rho = \sum_j a_j \sum_n \omega^{\gamma_j d^{-n}}$.

Theorem 24.8 (Support rank ρ , autonomous order $\rho + 1$ [02]). *The series $\mathcal{Z}_\rho$ has infinite support, $\rho(\mathcal{Z}_\rho) = \rho$, and $\text{aord}_{d,k}(\mathcal{Z}_\rho) = \rho + 1$. For real nonzero a_j and positive surreal γ_j it is an omnific integer, and its order is $\rho + 1$ over $\mathbb{C}$ as well.*

Proof. Strong reindexing of the supports gives

$$w_\rho := (S_d - 1)\mathcal{Z}_\rho = \sum_{j=1}^{\rho} a_j t^{-d\gamma_j}, \quad (24.4)$$

a finite-support element of rank ρ , so $w_\rho, S_d w_\rho, \dots, S_d^{\rho-1} w_\rho$ are independent by theorem 21.2. Let $\Lambda = \sum_j \mathbb{Z} \gamma_j$ and $K_0 = k(t^{\gamma_1}, \dots, t^{\gamma_\rho})$. The support classes of $\mathcal{Z}_\rho$ modulo Λ include $\gamma_j/d^n + \Lambda$, of exact order

d^n , so the group $\mathcal{D}_{\mathcal{Z}_\rho}$ of theorem 24.5 is infinite and $\mathcal{Z}_\rho$ is transcendental over K_0 . All the listed dilates of w_ρ lie in K_0 , so $\mathcal{Z}_\rho, w_\rho, S_d w_\rho, \dots, S_d^{\rho-1} w_\rho$ are $\rho + 1$ independent elements. The invertible triangular change of variables taking successive differences identifies the field they generate with $k(\mathcal{Z}_\rho, S_d \mathcal{Z}_\rho, \dots, S_d^\rho \mathcal{Z}_\rho)$, so the first $\rho + 1$ transforms are independent. Every nonnegative transform lies in $K_0(\mathcal{Z}_\rho)$ by (24.4) and its iterates, a field of transcendence degree $\rho + 1$, so the first $\rho + 2$ are dependent. Membership in **Oz** follows from positivity of all growth exponents and zero constant term, and the argument is unchanged after extending real coefficients to $\mathbb{C}$. $\square$

The case $\rho = 1, a_1 = 1, \gamma_1 = 1$ is u_d of theorem 13.2, so this is a third route to that theorem. The decreasing-denominator construction itself is in Chyzak–Dreyfus–Dumas–Mezzarobba [10, Remark 2.18], with a shifted starting index; source 02 claims only the exact-order calculation for every finite number of independent scales.

Example 24.9 (An explicit third-order equation [02]). Take $\rho = 2, d = 2, a_1 = a_2 = 1$, write $\mathcal{Z} = \mathcal{Z}_2$, and put $w_1 = S_2 \mathcal{Z} - \mathcal{Z}$, $w_2 = S_2^2 \mathcal{Z} - S_2 \mathcal{Z}$, $w_3 = S_2^3 \mathcal{Z} - S_2^2 \mathcal{Z}$. With $A = t^{-2\gamma_1}$ and $B = t^{-2\gamma_2}$ these are $A + B$, $A^2 + B^2$ and $A^4 + B^4$, so

$$2w_3 - w_2^2 - 2w_1^2 w_2 + w_1^4 = 0. \quad (24.5)$$

Source 02's verifier expands the left side to zero. theorem 24.8, not this identity, proves that no lower-order relation exists.

The finite-truncation boundary [02]. For $U_N = \sum_{n=0}^N t^{-\gamma d^{-n}}$,

$$(S_d - 1)U_N = t^{-d\gamma} - t^{-\gamma d^{-N}}. \quad (24.6)$$

Dropping the last term at a finite stage is wrong. The infinite identity follows from strong cancellation of the reindexed supports, not from the last monomial tending to zero in a valuation topology; the exponents are bounded, so it does not. This is the identity of Appendix B at $\gamma = 1$.

24.5 A family with a candidate gap ℓ [03]

For $\ell \geq 1$ put

$$u_{d,\ell} = \sum_{n \geq 0} \omega^{d^{-\ell n}}. \quad (24.7)$$

The same support argument makes it an omnific integer of rational rank one. Directly $S_d^\ell u_{d,\ell} - u_{d,\ell} = \omega^{d^\ell}$, hence

$$S_d^{\ell+1} u_{d,\ell} - S_d u_{d,\ell} = (S_d^\ell u_{d,\ell} - u_{d,\ell})^d, \quad (24.8)$$

an upper bound $\ell + 1$ on its order. Equality for general ℓ is not proved; proposition 24.7 is the case $\ell = 1$. It is a concrete target for the gap between rank and order on one rational exponent line, without the false claim that rank classifies infinite supports.

Remark 24.10 (The two hierarchies are different). [merge] Source 02's theorem 24.8 does not settle the order of $u_{d,\ell}$. It realizes the gap order – rank = 1 at every rank ρ , using ρ rationally independent scales, whereas $u_{d,\ell}$ has rank one and would realize the gap ℓ . Its dilates $S_d^i u_{d,\ell}$, $0 \leq i < \ell$, live on one rational line, so the transcendence argument over a lattice field used there does not separate them. Source 02 asks the parallel question for its iterated telescopes (Research question 16). Whether the order at fixed rank can exceed rank plus one on an omnific integer is open.

25 Joint independence and fresh parameters

Theorem 25.1 (Joint independence for separated support spans [03, 02]). *Let $y_1, \dots, y_s \in K_\Gamma(k)$ and $V_j = (H + \text{span}_{\mathcal{F}} \text{supp } y_j)/H \subseteq \Gamma/H$, and suppose $V_1 + \dots + V_s$ is a direct sum. For each j choose $m_j \leq \dim_{\mathcal{F}} V_j$ distinct positive $a_{j1}, \dots, a_{jm_j} \in \mathcal{F}$. Then all $S_{a_{j\nu}} y_j$ ($1 \leq j \leq s$, $1 \leq \nu \leq m_j$) are jointly algebraically independent over $K_H(k)$.*

Proof. In each support choose the first m_j greedy exponents modulo H . Because the sum is direct, the corresponding coordinate functionals can be chosen to annihilate H and every V_l , $l \neq j$, while being dual to the selected exponents of y_j ; extend them to Γ by a set-sized basis extension. The derivative matrix with rows all chosen dilates and columns all these Euler derivations is block diagonal: a functional chosen for V_j kills every exponent of y_l , $l \neq j$, and its positive multiples. Each diagonal block has nonzero determinant by the proof of theorem 20.6, so the whole determinant is nonzero, and lemma 20.2 applies. $\square$

Source 02 states this for $\mathcal{F} = \mathbb{Q}$ with $y_\alpha \in K_{V_\alpha}(k)$, $\text{span}_{\mathbb{Q}} \text{supp } y_\alpha = V_\alpha$ and $H + \sum_\alpha V_\alpha$ direct, for an arbitrary index family of which finitely many members are used; its functionals are extended by zero on H and the other V_β and each block is theorem 20.8. Since a relation involves finitely many members, the formulations are equivalent. Disjoint supports do not suffice: two disjoint sets of exponents can span the same direction, and the hypothesis is independence of the *spans modulo the core*.

25.1 An explicit ordinal family [03]

For each ordinal α and $n \geq 0$ put

$$g_{\alpha,n} = \omega^{\alpha+1/(n+2)}, \quad \Xi_\alpha = \sum_{n \geq 0} \omega^{g_{\alpha,n}}, \quad (25.1)$$

with surreal addition in $\alpha + 1/(n+2)$, a value strictly between α and $\alpha + 1$. For fixed α the $g_{\alpha,n}$ are positive and strictly decreasing, so $\Xi_\alpha \in \mathbf{Oz}$. The inner exponents are distinct even across ordinals: for $\alpha < \beta$, $\alpha + 1/(n+2) < \alpha + 1 \leq \beta < \beta + 1/(m+2)$. So the $g_{\alpha,n}$ are distinct Conway monomials, and every finite set of them is linearly independent over $\mathbb{R}$.

Theorem 25.2 (A class of independent real-dilation orbits [03]). *The family $\{S_a \Xi_\alpha : \alpha \in \mathbf{On}, a \in \mathbb{R}_{>0}\}$ is algebraically independent over $\mathbb{C}$: every finite subfamily with distinct pairs (α, a) is.*

Proof. A proposed relation uses finitely many ordinals and finitely many dilations of each. The $\mathbb{R}$ -spans of those supports are in direct sum, each infinite dimensional, with set-sized total span. Apply theorem 25.1 with $\mathcal{F} = \mathbb{R}$, $k = \mathbb{C}$, $H = 0$. $\square$

Source 03 proposes the explicit orbitwise and relative structure as new, not the bare existence of many algebraically independent surreals. The family is not asserted to be a transcendence basis, or to generate $\mathbf{No}[i]$.

Theorem 25.3 (Freshness beyond every set-sized Hahn core [03]). *Let $\mathfrak{P} \subseteq \mathbf{No}[i]$ be a set. There are a set-sized $\mathbb{R}$ -subspace $H \subseteq \mathbf{No}$ and an ordinal α_0 with $\mathfrak{P} \subseteq \mathbb{C}((t^H))$, under (2.4), such that*

$$\{S_a \Xi_\alpha : \alpha \geq \alpha_0, a \in \mathbb{R}_{>0}\}$$

is algebraically independent over the entire Hahn field $\mathbb{C}((t^H))$, hence over the field generated by $\mathbb{C}$ and $\mathfrak{P}$ and over all positive real dilates of the parameters.

Proof. Let H be the $\mathbb{R}$ -span of the union of the normal-form supports of the members of $\mathfrak{P}$; it is a set and places $\mathfrak{P}$ in the stated field. Every $h \in H$ has a set-sized normal form, and the union of all these second-level supports is a set. Ordinals are cofinal in the surreals, so choose α_0 above every exponent of that union (equivalently, bound the birthdays first). For $\alpha \geq \alpha_0$ the monomial $g_{\alpha,n}$ has its single normal-form exponent above α_0 , and distinct such monomials have distinct exponents; a nontrivial real combination of them has a nonzero normal form supported above α_0 and is not in H . So these $g_{\alpha,n}$ are linearly independent modulo H , with the ordinal blocks in direct sum. For a proposed finite relation let Γ be the $\mathbb{R}$ -span of H and the finitely many countable supports involved, and apply theorem 25.1 over $K_H(\mathbb{C})$. As H is an $\mathbb{R}$ -subspace, positive real dilations preserve its Hahn field, which contains the dilates of $\mathfrak{P}$. $\square$

Corollary 25.4 (One countable-support witness over any set of parameters [03]). *For every set $\mathfrak{P} \subseteq \mathbf{No}[i]$ some $z \in \mathbf{Oz}$ of countable support has its whole positive real dilation orbit algebraically independent over $\mathbb{C}(\mathfrak{P})$: take $z = \Xi_{\alpha_0}$.*

The parameter set may be of any size as long as it is a set. There is no claim of independence over the proper class $\mathbf{No}[i]$, which contains z ; and there is no single ordinal threshold for every parameter set: the explicit family is fixed, but its independent tail depends on the prescribed set.

25.2 Set-sized families over a prescribed base [02]

Theorem 25.5 (Arbitrary set-sized families over a prescribed base [02]). *Let $\mathfrak{B}$ be a set-sized subfield of $\mathbf{No}[i]$ and κ a set cardinal. There are $Y_\alpha \in \mathbf{Oz}$, $\alpha < \kappa$, such that*

$$\{S_q Y_\alpha : \alpha < \kappa, q \in \mathbb{Q}_{>0}\} \text{ is algebraically independent over } \mathfrak{B}, \quad (25.2)$$

each with positive support of order type ω and all coefficients one.

Proof. Let H be the rational span of the union of the outer normal-form supports of the elements of $\mathfrak{B}$: a set-sized rational subspace of $\mathbf{No}$ with $\mathfrak{B} \subseteq \mathbb{C}((t^H))$. Let $\mathcal{I}$ be the union of the inner normal-form supports of all $h \in H$, also a set. Choose an ordinal ν_0 with $-\nu_0 < \tau$ for every $\tau \in \mathcal{I}$; it exists because a set of surreals is bounded above by an ordinal (bound the lengths of their sign sequences and apply this to the negatives), and any positive ordinal will do if $\mathcal{I}$ is empty. For $\alpha < \kappa$ and $n \geq 0$ put

$$\beta_{\alpha,n} = \nu_0 + \omega\alpha + n + 1, \quad h_{\alpha,n} = \omega^{-\beta_{\alpha,n}}, \quad Y_\alpha = \sum_{n \geq 0} \omega^{h_{\alpha,n}}. \quad (25.3)$$

The additions and the product in $\beta_{\alpha,n}$ are *ordinal* operations; the minus sign and the two monomials are surreal. The $\beta_{\alpha,n}$ are pairwise distinct and increase with n for fixed α , so Y_α has the claimed omnific support. A nonzero finite rational combination of the $h_{\alpha,n}$ has inner support at the distinct exponents $-\beta_{\alpha,n}$, all below $-\nu_0$, while every $h \in H$ has inner support in $\mathcal{I}$, above $-\nu_0$; uniqueness of normal forms shows that the whole family is rationally independent modulo H , and that a combination vanishes only when all coefficients do. With $V_\alpha = \text{span}_{\mathbb{Q}}\{h_{\alpha,n} : n \geq 0\}$, the sum of H and the V_α is direct and each V_α is infinite dimensional. Apply theorem 25.1 to each finite subfamily, in the set-sized Hahn field of that direct sum; independence over $K_H(\mathbb{C})$ implies independence over $\mathfrak{B}$. $\square$

Corollary 25.6 (Free omnific difference algebras [02]). *The assignment $X_{\alpha,q} \mapsto S_q Y_\alpha$ gives injective ring homomorphisms*

$$\mathbb{Z}[X_{\alpha,q} : \alpha < \kappa, q \in \mathbb{Q}_{>0}] \hookrightarrow \mathbf{Oz}, \quad (25.4)$$

$$\mathbb{Z}[i][X_{\alpha,q} : \alpha < \kappa, q \in \mathbb{Q}_{>0}] \hookrightarrow \mathbf{Oz}[i]. \quad (25.5)$$

For each $r \in \mathbb{Q}_{>0}$ they are equivariant for the permutation $X_{\alpha,q} \mapsto X_{\alpha,rq}$ and S_r . Restricting to $q = d^n$, $n \in \mathbb{Z}$, gives a free inversive difference polynomial ring for S_d .

Proof. Algebraic independence gives injectivity. Positive support and integral, respectively Gaussian integral, constant coefficients are preserved by these polynomial operations and by every S_r ; equivariance is $S_r S_q = S_{rq}$. $\square$

There is no embedding of $\mathbb{C}$ into $\mathbf{Oz}$ here. With coefficients in $\mathbb{C}$ the polynomial ring embeds into the nonnegative-growth-support surcomplex ring (the ring $A_\Gamma(k)$ for $k = \mathbb{C}$), and its fraction field into $\mathbf{No}[i]$. With coefficients in the prescribed $\mathfrak{B}$, independence gives ordinary polynomial and rational function field embeddings into $\mathbf{No}[i]$; equivariance with a fixed coefficient field additionally needs a specified compatible action on $\mathfrak{B}$. The theorem quantifies over set-sized bases and families; it constructs no set of proper-class transcendence degree and asserts no independence over all of $\mathbf{No}[i]$, which is impossible. It also does not make $(\mathbf{No}[i], S_d)$ existentially difference closed: $S_d x - x = 1$ has no solution, since the left side has zero constant coefficient. Large free subalgebras and solvability of difference equations are different properties.

[merge] The two constructions differ: source 03 places its fresh exponents *above* an ordinal bound (large inner exponents, real dilations, one ordinal-indexed family), source 02 places them *below* the negative of one (infinitesimal inner exponents, rational dilations, κ many orbits with coefficients one, and ring embeddings). Both separate a set-sized base by bounding the second-level supports of its exponent span, the localization also used by the independent-copies report (`isc:cor:parameters`); neither this report nor that one uses the other's results.

26 Prime characteristic

Characteristic zero is essential to theorem 20.7. For coefficients in $\mathbb{F}_p$, Frobenius gives $S_p f = f^p$; for $y = T_1 + T_2$ with independent monomials, $S_p y = T_1^p + T_2^p = y^p$, a dependence at rank two (section 14). Both the factor $q = p$ in the logarithmic Jacobian and the minimal-degree argument fail. There is nevertheless an exact finite-support formula after grouping dilations by their p -free parts.

Theorem 26.1 (Prime-field finite-support formula [03]). *Let Λ be a finitely generated ordered abelian group and $f = \sum_{a \in \Sigma} c_a t^a \in \mathbb{F}_p[t^\Lambda]$ with nonzero coefficients and Σ generating Λ as an abelian group; let $\rho = \text{rank}_{\mathbb{Z}} \Lambda$. For positive integers $n_1, \dots, n_m$ put $n_j^{(p)} = n_j / p^{v_p(n_j)}$ and $s = |\{n_1^{(p)}, \dots, n_m^{(p)}\}|$. Then*

$$\text{trdeg}_{\mathbb{F}_p} \mathbb{F}_p(S_{n_1} f, \dots, S_{n_m} f) = \min\{\rho, s\}. \quad (26.1)$$

Proof. The torsion-free group Λ has a $\mathbb{Z}$ -basis, identifying its group algebra with a Laurent polynomial ring in ρ variables; the images of Σ span $\mathbb{F}_p^\rho$, since Σ generates Λ . The coordinates modulo p give derivations $\partial_i(t^a) = (a_i \bmod p)t^a$. For distinct positive integers $u_1 > \dots > u_\ell$ prime to p , $\ell \leq \rho$, select a greedy independent support tuple using the projected vectors over $\mathbb{F}_p$ and dual functionals there. Dominance uses only linear rank, so it holds over $\mathbb{F}_p$; rearrangement uses the ordered group Λ and positive integer weights u_j , not an order on $\mathbb{F}_p$, each exchange moving the exponent by a positive integer multiple of a positive element of Λ . The unique extremal determinant term has nonzero coefficient: each u_j is nonzero modulo p , each selected c_a is nonzero, and the projected determinant is nonzero. So the derivative matrix is invertible. The minimal-relation argument of lemma 20.2 still applies over the perfect field $\mathbb{F}_p$: if all formal partial derivatives of a minimal relation vanished, every exponent in it would be divisible by p , and since coefficients have p th roots in $\mathbb{F}_p$ it would be F_1^p for a smaller relation F_1 ; otherwise a nonzero partial derivative gives the usual contradiction. So the ℓ observations are independent. The bound ρ comes from the Laurent monomial

field, the bound ℓ from the number of observations. Finally $S_{n_j}f = (S_{n_j^{(p)}}f)^{p^{v_p(n_j)}}$, so the field generated by one observation for each distinct p -free part contains the original observation field and is algebraic over it; the transcendence degrees agree. $\square$

The lattice must be the one *generated by the support*, not an arbitrary larger one: $T_1^p + T_2^p$ has zero ordinary coordinate derivatives in $\mathbb{F}_p(T_1, T_2)$, but in its support lattice, with monomials T_1^p, T_2^p , it has full rank two. The theorem is for prime-field coefficients: other characteristic- p coefficients bring Frobenius twists and are not covered. Source 03 includes this complement with proof as a structural comparison, not as a separately certified priority claim; source 02 asserts no positive-characteristic version.

27 Scope, questions and non-claims of the additions

27.1 Boundaries

Positivity and distinctness [03, 02]. Negative dilation does not preserve general Hahn supports, and even for finite support mixed signs destroy the conclusion: $f = T_1 + T_1^{-1} + T_2 + T_2^{-1}$ has rank two but $S_{-1}f = f$, so the multipliers 1, -1 give a dependent pair. Strict positivity is used in the coordinatewise comparison and in the rearrangement. Repeated multipliers repeat an observation, and $d = 1$ makes S_d the identity, so the exact order fails except accidentally in the smallest ranks. A zero multiplier is not in the dilation group and would collapse all monomials to one, generally incompatible with infinite summation.

Rational rank and real rank [03]. The lower bound of theorem 20.7 holds over any $\mathcal{F}$, including $\mathbb{R}$, but the exact upper bounds of theorems 21.2 and 22.2 use rational dilations and rational rank, and do not become exact with real rank and real dilations. The finite-support $f = \omega$ has real rank one, yet f and $S_{\sqrt{2}}f = \omega^{\sqrt{2}}$ are independent over $\mathbb{C}$: a relation would need two distinct nonnegative integer pairs with equal exponents $i + j\sqrt{2}$. The real-rank bound promises only transcendence degree one there; theorem 24.2 uses *infinite* real rank on purpose.

The coefficient field is part of the statement [02]. A nonconstant coefficient can create a relation absent over k : $S_d u_d - u_d = \omega^d$ is linear once ω^d is a coefficient. The relative theorem controls directions outside the core's exponent space, but an absolute-order calculation is not preserved under arbitrary enlargement of the base. In theorems 25.3 and 25.5 the base is fixed first and the fresh scales chosen afterwards; exchanging these quantifiers would give an impossible statement.

No conclusions about other structures [03, 02]. The Euler derivations prove nothing about differential transcendence under the natural surreal derivation. Exponent dilation is not the surreal exponential, and no exponential-field automorphism is classified. The omnific statements concern normal-form support and algebraic difference equations, not factorization, primality, Diophantine decidability, integral-closure or normalization fibres, holonomic functions under argument dilation, analytic convergence on **No**, birthdays, or the first-order theory of the surreals with dilations; source 03's exploratory reading of quotient and normalization reports supplies no theorem on those subjects. The finite algorithms assume exact finite exponent coordinates and supply no notation system for all surreals; the ordinal family is not a maximal transcendence basis; and written proofs and exact experiments are different kinds of evidence.

27.2 The questions of this report

Section 19.1 gives the disposition, repeated as status notes in section 16: Research question 1 is answered in every rank (corollary 21.4); the rank component of Research question 2 is answered (corollary 20.11), its order-type and denominator components are open, and (24.2) and theorem 24.8 show unbounded denominators at order two and gaps of one at every rank; Research question 5 gains the rank datum of remark 20.12; Research question 8 gains its relative part (remark 20.13). The questions are the repository's own; no named published problem is claimed solved.

27.3 Proof dependencies and a formalization plan

Layer or conclusion	Obligation and sources
Hahn arithmetic	Support-preserving operations, finite coefficient convolutions, finite products and strong summability (both).
Linear algebra	Greedy extrema in a well-ordered support; dual functionals on a set-sized quotient, which is never ordered (both).
Ordered comparison	Coordinatewise greedy dominance; strict rearrangement for positive distinct weights (both).
Differential algebra	Euler derivations annihilating the whole core; a nonzero derivative determinant forbids a minimal relation (both).
Noncancellation	The unique extremal support tuple and its explicitly nonzero coefficient (both).
Finite and rational upper bounds	Independent monomial coordinates and denominator clearing (both); support-linear projection and the reduced quotient (03); coefficient-vector annihilators and finite Jacobian rank (02).
Free families	Blocks in direct sum; separation of fresh second-level supports from a set-sized base; normal-form uniqueness (both).
Exact telescoping hierarchy	Finite-support exactness; character-orbit denominator obstruction; successive differences (02).
Surreal transfer	Conway normal forms and set-sized localization (both).

No part of this chain uses theorem 6.1, the Böttcher coordinate, Newton–Puiseux factorization, a quotient or normalization theorem of another report, or a global class basis, and no unproved repository result is a premise; the bounded-denominator lemmas are proved separately and are not premises of the main theorem. Both sources propose formalizing without surreals first. Source 03's milestones: finite sets of integer exponent vectors and Laurent polynomials, greedy dominance and the unique leading term, giving independence over any characteristic-zero field with a finite certificate format; then abstract well-ordered supports, where the new obligation is the legitimacy of collecting the determinant expansion by ordered tuples, with the product rule for ∂_λ and the derivative criterion proved separately; then Conway normal forms, with set-sized localization explicit and proper-class conclusions as finite-instance schemes; and the rational-function extension kept separate from the core, so that a correction to a reconstruction claim cannot touch the rank obstruction. Source 02's five stages are the same in substance: greedy basis and dominance on a well-ordered index set; rearrangement and the unique leading coefficient; Euler derivations with their product and commutation laws; the relative theorem by the minimal-degree argument; and specialization to finite support and rational functions, with proper-class applications by set-sized localizations rather than one universe-sized type. Points to keep: Γ/H is used only as a vector space, and all comparisons are made in Γ ; k need not be ordered; the functionals annihilate *every* exponent of H ; projecting a product is justified only when one factor is supported in the subgroup; positivity and distinctness of the weights are both invoked. Neither source supplies Lean code, and neither chain has been checked by the repository's build.

27.4 Recorded finite checks

Source 03's standard-library program `code/03-support-rank-verify.py` uses exact fractions and finite-field arithmetic, with seed 20260923; source 02's `code/02-dilation-rank-verify.py` uses exact rational arithmetic for its determinants and SymPy 1.14.0 for a few polynomial identities, with the same seed. Their recorded runs are:

Source 03	Checks	Source 02	Cases
Full-rank Jacobians, char. 0	85	Vector-valued greedy determinants, ranks 1–4	257
Partial-rank Jacobians, char. 0	60	Logarithmic Jacobians, ranks 1–4	133
Char. p Jacobians	72	Support bases compared with greedy	1,918
Rational-function Jacobians	4	Truncated telescopes with boundary	140
Frobenius identities	72	Rational-profile ranks	10
Negative-dilation example	1	Worked determinant and relation	2
Rank and inverse-matrix entries	528	Frobenius counterexamples	6
Power-sum reconstructions	668	Negative-dilation example	1
Total check units	1,490	Named cases (with 1,918 comparisons)	549

All passed. Source 03's 217 polynomial Jacobian cases compare the exact expanded determinant with the predicted unique extremal term, including signed coefficients, negative exponent coordinates, rational dilations and dependent supported vectors; four further cases check cleared rational-function Jacobians, including a rank-one quotient whose two-observation determinant vanishes. Source 02's 390 determinant cases compute the full determinant by an independent Leibniz expansion and compare its actual leading term with the greedy prediction, so signs, exponent assignments and cancellations are checked without using the claimed formula. Counts are recorded check units, not counts of verified theorems. None of these computations proves an infinite-support theorem, a summability justification, a basis extension, the character extension lemma, a class-sized construction or a priority claim. Both suites were rerun for this report (Appendix C.6).

27.5 Research questions of the additions

The questions below continue the numbering of section 16. They are the sources' proposed directions, not previously published open problems. Where the two sources ask essentially the same question it is printed once, with both formulations.

Research question 13 (Exact order of the cyclic-denominator family; high priority; source 03). For $u_{d,\ell}$ in (24.7), is the order under S_d exactly $\ell + 1$? Equation (24.8) gives the upper bound, and proposition 24.7 settles $\ell = 1$. A first target is independence of $u_{d,2}$, $S_d u_{d,2}$, $S_d^2 u_{d,2}$ over $\mathbb{C}$, which would begin to measure the order-rank gap on one rational exponent line.

Research question 14 (Difference transcendence on one exponent line; source 03). Construct, with a transparent proof, an omnific integer of rational rank one that is difference transcendental under a fixed S_d , or show that a proposed broad class of rank-one supports cannot be. The infinite-rank constructions do not settle this. A coefficient-twist or lacunarity argument controlling several transforms at once, rather than algebraicity over one monomial field, may help.

Research question 15 (The order spectrum at fixed rank; source 02). For fixed rational rank ρ , which autonomous orders occur among infinite-support Hahn series, and among omnific integers? Rational monomial profiles have order ρ and the telescopes of theorem 24.8 order $\rho + 1$; an infinite rational profile need not be omnific, so these are not yet realizations in one class. Can every integer at least ρ occur? Can infinite order occur at rank one? A classification must separate the support-lattice index from the rank.

Research question 16 (Iterated telescopes on one rational scale; source 02). For $s \geq 0$ let $u_d^{[s]} = \sum_{n \geq 0} \binom{n}{s} \omega^{d^{-n}}$, with $\binom{n}{s} = 0$ for $n < s$. Reindexing gives $(S_d - 1)u_d^{[s]} = u_d^{[s-1]}$ for $s \geq 1$ and $(S_d - 1)u_d^{[0]} = \omega^d$, so the order is at most $s + 2$. Is it exactly $s + 2$? The character obstruction gives transcendence over the monomial field but not joint independence of the intermediate $u_d^{[j]}$, which is not proved.

Research question 17 (Support structure at fixed order; sources 03 and 02). Source 03: with $\rho \leq 2$ now forced for order-two equations, which well-ordered supports in a fixed two-dimensional rational exponent space occur? Does a fixed relation bound any support invariant beyond rank? Separate order type, denominator growth and accumulation at nonprincipal cuts; this is the remaining geometric and combinatorial part of Research question 2. Source 02: a relation in m dilates forces rank at most $m - 1$; what further restrictions follow from a bound on the degree? Candidate invariants are denominator growth, the subgroup generated under d and d^{-1} , and ordinal type. The rank theorem is not a claim that supports lie in finitely many arithmetic progressions; find a structure theorem or families ruling one out.

Research question 18 (Degrees of observation fields; high priority; sources 03 and 02). Source 03: for reduced $f = P/Q$ of rank ρ and distinct positive integers $n_1, \dots, n_\rho$, compute or sharply bound $[k(\Lambda_f) : k(S_{n_1}f, \dots, S_{n_\rho}f)]$, with $\Lambda_f = \Lambda \cap \text{span}_{\mathbb{Q}} \text{supp } f$ for a fixed finite monomial lattice Λ presenting f . Rank proves finiteness once a common lattice is fixed, but not the degree or Galois group; how much is explained by finite monomial symmetries and how much by other coincidences? Power-sum fibres [18] show that dimension and degree are different problems. Source 02: for a finite profile of rank ρ determine $[k(t^\Lambda) : k(S_{q_1}y, \dots, S_{q_\rho}y)]$ for the support-generated lattice Λ when the parameters make the smaller field a subfield of $k(t^\Lambda)$ (for nonintegral parameters first choose a common lattice containing all scaled supports). When is the extension trivial? How do lattice and coefficient symmetries and Newton polytopes determine its degree and ramification? Uniform independence alone gives no birational recovery.

Research question 19 (The full field of observations; source 03). For fixed f and $d \geq 2$, describe the field generated by all $S_d^j f$, $j \geq 0$. Does a finite initial segment generate it, and when can the monomial field be recovered modulo an explicitly computable finite symmetry group? The theorem gives its transcendence degree, not its position in a rational monomial field. Distinguish integer d , where one lattice contains all forward transforms, from nonintegral rational d , where unbounded denominators can matter.

Research question 20 (Small relations from certificates; sources 03 and 02). Source 03: design an algorithm taking a sparse reduced rational function to a low-degree relation among its first $\rho + 1$ transforms, with a compressed elimination certificate, degree bounds in each observation and a proof of minimality in a chosen sense; the greedy certificate may organize an elimination order without pretending to supply the relation. Source 02: given a finite profile of rank ρ and $\rho + 1$ positive integer parameters, construct a small relation among the dilates; can mixed volumes or sparse elimination control its multidegrees and heights? For rational profiles, can a canonical minimal monomial quotient be recovered from the annihilator of (22.3)?

Research question 21 (Overlapping support spans; sources 03 and 02). Find a necessary and sufficient support-and-coefficient criterion for joint independence of several different series and their dilates when their support spans overlap; theorem 25.1 is only sufficient. A support-only answer is impossible: $T_1 + T_2$ and $T_1 - T_2$ are independent in characteristic zero, two copies of $T_1 + T_2$ are not, and merely disjoint supports do not suffice. Source 03 suggests a leading-term test for the coloured Jacobian whose rows come from different coefficient patterns and dilation lists; source 02 suggests mixed matroid data.

Research question 22 (Finite supports under real dilations; source 03). For finite-support surcomplex f and distinct positive reals $a_1, \dots, a_m$, determine the dependence pattern from the rational relations among all products $a_j g$, $g \in \text{supp } f$. The rational theorem and the real-rank lower bound are two constraints, not a classification; one monomial already detects rational independence of the scalars.

Research question 23 (Beyond scalar exponent dilation; source 02). For an ordered additive automorphism θ of the exponent space, when can the leading-determinant method control $y, S_\theta y, \dots$? Scalar dilation admits strict rearrangement, which a general shear need not. Find hypotheses replacing positive scalar ordering and classify the simplest resonance obstructions.

Research question 24 (Character twists and moving coefficients; source 02). Replace S_q by $\sigma(\sum c_\gamma t^\gamma) = \sum \varphi(c_\gamma) \chi(\gamma) t^{q\gamma}$, with φ acting on coefficients and χ a multiplicative character. When do iterates keep the uniform algebraic matroid? Which parts of the proof survive when the coefficient vectors of different rows are no longer equal up to scalars?

Research question 25 (Frobenius-twisted coefficients; source 03). Extend theorem 26.1 to coefficients in $\mathbb{F}_{p^e}$ or a general perfect field. Dilation by p then interacts with the inverse Frobenius twist on coefficients, and p -free parts alone need not keep all information; seek an invariant combining lattice rank with coefficient Frobenius orbits.

Research question 26 (Positive characteristic with inseparability separated; source 02). Can a useful rank bound be recovered after quotienting out Frobenius relations, or with Hasse derivatives and p -independence? Even parameters prime to p may interact through inseparability, and a valid result must separate that from Hahn support cancellation.

Research question 27 (Effective infinite-support certificates; source 03). Specify representations of infinite supports for which the greedy choices and their dominance have finite, mechanically checkable certificates, for example recursive supports with an effective rank oracle, or finite grammars for nested monomial scales. Separate summability, equality of exponents, and finding the least support element outside a span; none should be treated as decidable merely because the theorem is finite dimensional.

Research question 28 (Formalization and proof certificates; high priority; sources 03 and 02). Formalize the finite greedy-dominance lemma, strict rearrangement in an ordered vector space, the derivative-matrix criterion, and then the Hahn summability argument (section 27.3). A first milestone is the finite Laurent polynomial theorem with an executable certificate checker that records a rational support basis, the selected greedy exponents and the unique nonzero leading coefficient, avoiding the full Jacobian; a second is the relative well-ordered-support theorem, without postulating its nonvanishing conclusion as an axiom. Finite tests are not formal verification.

Research question 29 (Orbit-free extensions and further surreal operations; source 03). Use the tail of theorem 25.3 to study extensions of set-sized difference fields inside $\mathbf{No}[i]$ with all positive rational or real dilations named. Which free orbit extensions can also satisfy prescribed conditions involving Conway simplicity, the natural surreal derivation, or the surreal exponential? The construction controls exponent-linear algebra and algebraic relations only.

Research question 30 (Birthday bounds and restricted universes; source 02). Theorem 25.5 works over every set-sized base but gives no optimized birthday bound. For a birthday-bounded base and a cardinal κ , how low in the surreal hierarchy can such a family be placed? Which initial subfields contain enough fresh inner scales for the construction?

27.6 What the additions do not claim

Every limitation stated by source 03 or source 02 is kept, with its source; items marked [merge] were added in the merge.

- R1. Neither source is refereed or independently reviewed; source 02's audit states that its proofs and audit were prepared by the same assistant and that no external reviewer was consulted, and source 03's proofs were reviewed only during preparation. Both are AI-assisted drafts, and no theorem is Lean-verified or checked by another proof assistant (both).
- R2. No named published conjecture is claimed solved. The questions answered are this report's research questions, identified at fixed pins. Novelty is proposed and priority is not certified; the literature comparisons were focused, not exhaustive, and an absence of search results is not proof of priority (both).
- R3. Classical inputs are credited, not claimed: Hahn arithmetic and Conway normal forms [15, 14, 4], greedy selection [16], the Jacobian criterion [19], series derivations [17], power-sum independence [18, Proposition 3], the decreasing-denominator series [10, Remark 2.18], and the character method for ramification. No priority is claimed for Hahn summation, the Jacobian criterion, power-sum independence, or the bare existence of large algebraically independent families of surreals (both).
- R4. Characteristic zero is required; there is no unrestricted positive-characteristic version (Frobenius), and the prime-field formula covers only $\mathbb{F}_p$ coefficients and the support-generated lattice (both; the formula is source 03's).
- R5. Distinct positive parameters are required: repeated parameters, $d = 1$, negative dilations and zero are excluded, with the rank-two counterexample $T_1 + T_1^{-1} + T_2 + T_2^{-1}$ (both).
- R6. There is no classification of infinite supports of finite rank. For infinite support the rank bound is necessary, not sufficient, and the lower bound is not an equality; nothing bounds support order type, denominator growth, accumulation at cuts, or the support module or lattice generation asked for in Research questions 2 and 8 (both).
- R7. The Euler derivations are auxiliary algebraic probes: not the Berarducci–Mantova derivation, not asserted to be Hardy-type or exponential-compatible, and nothing is claimed about differential transcendence under a natural surreal derivation (both).
- R8. S_q is exponent dilation: not a power map, not the surreal exponential, not argument dilation. Nothing is claimed about factorization, primality, Diophantine decidability, integral-closure or normalization fibres, analytic convergence, birthdays, exponential-field automorphisms, or the first-order theory of the surreals with dilations (both).
- R9. No independence over all of $\mathbf{No}[i]$ is claimed, which is impossible; the base is fixed before the fresh scales; the families are not transcendence bases and do not generate $\mathbf{No}[i]$; no global class basis is used (both).
- R10. Finite tests are not proofs: the recorded counts are check units, and they certify no infinite, summability, basis-extension, character-extension or proper-class statement and no priority (both).
- R11. The algorithms assume exact finite input (exponent coordinates, or a certified rationally independent basis, over an effective field); they do not decide equality or rational dependence of arbitrary surreal expressions, and the greedy choices for infinite supports are existence arguments, not terminating procedures (both).
- R12. Exact order is not reconstruction: the uniform matroid computes no minimal equation and no extension degree, and symmetric profiles cannot be recovered beyond their symmetries (both).
- R13. $(\mathbf{No}[i], S_d)$ is not existentially difference closed: $S_d x - x = 1$ has no solution. The embeddings of corollary 25.6 do not embed $\mathbb{C}$ in $\mathbf{Oz}$, and equivariance with a coefficient field needs a specified action (source 02).

- R14. Theorem 24.5 and lemma 24.6 are necessary conditions only; a series on a fixed lattice can be transcendental (both). The exact order of $u_d^{[s]}$ is not proved (source 02).
- R15. The auxiliary weight in example 23.2 is not the actual order of the surreal exponents; it proves only that a Laurent polynomial is nonzero (source 02). The upper bound of theorem 22.2 by Jacobian rank is proved only in a finitely generated rational function field (source 02).
- R16. Real rank is not an exact upper bound for real dilations (ω against $\omega^{\sqrt{2}}$) (source 03).
- R17. The certificate certifies independence and produces no relation; its polynomial-time claim is for encoded finite input; the program's full expansions are not the proposed method (source 03).
- R18. The ordinal family is not a transcendence basis, and there is no uniform threshold for all parameter sets (source 03). The exact order of $u_{d,\ell}$ for $\ell \geq 2$ is not proved (source 03).
- R19. The source comparisons are pinned: source 03 read, through a GitHub connector at efc5446, the root and documentation catalogues and this report's README, source and directory, and selected quotient and normalization material; source 02 read, at 958b5c4, this report's source (its Sections 13 and 16) and the single-dilation report's README, and surveyed the root and documentation guides. Neither is a line-by-line review of the collection, and neither made a repository Lean build. Source 02's calculations were run locally with Python and SymPy, not through a Wolfram connector; its sources Edmonds and Beecken–Mittmann–Saxena were consulted at metadata or theorem level; source 03 read Melánová–Sturm–Winter in full HTML, Edmonds, Gonshor and L'Innocente–Mantova by publisher record, and Kuhlmann–Matusinski by abstract only.
- R20. [merge] No `adr:src` label has a Lean mapping in `FORMALIZATION.md`.
- R21. [merge] Remark 20.12 is a rank datum for Research question 5, not an answer, and remark 20.13 answers only the relative-rank part of Research question 8.
- R22. [merge] Source 02 does not answer source 03's question on $u_{d,\ell}$ (remark 24.10).
- R23. [merge] Source 03's text (its source-audit appendix) says that page count, build diagnostics and file hashes are in the delivery's build record; the shipped `data/03-support-rank-BUILD_RECORD.json` contains no file hashes (Appendix C.6).

A A worked elimination identity

Let $X = T^2 + T$ and $Y = T^4 + T^2$. From $T^2 = X - T$ one obtains

$$T^4 = (X - T)^2 = X^2 + X - (2X + 1)T,$$

and hence

$$Y = X^2 + 2X - 2(X + 1)T.$$

Therefore

$$T = \frac{X^2 + 2X - Y}{2(X + 1)}.$$

The denominator is nonzero in $k(T)$ in characteristic zero. Substitution into $T^2 + T - X = 0$ and clearing denominators gives

$$Y^2 - (2X^2 + 6X + 2)Y + X^4 + 2X^3 + 2X^2 = 0.$$

This simultaneously exhibits the relation and reconstructs the primitive monomial from the pair. The degree theorem proves minimality; the calculation alone would not rule out a hidden linear relation.

B Reproducing the finite checks

The file `verify.py` requires Python 3.9 or later and SymPy. It uses only exact arithmetic. Run

```
python -m pip install sympy
python verify.py
```

The second command writes `verification_results.json` beside the script and prints the same report. The recorded run used Python 3.13.5 and SymPy 1.14.0. The script is not an implementation of infinite Hahn series and does not pretend that truncation is an infinite proof. [write] In this directory the script is `code/verify.py` and the recorded run is `data/verification_results.json`; see Appendix C.4 before rerunning.

For the inverse Böttcher checks, set $L(z) = zF(1/z)$ for a monic centered polynomial $Q(X) = \sum q_j X^j$ of degree d . The checked identity is

$$L(z^d) = \sum_{j=0}^d q_j z^{d-j} L(z)^j.$$

The coefficient of degree m on the right contains $d[z^m]L$ plus already known coefficients. Since $d \neq 0$, this is an exact recursion. The program independently recomputes the two sides after constructing the coefficients and verifies equality through degree 32.

For the finite sums $u_{d,N} = \sum_{n=0}^N \omega^{d-n}$, the checked identity is deliberately the correct truncated one,

$$S_d u_{d,N} - u_{d,N} = \omega^d - \omega^{d-N},$$

not the infinite identity with its boundary term silently deleted. The infinite identity follows separately from coefficientwise comparison of the full well-ordered supports.

C Provenance of this report

C.1 One manuscript

This report is built from *one* manuscript: the archive `surreal_dilation_rigidity`, manuscript 08 of batch 30 of the collection, dated 23 September 2026 and pinned to repository commit `b895e8672990e8b5a56f97dd7246f9dd5f86c771`. It was placed in commit `21375f8` as a new report: no report of the collection names its question, and it continues the single-dilation report's subject without answering any of that report's questions (section 1.5). There was no second source; nothing was merged and nothing was selected away. Every result, proof, example, question and limitation of the manuscript is printed here. [merge] This subsection describes the report as first written. Since batch 33 it also prints two later manuscripts, sources 02 and 03, described in Appendix C.5; the sentences above about one source hold for Sections 1–18 and Appendices A and B.

The delivered verification program `code/verify.py`, its recorded run `data/verification_results.json`, the dependency pin `data/requirements.txt` and the author-side audit `SOURCE_AUDIT.md` are kept byte-identical. The manuscript's article (delivered as `dilation_rigidity.tex`) and `README` were rewritten; the delivered PDF and checksum list are not shipped, and `article.pdf` is a build of this text.

The editorial changes are these.

- (1) Every label carries the prefix `adr:`. The source's 83 labels are kept, unchanged after the prefix, and eight were added (`adr:sec:conventions`, `adr:sec:collection`, `adr:sec:nonclaims`, `adr:app:report`, `adr:app:onesource`, `adr:app:pinned`, `adr:app:literature`, `adr:app:files`).

- (2) The title page names the report as a single-source report of the collection, gives the batch and pin, and extends the status box. Sections 1.4 and 1.5, the ledger of section 18 and this appendix are new, and the bibliography gained the entry [8]. Sentences marked [write] were added at the end of the source's subsection on the repository, at the end of its literature paragraph on Mahler equations, after lemma 3.3, after theorem 13.2, in section 14 (two places), after the question on nonintegral rational dilation, in section 17 and in Appendix B. The bibliography entry [7] gained one sentence.

No mathematical statement was changed, and no symbol was renamed: the collisions are between reports, and section 1.4 fixes the meanings. Where the edit had to choose: the manuscript was placed as a new report rather than as an addition to the single-dilation report, because it answers no question or named gap of that report; it was placed among the surcomplex reports, because its main theorem is uniform in the characteristic-zero coefficient field, its omnific corollary treats $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ together, and its nearest neighbour is there; the title is kept; and the directory name records the subject, first-order autonomous relations between an element and its exponent dilate.

C.2 Repository statements at the pin

The source compared itself with the collection at its pin. Checked against the placement commit 21375f8, its statements stand.

- The single-dilation report uses the recognizer $S_2w = w^2$ (`dsup:lem:monomialrecognition`), treats linear and multiplicative dilation equations (`dsup:thm:dilationcohom`), proves undecidability of the expanded field (`dsup:thm:undecidable`), and recovers the normal-form data from one dilation (`dsup:thm:definability`).
- No report of the collection supplies the classification of theorems 6.1, 8.1 and 10.1; superattracting germs and Böttcher coordinates occur in no other report.
- The three reports the source inspected (single dilation, dynamics, entire functions at arbitrary rank) are unchanged between the pin and the placement commit, as are the surcomplex-automorphisms and exponential-automorphism reports compared in section 1.5. The holonomic-rigidity report gained material after the pin; none of it concerns exponent dilations.

C.3 References checked for this report

The placement audit confirmed the references to Neumann [4], Mannaa–Coquand [5], Salerno–Silverman [6], Chyzak–Dreyfus–Dumas–Mezzarobba [10] and Faverjon–Roques [11]. The following were checked when the report was written.

- *L'Innocente–Mantova* [14]. In the introduction of arXiv version 5, Theorem B states that the omnific integer $\omega^{\sqrt{2}} + \omega + 1$ is prime in $\mathbf{Oz}$; the abstract describes this as Gonshor's conjecture. The attribution in section 10 is therefore correct. The arXiv page lists the journal DOI 10.1016/j.aim.2024.109513; the volume number was not checked.
- *Nishioka–Nishioka* [7]. The bibliographic data were confirmed on Project Euclid, and the full text (seven pages, open access) was read. It works with ordinary convergent power series over a number field or $\mathbb{C}$ and proves transcendence of values. It records Mahler's 1983 criterion [8] (Theorem 2 there): a convergent power series $f(z) = f_0 + f_r z^r (1 + \sum_{j \geq 1} f_j z^j)$ satisfying $P(f(z), f(z^d)) = 0$, for an irreducible P with algebraic-number coefficients that is not a constant multiple of $u - v$, exists if and only if three conditions

on a Puiseux expansion of the curve $P = 0$ at a point (f_0, f_0) hold. It also records S. Nishioka’s algebraic independence theorem for solutions of first-order rational Mahler equations of distinct degrees; the example $f(z^2) = f(z)^2/(1 - 2f(z)^2)$ with solution $(z^r + z^{-r})^{-1}$, whose reciprocal is the identity $y = T + T^{-1}$, $S_2y = y^2 - 2$, of section 12; and a computation showing that $g(z^d) = g(z)^d + c$ with $d \geq 3$ and $c \neq 0$ has no solution $g \in \mathbb{C}(z)$ (the proof first reduces to Laurent polynomials), a rank-one relative of theorems 11.2 and 12.1. It contains nothing on Hahn series of higher rank, support collapse, negative-support rings, relation degrees or omnific integers. Mahler’s criterion is a rank-one antecedent of the existence statements in theorem 7.3 and corollary 7.4 and of theorem 7.1, restricted to ordinary power series (non-claim W2). Mahler’s paper itself was not read.

- *Gontsov–Goryuchkina* [9]. Title, journal, volume, year and article number were confirmed through the DOI registry; the paper was not read, so the source’s use of it for the decreasing-denominator pattern is unverified here.
- *Faverjon–Poulet* [12] and *Faverjon–Roques* [13]. Titles, authors and arXiv numbers were confirmed on arXiv (submitted 24 November and 11 December 2025). Only the abstracts were read, as by the source. The first abstract describes an algorithm computing a basis of solutions of any Mahler equation, with a decomposition over Puiseux series. The second describes a purity theorem, that certain arithmetic properties of a Mahler function propagate to all solutions of its minimal equation; the source’s summary of it in section 1 as developing generalized Mahler-series structure could not be confirmed from the abstract. Neither is a proof input.
- Basu–Pollack–Roy [20] and the encyclopedia page [21] were not checked.

C.4 Files as placed and rerun

File	Role
code/verify.py	the delivered finite checks (delivered as verify.py)
data/verification_results.json	the delivered recorded run
data/requirements.txt	the delivered pin sympy==1.14.0
SOURCE_AUDIT.md	the delivered author-side source and claim audit

The program writes `verification_results.json` beside itself, overwriting any file of that name, and prints the same report. In this layout a run in place would create `code/verification_results.json` inside the report directory. Appendix B and the audit describe the flat delivered layout. Run it on a copy, from the report directory, with a fresh scratch directory `$T`:

```
python -m pip install -r data/requirements.txt
cp code/verify.py $T/ && (cd $T && python verify.py)
```

and compare `$T/verification_results.json` with `data/verification_results.json`. This was done for this report under Python 3.14.4 with SymPy 1.14.0: exit code 0, status PASS, and the output agrees with the recorded run except for the field recording the Python version (3.13.5 in the recorded run). [merge] These are source 01’s files. The files of sources 02 and 03 are listed in Appendix C.6; source 03’s verifier, run in place without `-output`, would overwrite `data/verification_results.json` above, so it too is run only on a copy.

C.5 Sources 02 and 03 (batch 33)

[merge] Since batch 33 the report is built from three manuscripts. In its local numbering, source 01 is the original manuscript (section C.1); the two additions were placed in commit aa9c891 and keep the prefixes 02- and 03- of their staged files.

Local	Manuscript	Pin	Contributes
03	batch 33, manuscript 04; archive <code>surreal_dilation_rank_research</code> ; <i>Support Rank and Algebraic Independence of Dilation Orbits</i> ; 23 September 2026; 29 pages, US Letter	efc5446	The base of sections 19 to 27: the theorems over an ordered scalar field $\mathcal{F}$ relative to $K_H(k)$, rational descent and the reduced-quotient formula, the certificate, Ξ under real dilations, the ordinal family and uniform tail, the prime-field formula, the real-rank boundary, the $u_{d,\ell}$ family, twelve questions.
02	batch 33, manuscript 02; archive <code>surreal_dilation_rank</code> ; <i>Support Rank and Algebraic Freeness under Surreal Dilations: exact finite-profile order, higher-order support bounds, and free omnific difference algebras</i> ; 23 September 2026; 23 pages, A4	958b5c4	The same central theorems over $\mathbb{Q}$; the vector-valued greedy determinant; the coefficient-matrix route, Jacobian-rank lemma and algorithm; the second rank-three certificate; the $\mathbb{Z}^s$ -shift corollary; the character extension lemma and denominator index; the telescopes of rank ρ and order $\rho + 1$; κ -families and free difference algebras; the relative part of Research question 8; eleven questions.

Full pins: efc5446229dbf61a97bdd5edae91dba20af9d131 and 958b5c4865819bd55ea1f5ffc050282aea7ef570. Both are later than this report's original pin; at both, and at aa9c891, this report's `article.tex` is the file written in 260c143, so the sources' descriptions of it (proposition 13.4, Research questions 1, 2 and 8, theorems 13.2 and 13.3) were accurate and are not stale. Source 03 also read this report's README at its pin; it was unchanged at the placement commit, and is rewritten for this merge.

Where the merge chose. Source 03 is the base because it is more general: an ordered subfield $\mathcal{F}$ of k instead of $\mathbb{Q}$, real dilations, and further results. Source 02's proofs use only $\mathbb{Q}$, and were re-read for hidden uses of a stronger hypothesis; there are none. Theorems proved by both are printed once with both named: the Euler derivations, the Jacobian criterion, greedy dominance, rearrangement, the leading term and the support-rank obstruction (source 02's Theorem 4.3 is source 03's Theorem 1.1 at $\mathcal{F} = \mathbb{Q}$), the relation-rank corollary (source 02's Corollary 4.4, source 03's Corollary 5.2), the exact finite-support formula (source 02's Theorem 5.1, source 03's Theorems 1.2 and 5.1 with Corollary 6.6), the uniform matroid (source 02's Remark 5.3, source 03's Corollary 6.5), Ξ (source 02's Theorem 7.1 for $\Xi - \omega$ and rational dilations, source 03's Theorem 8.2), and the block theorem (source 02's Theorem 7.3, source 03's Theorem 9.1). Kept as marked second routes: the choice of functionals before the greedy basis (remark 20.9), the coefficient-matrix proof of exact order for rational profiles (section 22), the two character arguments (theorem 24.5 and lemma 24.6), and the two proofs of order two for u_d (proposition 24.7, theorem 24.8), which are second and third routes to theorem 13.2. Both rank-three certificates are printed. Near-identical questions of the two sources are printed once with both formulations.

Notation. Both sources use growth support; it is converted to this report's valuation convention (section 19.2), and the renamed symbols are listed there. In particular D_λ, D_ℓ become ∂_λ (this report reserves D for integers and auxiliary series), rank is ρ and order is aord (this report uses r for an order), and source 02's parameters are listed in decreasing order, with its original sign convention stated in remark 20.9.

Corrections made in printing. Source 03’s text has five unclosed quotation marks around the titles of Research questions 1 and 2 (two in its introduction, one in its scope section, two in its bibliography entry for this report); here the titles are not quoted, and the questions are cited by number. Source 03’s statement that its build record holds file hashes is false for the shipped record (non-claim R23). The sources’ “Section 13” and “Section 16” of this report are sections 13 and 16, unchanged. No mathematical statement of either source was found false.

Stale text in this report corrected in the merge. Status notes (batch 33) were added after proposition 13.4, after Research questions 1, 2, 5 and 8, to non-claim S22 and the closing recommendation of section 16, and to section C.1; the title page, abstract and status box now name three manuscripts; a credit to Melánová–Sturmfels–Winter follows theorem 13.3, which is their Proposition 3 when all $a_j = 1$ and which the original manuscript did not cite; section 1.5 gained a paragraph on the additions and on the independent-copies report; and the bibliography gained [15], [16], [17], [18] and [19]. These are marked [merge].

Literature checked in the merge. Proposition 3 of Melánová–Sturmfels–Winter was read in the arXiv HTML version: for m distinct positive integers $a_1, \dots, a_m$ and $m \leq n$, the map $\mathbb{C}^n \rightarrow \mathbb{C}^m$ given by the power sums $x_1^{a_1} + \dots + x_n^{a_n}$ is dominant, which is algebraic independence of those power sums; the proof shows that every maximal minor of the Jacobian is a Vandermonde determinant times a Schur polynomial, up to a constant. With $n = m = r$ and $a_j = d^{j-1}$ this is the independence statement of theorem 13.3 for $a_1 = \dots = a_r = 1$. The other new references ([15, 16, 17, 19]) are cited as the sources cite them and were not re-read for this report.

C.6 Files of sources 02 and 03, and their rerun

File	Role (delivery name)
03-support-rank-PROOF_AND_SOURCE_AUDIT.md	source 03’s proof and source audit
code/03-support-rank-verify.py	its exact checks, standard library (code/verify.py)
code/03-support-rank-build.sh, code/03-support-rank-build.ps1	its build scripts (build.sh, build.ps1)
data/03-support-rank-verification_results.json	its recorded run (data/verification_results.json)
data/03-support-rank-BUILD_RECORD.json	its build record (BUILD_RECORD.json)
02-dilation-rank-PROOF_AND_SOURCE_AUDIT.md	source 02’s proof and source audit
code/02-dilation-rank-verify.py	its exact checks, SymPy (verify.py)
code/02-dilation-rank-build.py	its build helper (build.py)
data/02-dilation-rank-verification_results.json	its recorded run (verification_results.json)
data/02-dilation-rank-BUILD_REPORT.json	its build report (BUILD_REPORT.json)
data/02-dilation-rank-requirements.txt	its pin sympy==1.14.0 (requirements.txt)

These files are byte-identical to the deliveries. The manuscripts, READMEs, PDFs and checksum lists of the two sources are not shipped. The delivered texts name their files by delivery names.

Source 03’s program rewrites its record in place. Its default output is the file data/verification_results.json one directory above the script, which in this layout is the *original* manuscript’s recorded run, not source 03’s. Never run it in place without `-output`. Its `build.sh` and `build.ps1` run it that way and then run `pdflatex` on `article.tex` beside them, so they must not be run in this directory either. Run it on a copy, from the report directory, with a fresh scratch directory \$T:

```
mkdir -p $T/code $T/data
cp code/03-support-rank-verify.py $T/code/verify.py
(cd $T && python code/verify.py)
diff $T/data/verification_results.json \
    data/03-support-rank-verification_results.json
```

This was done for this report under Python 3.14.4: exit code 0, status PASS, 1,490 checks, and the written record is identical to the shipped one up to line endings (it records no interpreter version or timing). The run takes about one second.

Source 02's program writes `verification_results.json` in the current directory unless `-output` is given, and its build helper writes `build-pass-N.txt` files and runs `pdflatex` beside itself. Run it on a copy:

```
python -m pip install -r data/02-dilation-rank-requirements.txt
cp code/02-dilation-rank-verify.py $T/
(cd $T && python 02-dilation-rank-verify.py --output out.json)
diff $T/out.json data/02-dilation-rank-verification_results.json
```

This was done for this report under Python 3.14.4 with SymPy 1.14.0: exit code 0, 549 named cases and 1,918 comparisons, the rank-three leading term $-96T_1^9T_2T_3^6$ with 45 terms, and the output agrees with the record except for the recorded Python version (3.13.5 in the delivery) and elapsed time (about 10 seconds here, 3.9 in the delivery). The build records state 29 and 23 pages for the delivered PDFs, which are not shipped.

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